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Bae

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(54) **LAUNDRY TREATING APPARATUS**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 782 days.

This patent is subject to a terminal disclaimer.

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(52) **U.S. Cl.**
CPC **D06F 58/24** (2013.01)

(58) **Field of Classification Search**
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USPC 34/73
See application file for complete search history.

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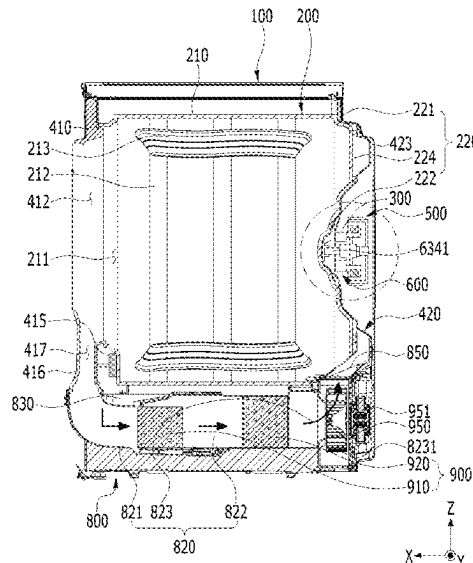
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(57) **ABSTRACT**

A laundry treating apparatus includes a cabinet, a drum, a base, and a motor. The base is disposed under the drum and guides air circulating to the drum. The base includes an air circulating channel, a heat exchanger, a water collector body, a cleaning water channel, a pump, and a channel switching valve. The channel switching valve is connected to the pump and receives water from the pump and delivers the water to the cleaning water channel. The channel switching valve includes a water receiving portion connected to the pump and a connective portion connected to the water receiving portion and coupled to the air circulating channel to deliver the water to the cleaning water channel. The connective portion is disposed at a side of the air circulating channel. At least a portion of the connective portion is positioned lower than a top surface of the air circulating channel.

20 Claims, 25 Drawing Sheets



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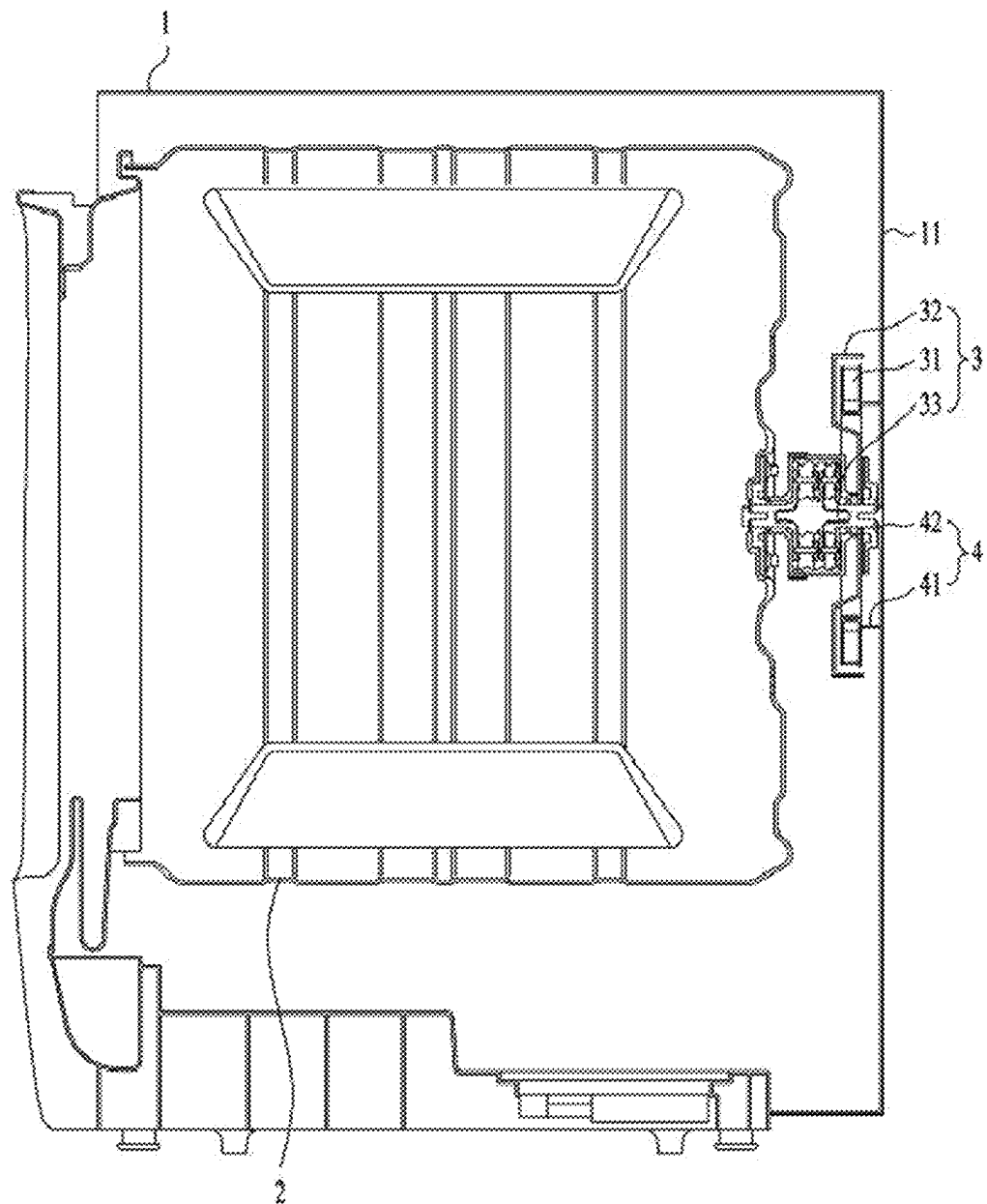
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FIG. 1



Related art

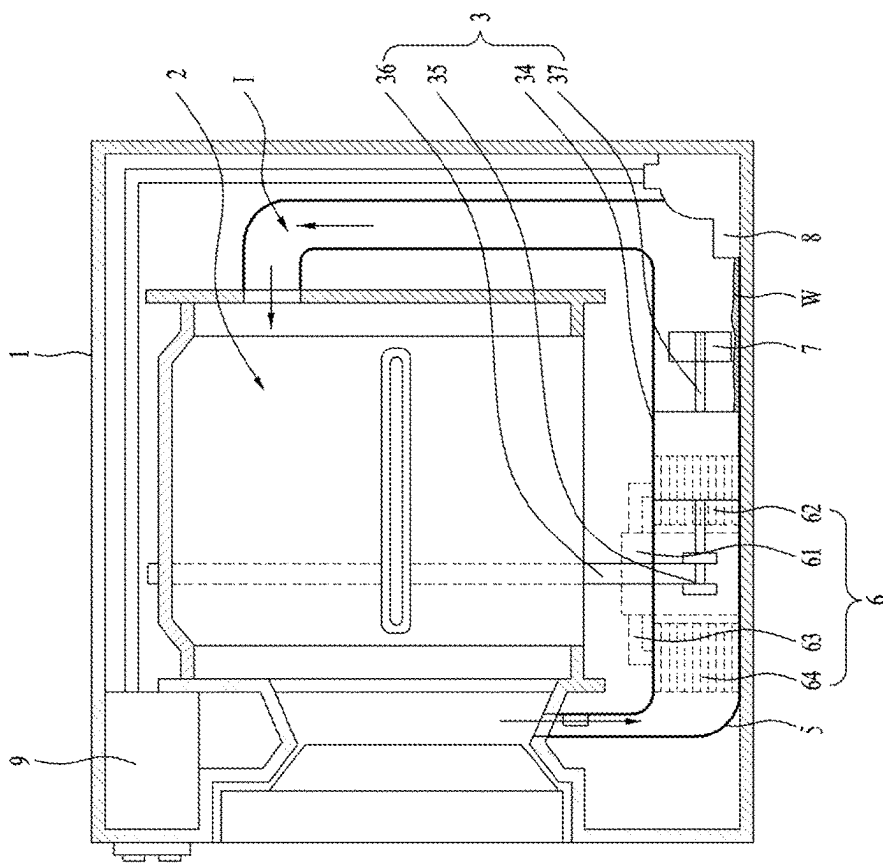


FIG. 2A

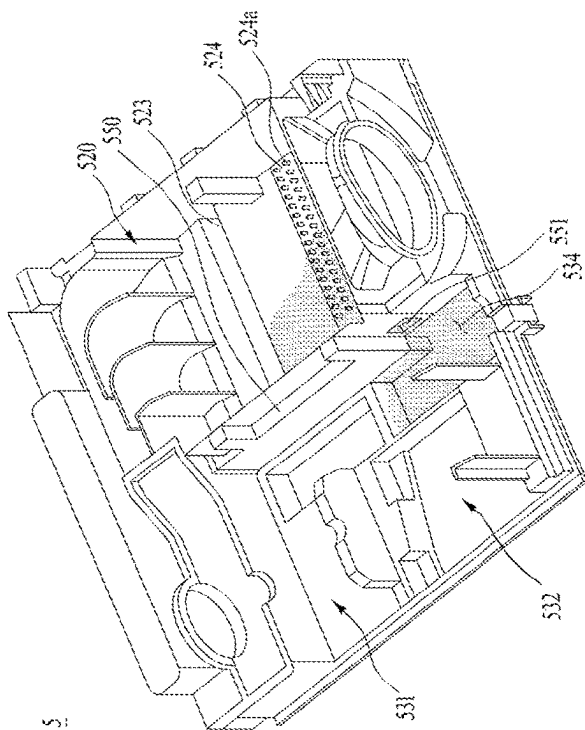


FIG. 2B

Related Art

FIG. 3

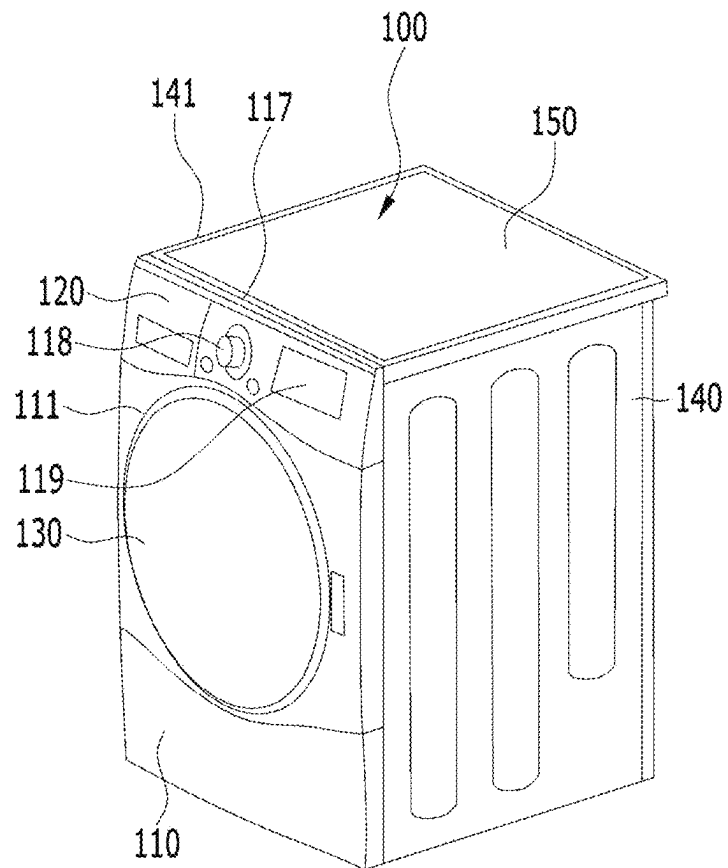


FIG. 4

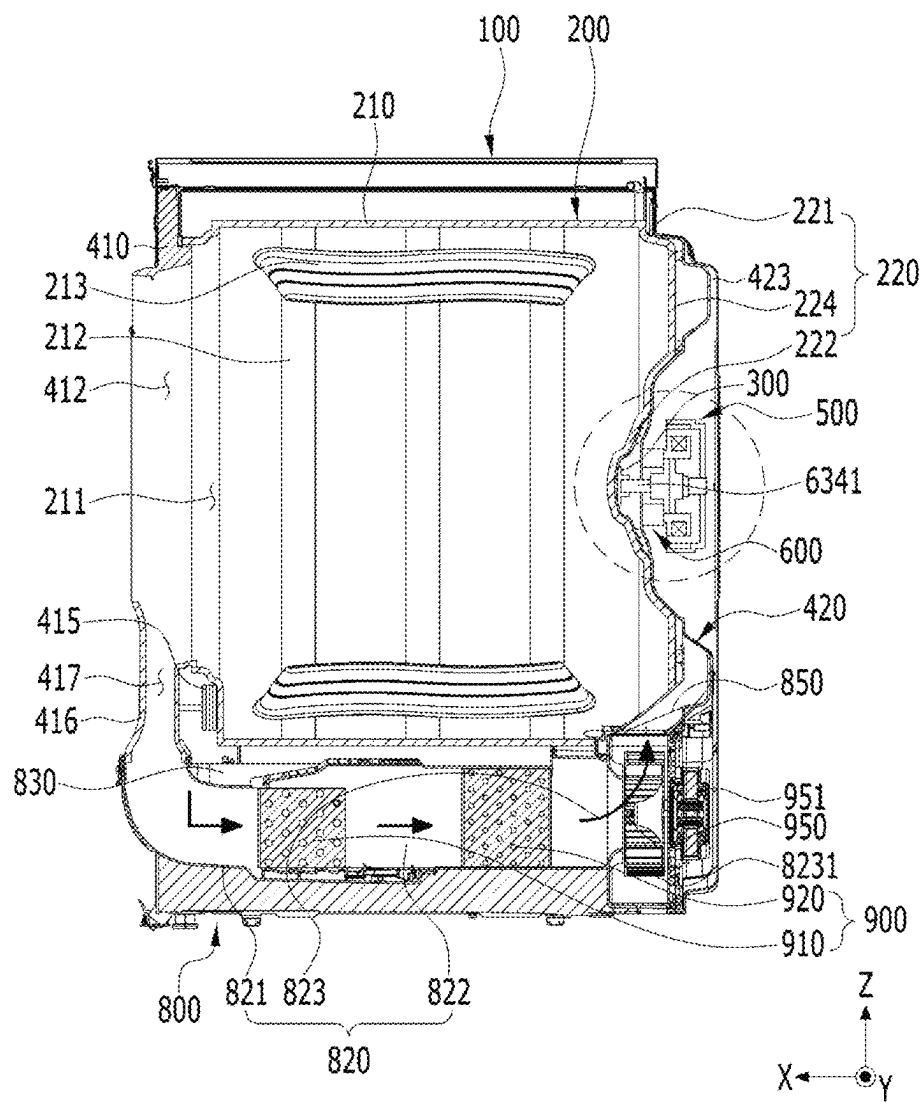
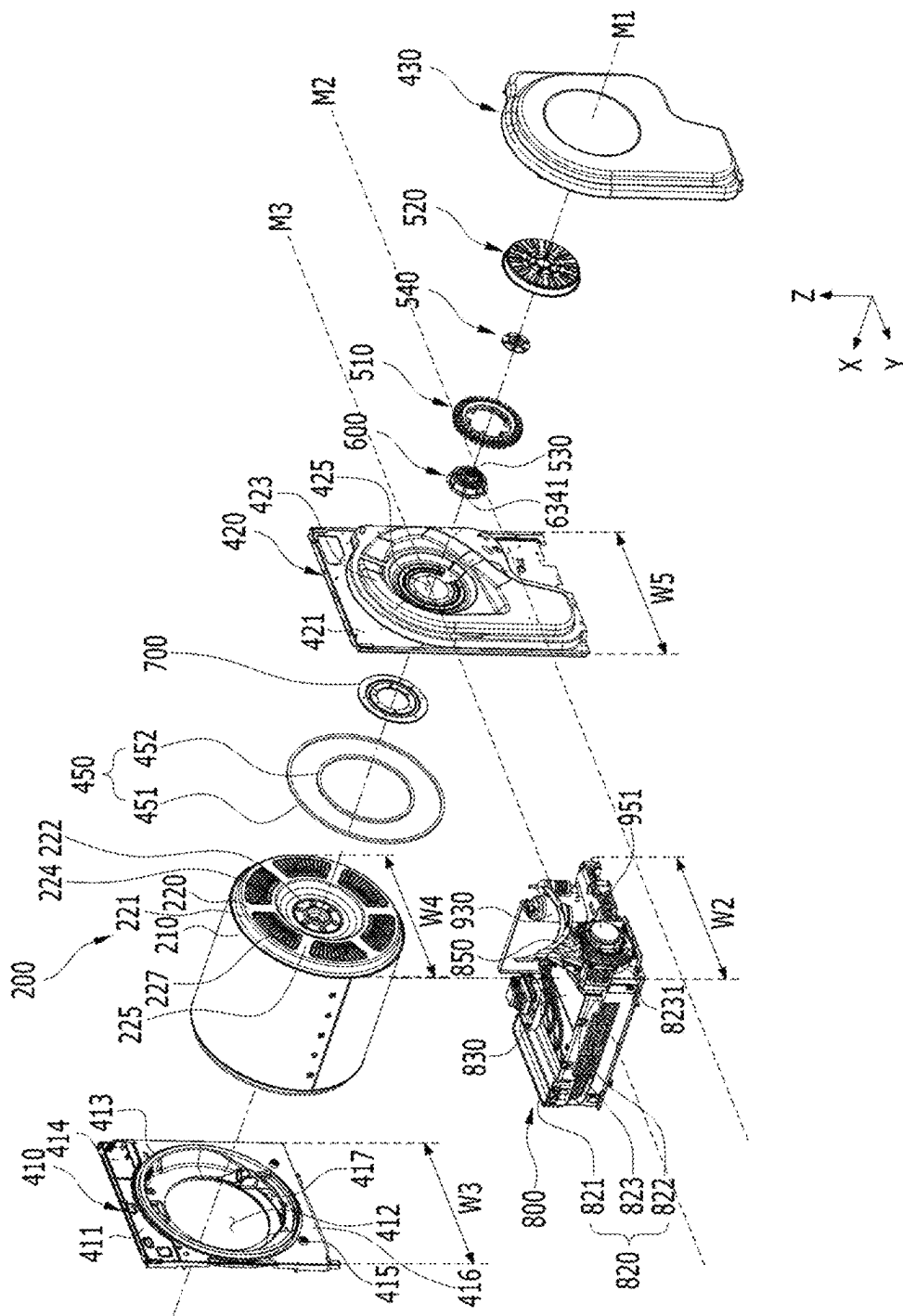


FIG. 5



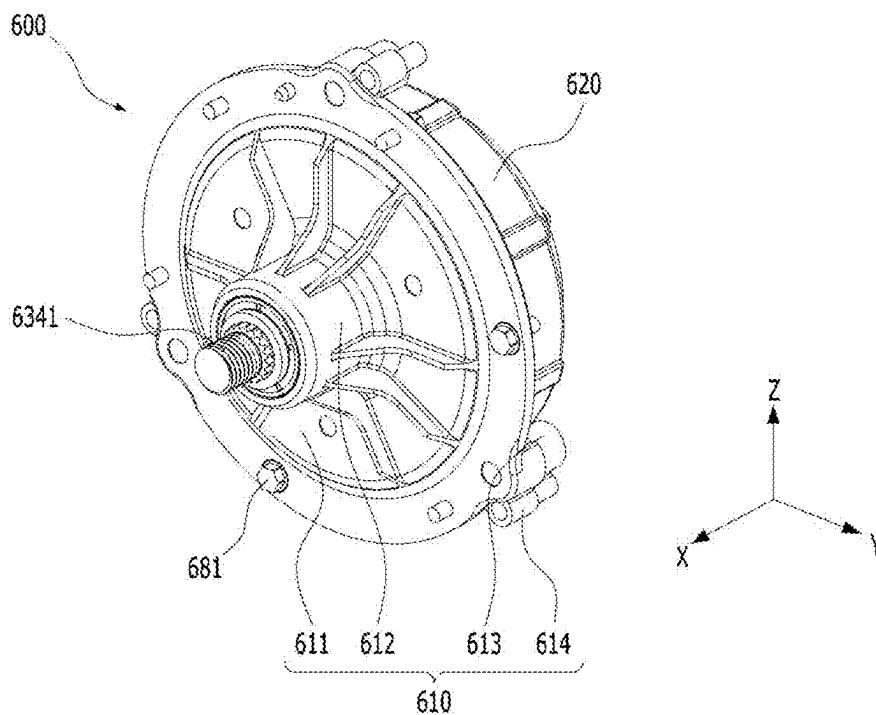


FIG. 6A

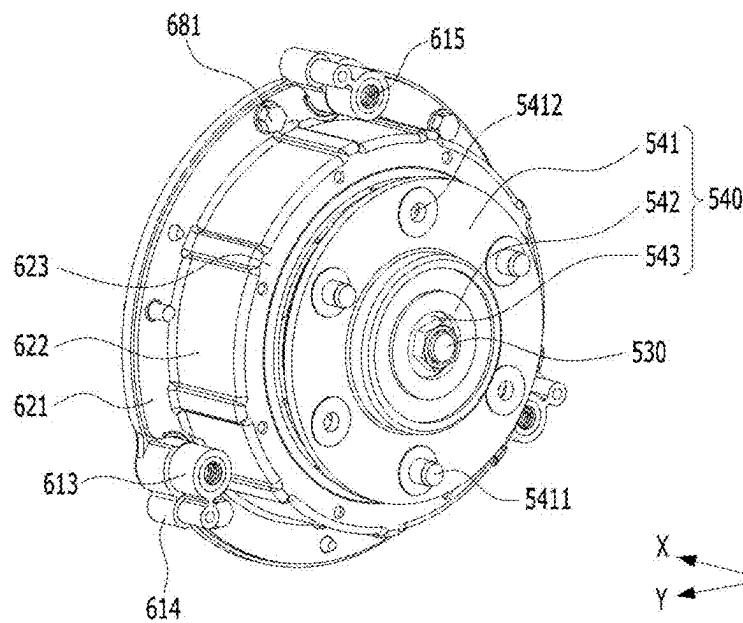


FIG. 6B

FIG. 7

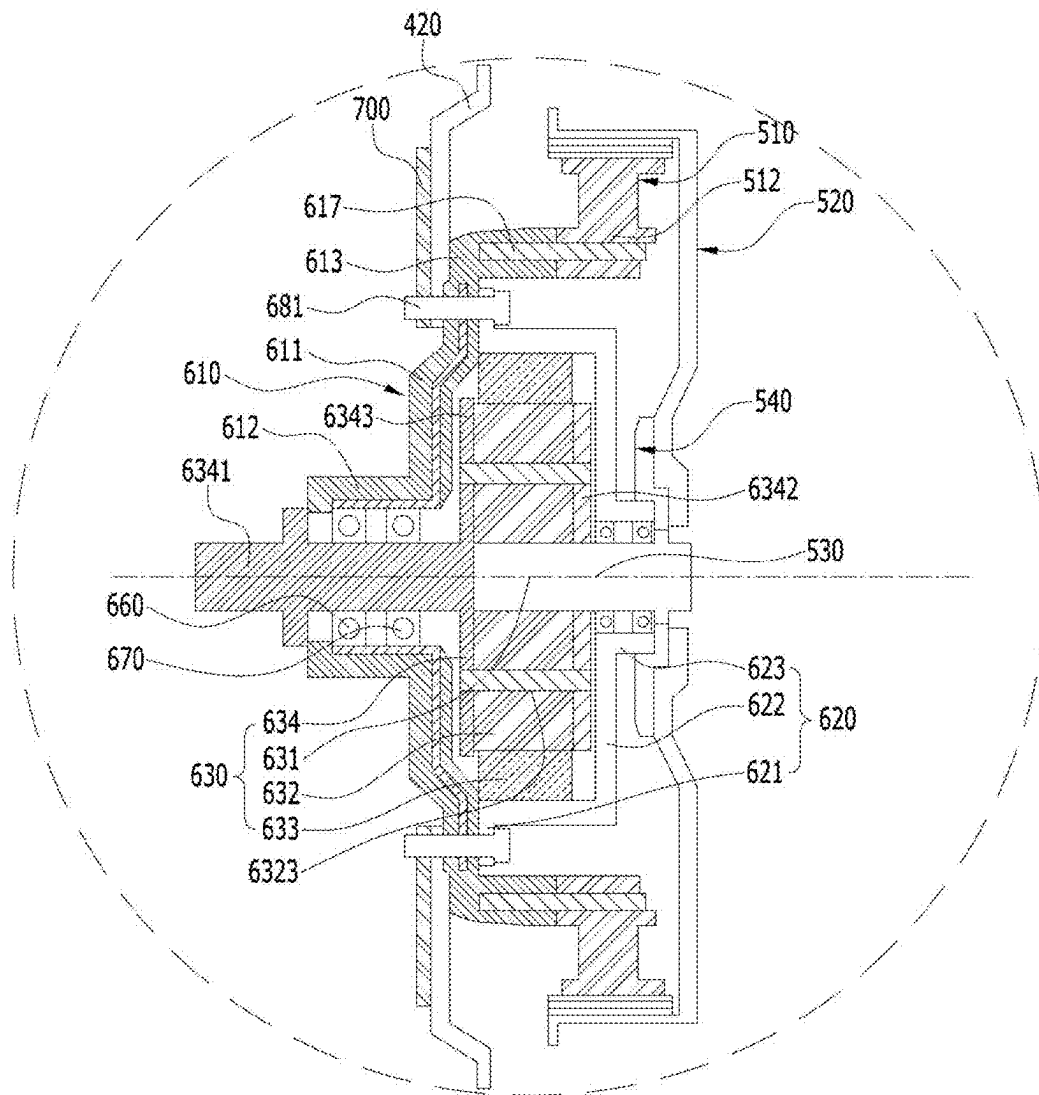


FIG. 8

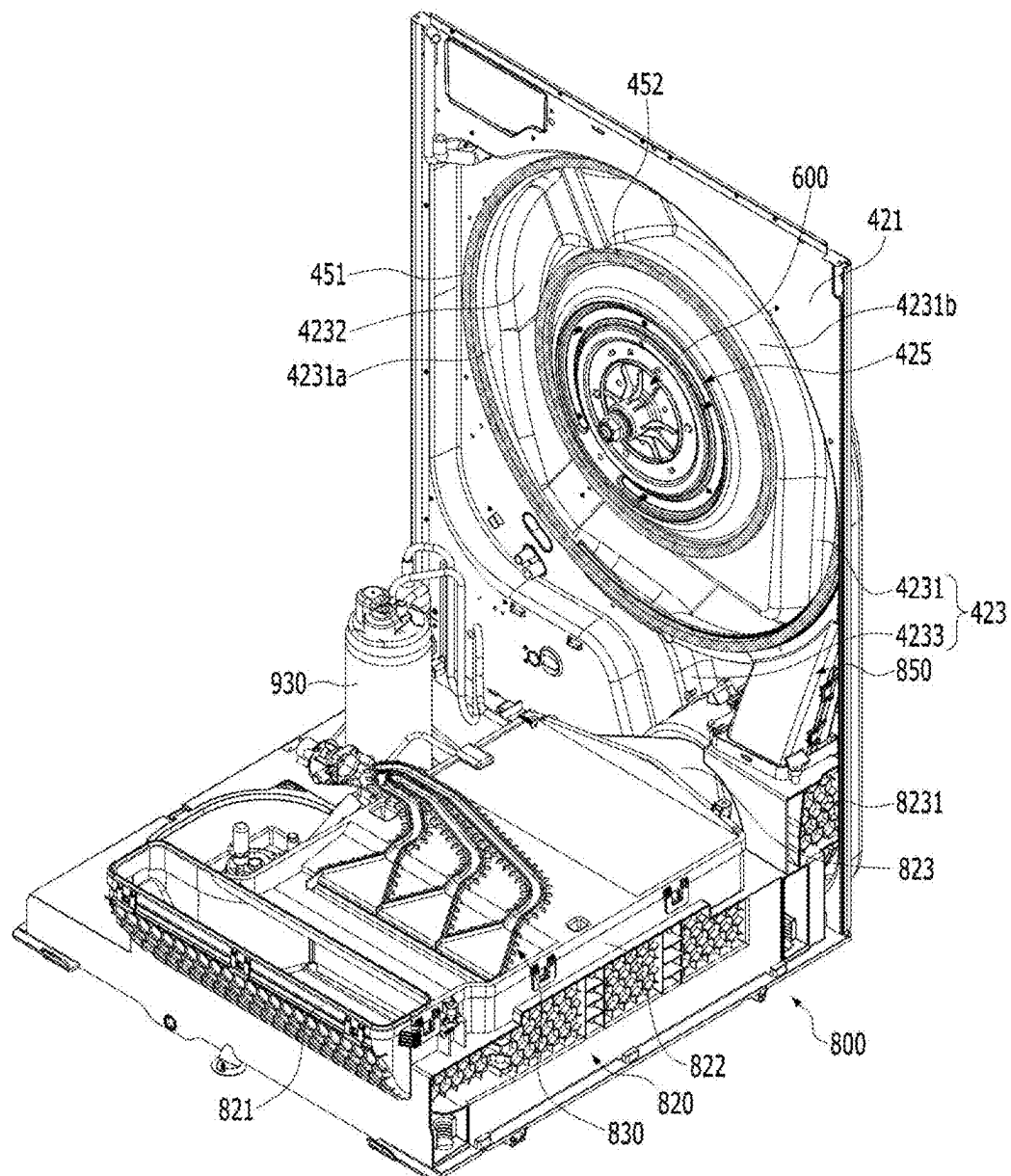


FIG. 9

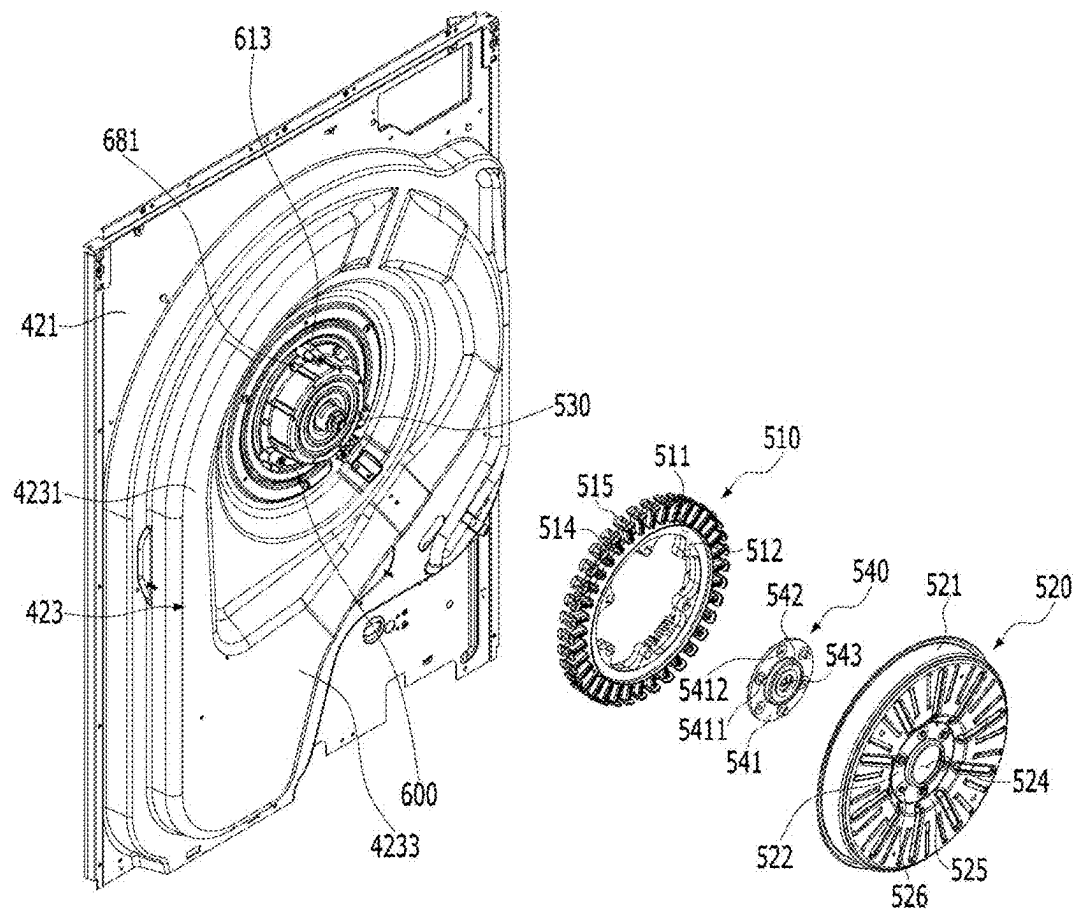


FIG. 10

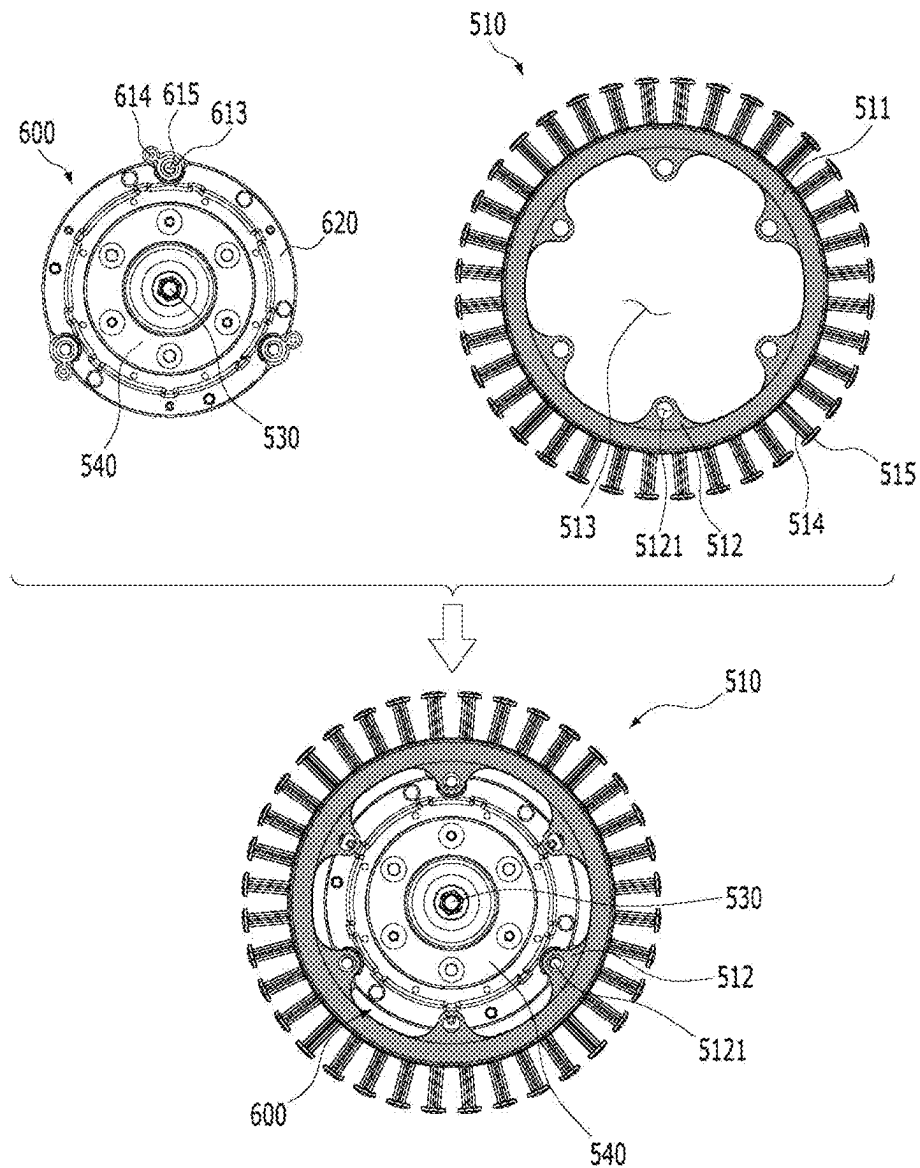


FIG. 11

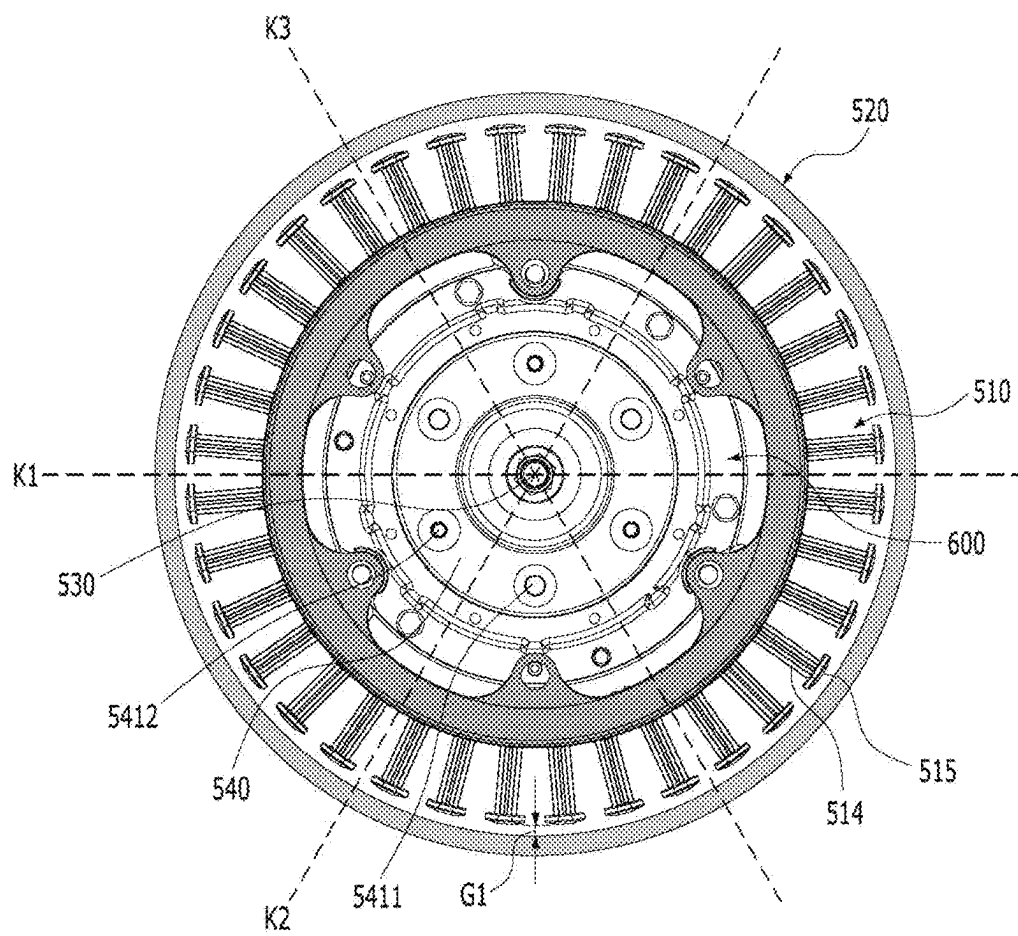


FIG. 12

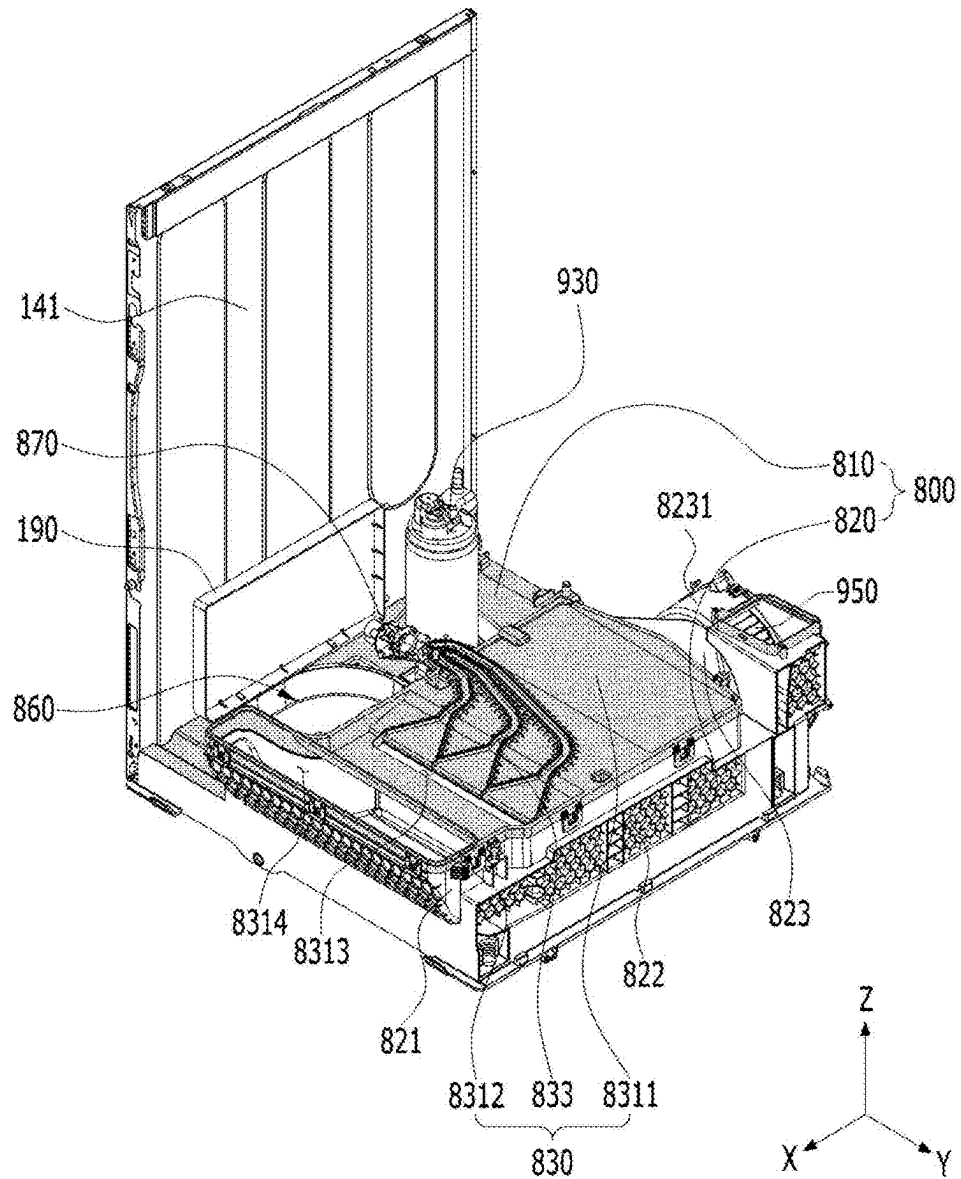


FIG. 13

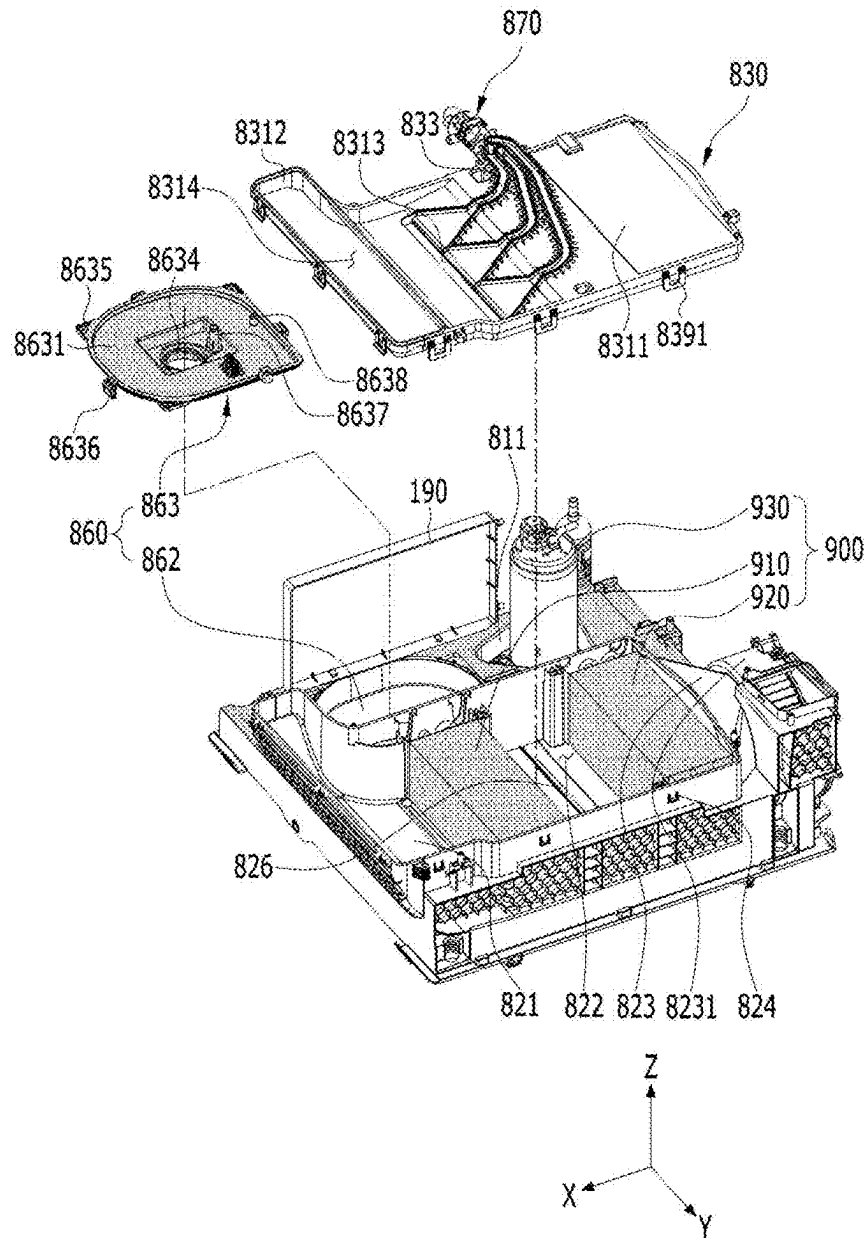


FIG. 14

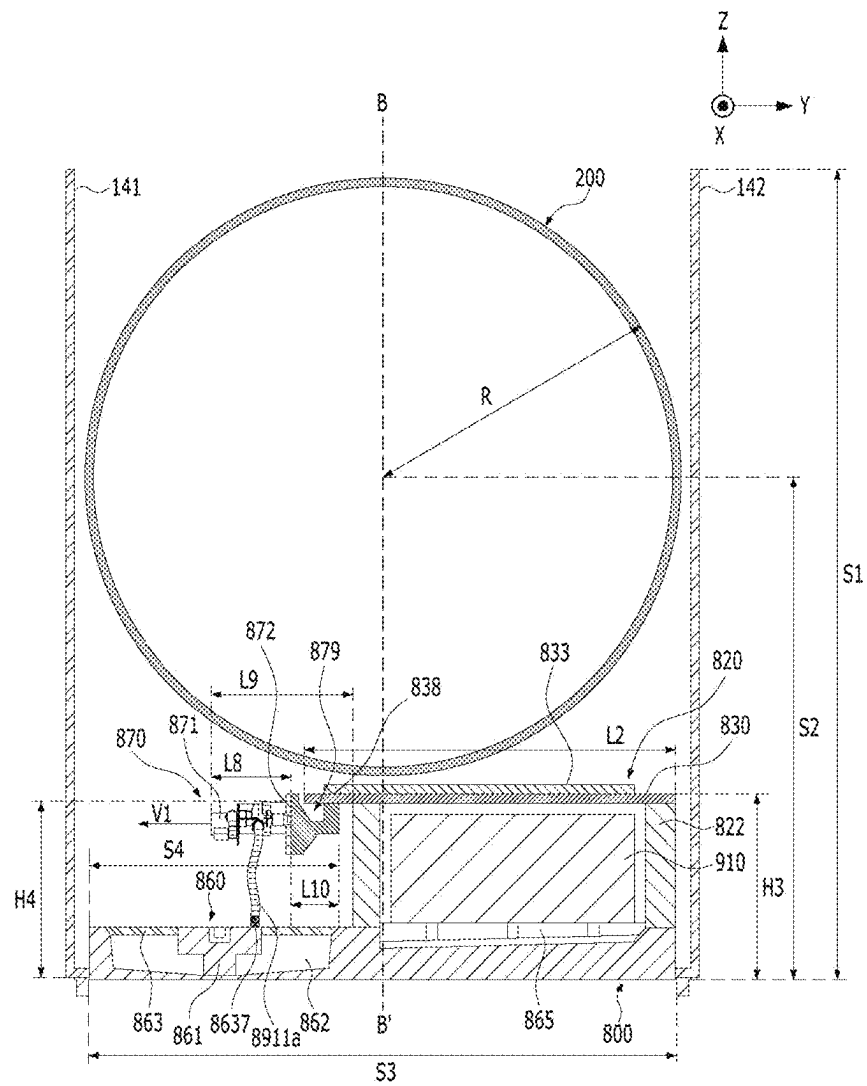


FIG. 15

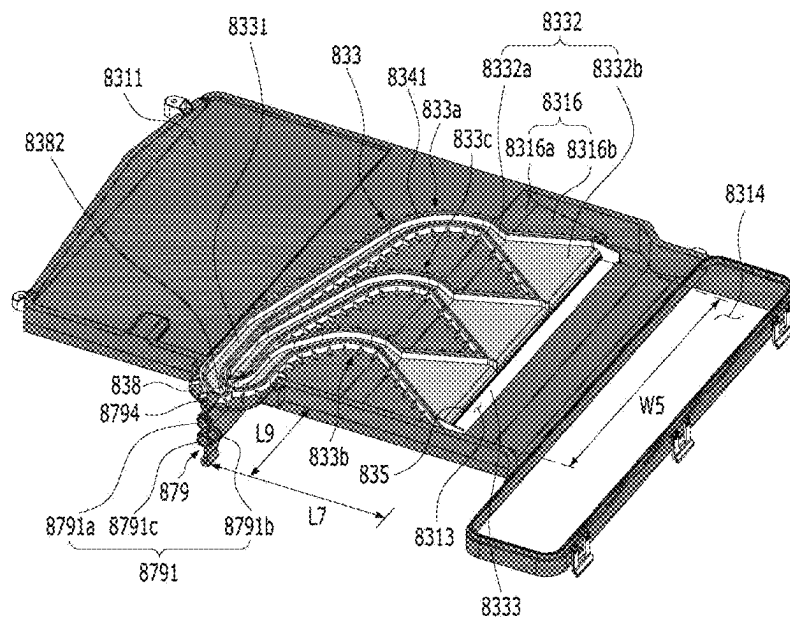


FIG. 16

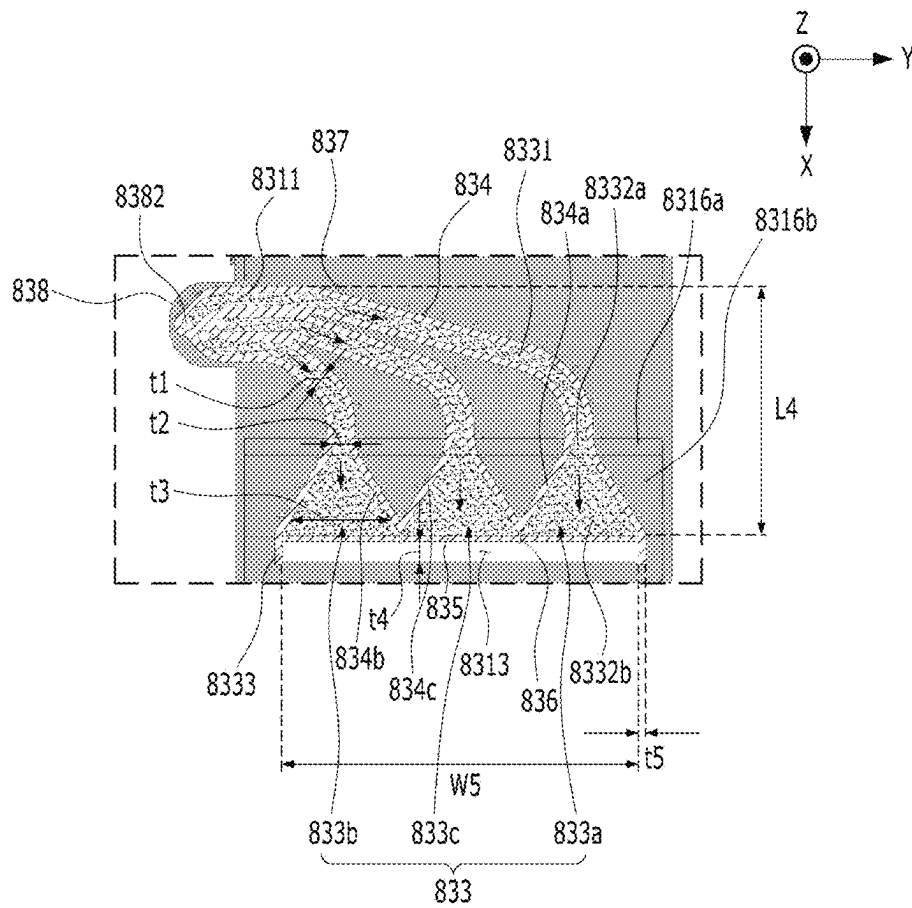


FIG. 17

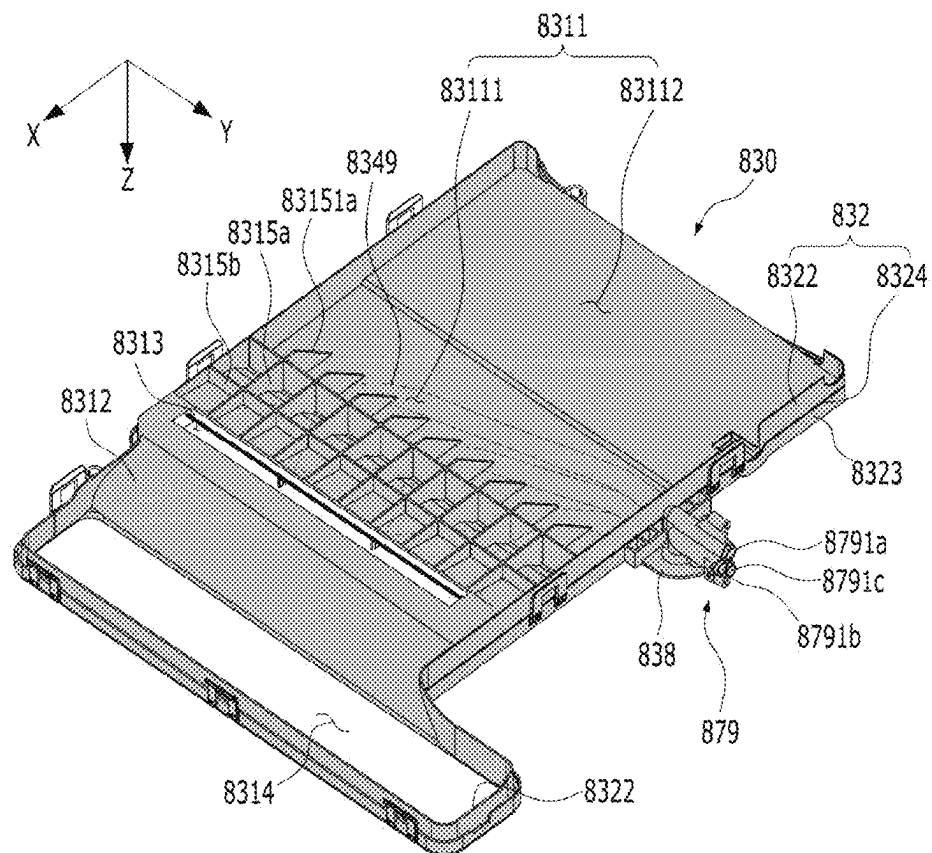


FIG. 18

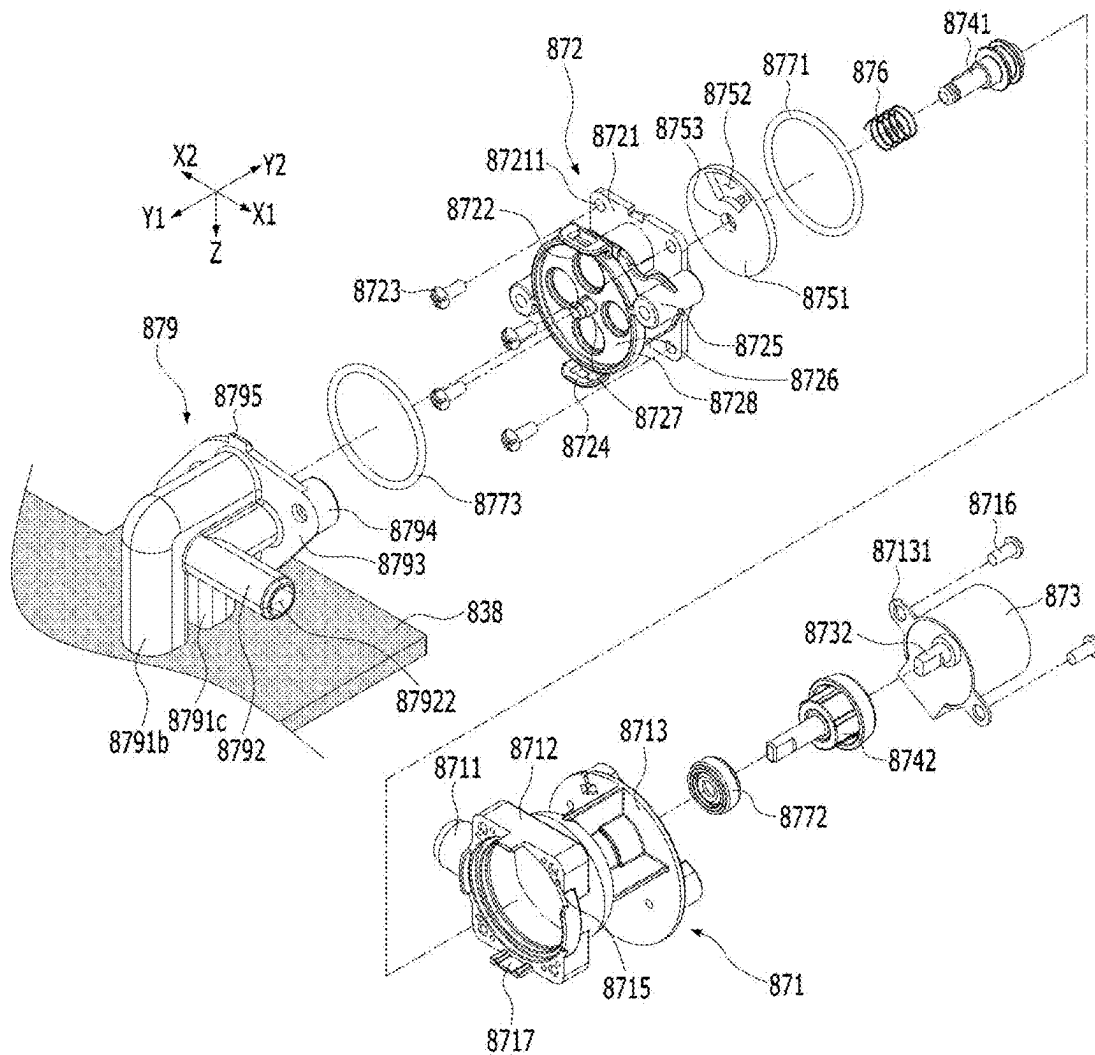


FIG. 19

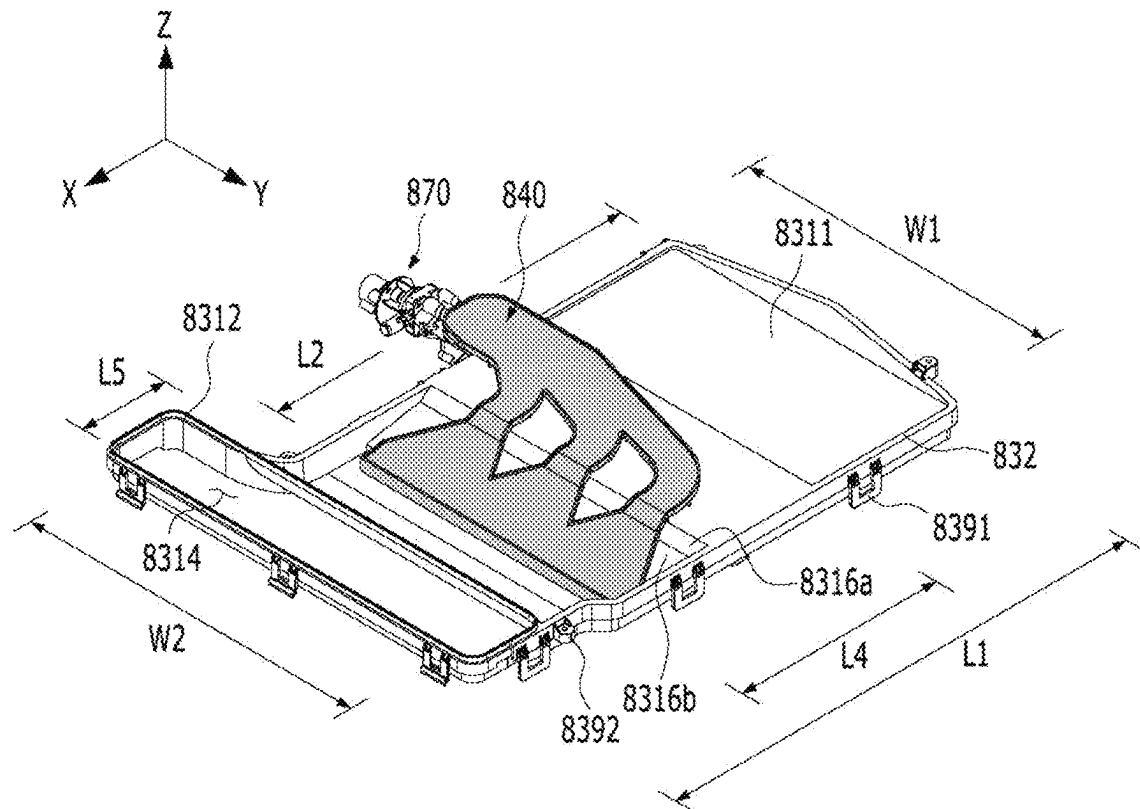


FIG. 20

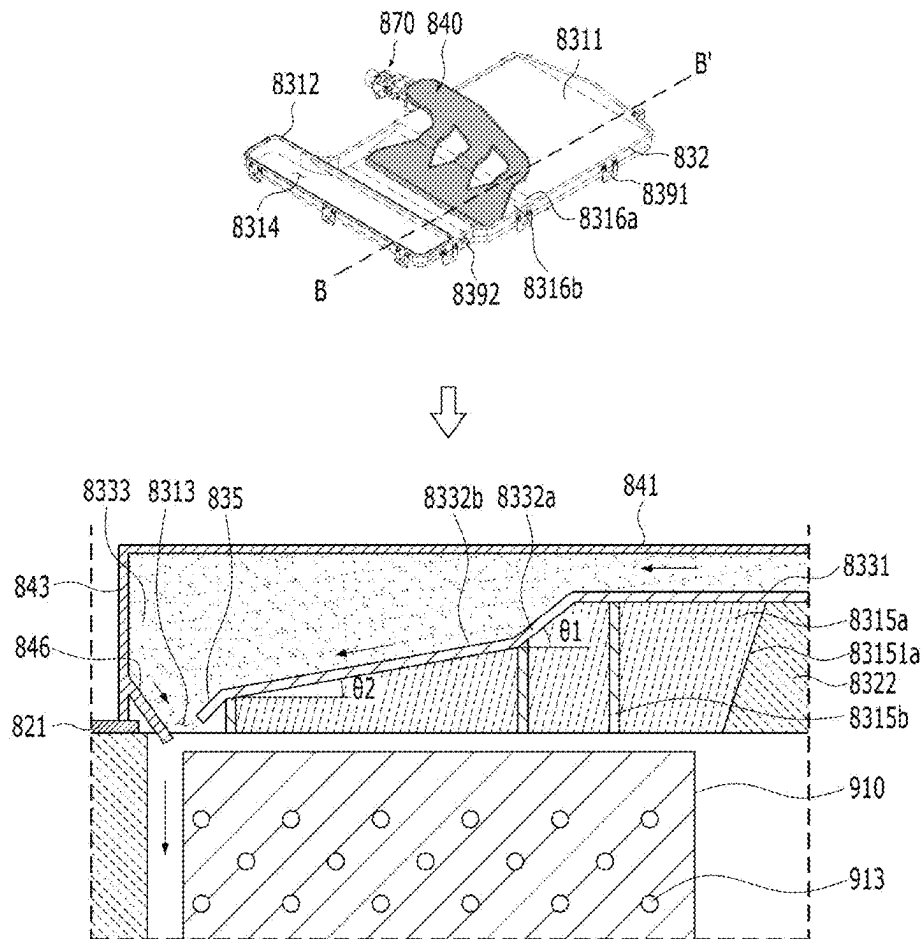


FIG. 21

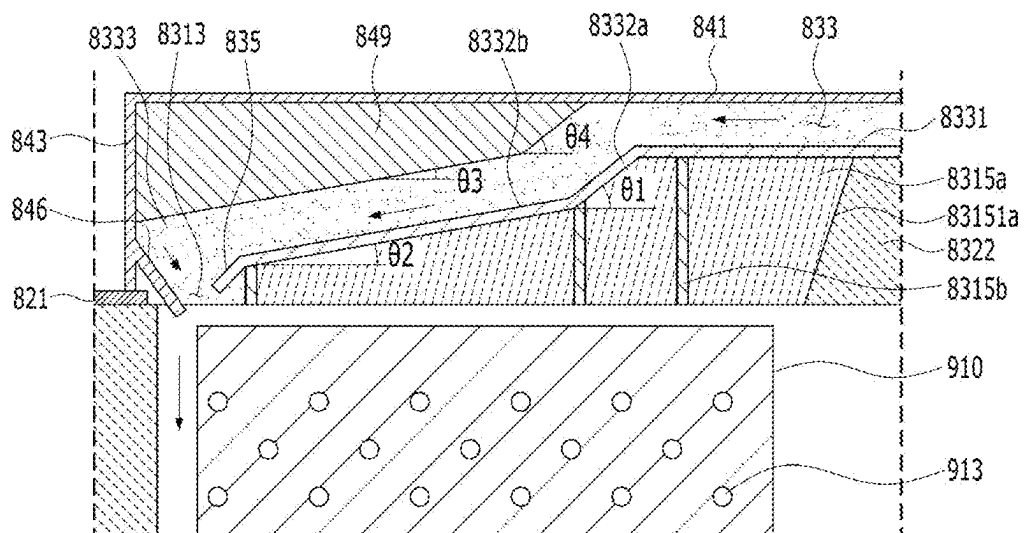
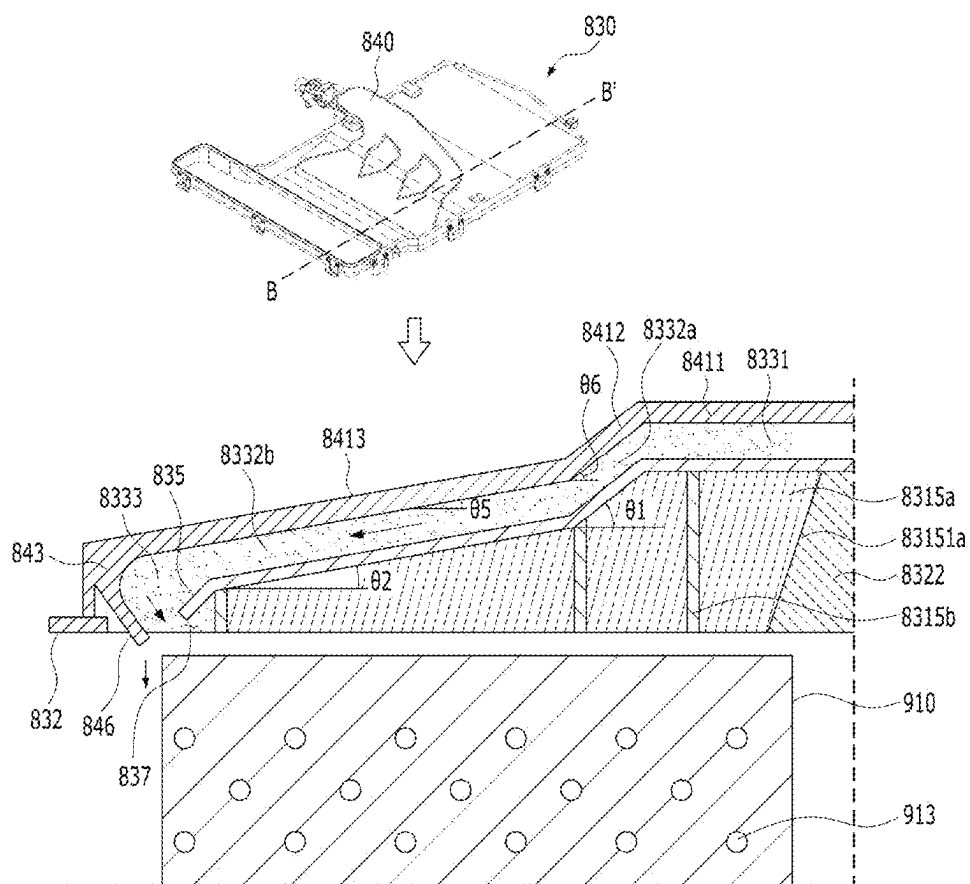


FIG. 22



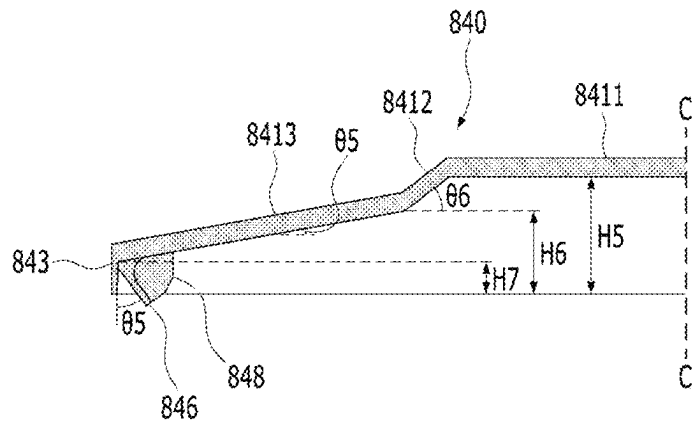


FIG. 23A

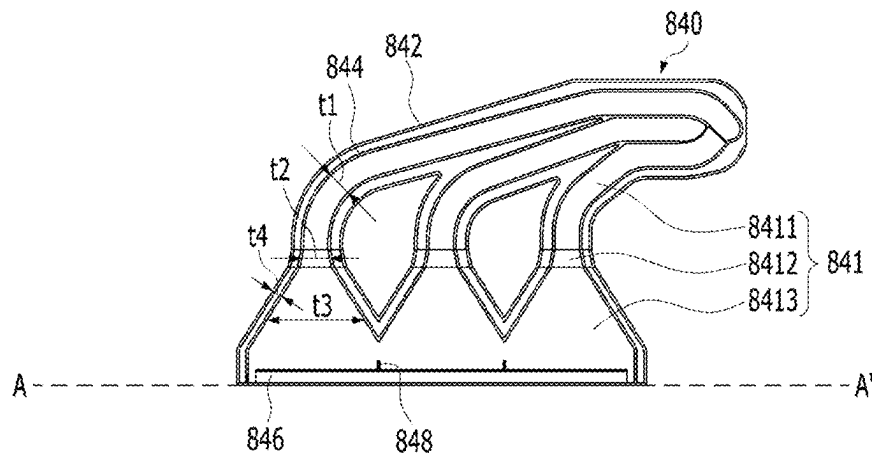


FIG. 23B

FIG. 24

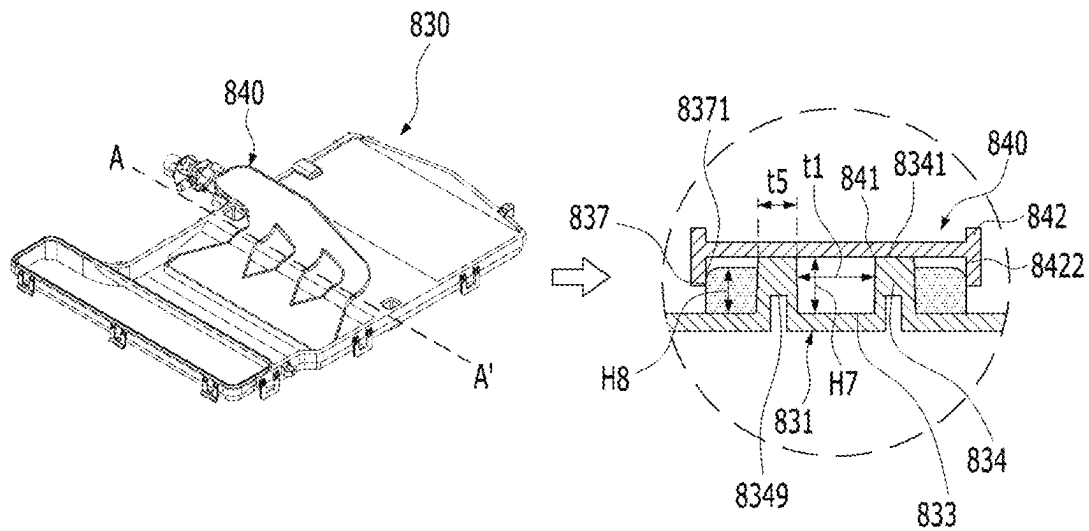
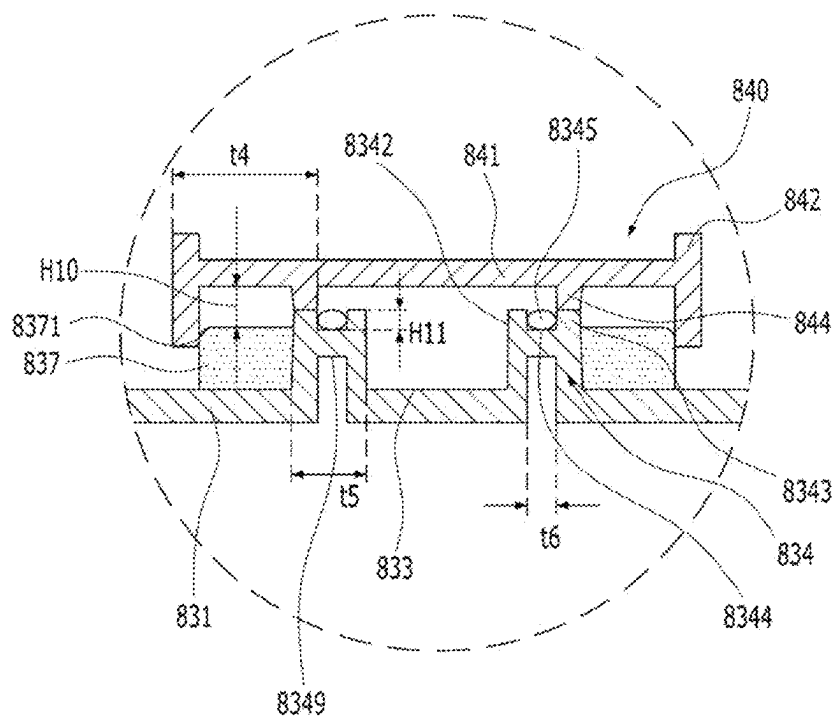


FIG. 25



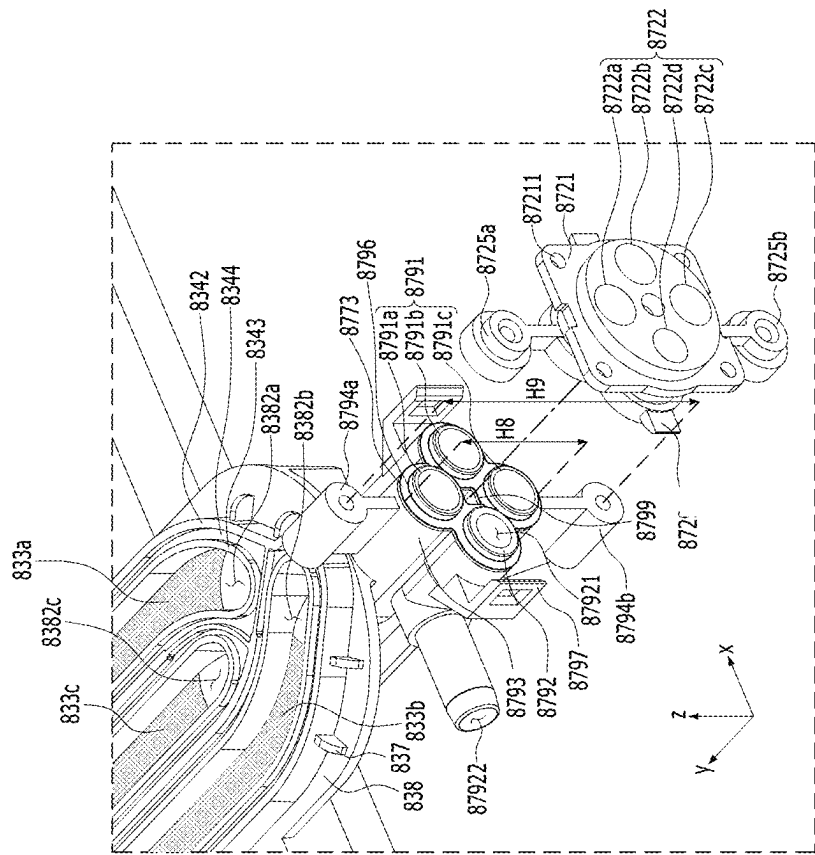


FIG. 26A

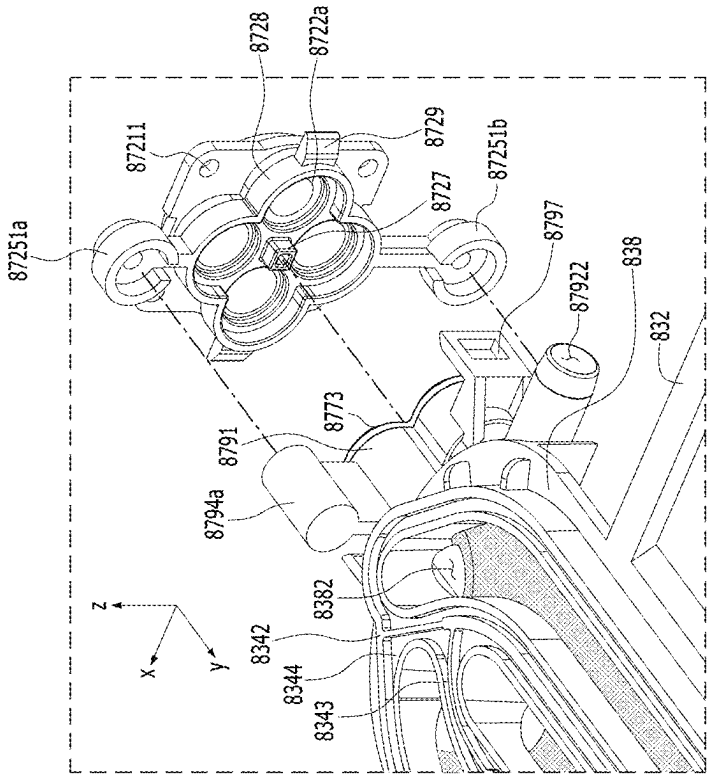
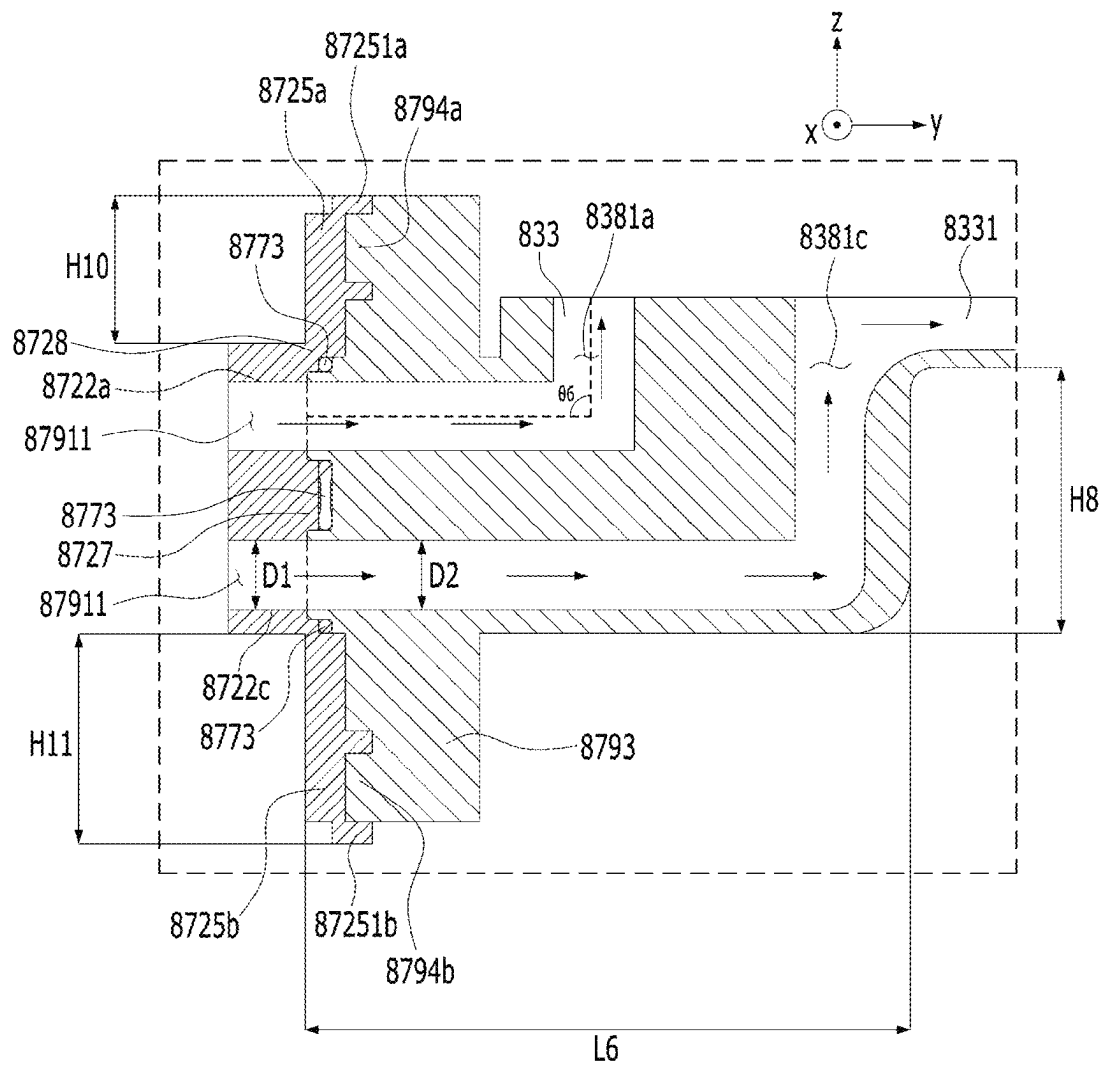


FIG. 26B

FIG. 27



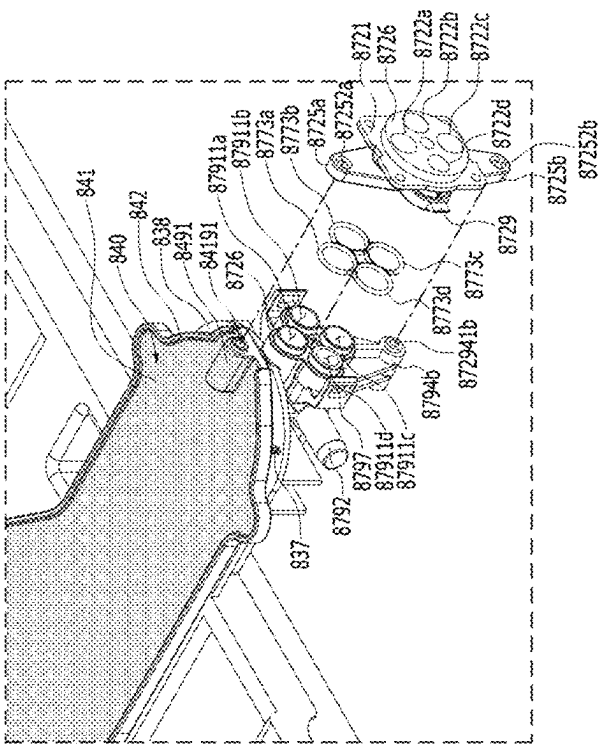


FIG. 28B

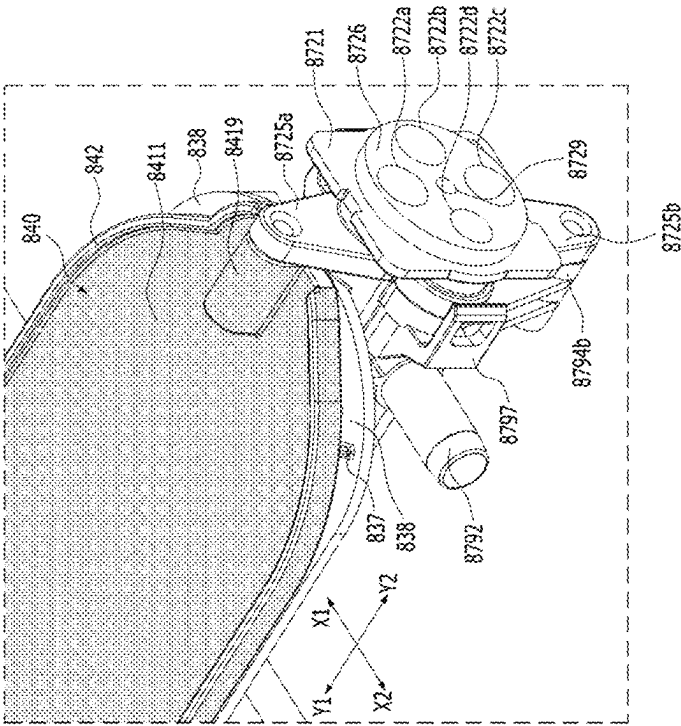


FIG. 28A

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LAUNDRY TREATING APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of Korean Patent Application Nos. 10-2021-0017559, filed on Feb. 8, 2021, the disclosure of which is hereby incorporated by reference as if fully set forth herein.

TECHNICAL FIELD

The present disclosure relates to a laundry treating apparatus.

BACKGROUND

A laundry treating apparatus may remove dust or foreign substances from laundry by applying physical force to the laundry. For example, the laundry treating apparatus may include a washing machine, a dryer, and a refresher (styler).

The washing machine may perform a washing cycle for separating and removing foreign substances from the laundry by supplying water and detergent to the laundry.

The dryer may be classified into an exhaust type dryer or a circulation type dryer which is configured to produce high-temperature hot air through a heater and perform a drying cycle to remove moisture contained in the laundry by exposing the hot air to the laundry.

In some cases, the dryer may omit a component for supplying or draining water to or from an inside of the laundry and a tub for accommodating the water so that a drying cycle may be performed without those components. This may simplify an internal structure of the laundry dryer, and may directly supply the hot air to the drum accommodating the laundry to improve drying efficiency.

For example, the dryer may include a drum to accommodate the laundry, a hot air supply to supply hot air to the drum, and a driver to rotate the drum. Accordingly, the dryer supplies hot air to the inside of the drum to dry the laundry accommodated in the drum, and rotates the drum such that a surface of the laundry may be evenly exposed to the hot air. As a result, an entire surface of the laundry may evenly contact the hot air to complete drying.

In some cases, the driver may be fixed inside the cabinet in order to rotate the drum. The driver may be configured to rotate a rotation shaft coupled to the drum, and the driver may be coupled to the rotation shaft in a parallel manner. In some cases, the dryer may not have a fixed tub inside the cabinet, and thus the driver may not be fixed to the tub, which is typically included in a washing machine.

In some cases, a dryer may include a driver that is fixed to a rear face of the cabinet.

FIG. 1 shows a structure of a conventional dryer in which the driver is coupled to the rear face of the cabinet in related art.

The dryer may include a cabinet 1 constituting an outer shape, a drum 2 rotatably disposed inside the cabinet 1 to accommodate laundry, and a driver 3 configured to rotate the drum 2.

The driver 3 may be disposed on a rear face of the drum 2 and configured to rotate the drum 2, and coupled to and fixed to a rear panel 11 constituting the rear face of the cabinet 1. In this way, the driver 3 is fixed to the cabinet 1 so that the drum 2 may be rotated.

The driver 3 of a conventional dryer as described above includes a stator 31 fixed to the rear panel 11, a rotor 32

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rotating by the stator 31, and a rotation shaft 33 coupled to the rotor 32 to rotate the drum 2. The dryer further includes a speed reducer 37 configured to rotate the drum 2 while increasing a torque by decreasing a RPM of the rotation shaft 33.

Further, the conventional dryers may include fixing means 4 for fixing the driver 3 to the rear panel 11. The fixing means 4 may include one or more of first fixing means 41 for fixing the stator 31 to the rear panel 11 and second fixing means 42 for fixing the rotation shaft 33 to the rear panel 11. As a result, in the conventional dryers, the rotation shaft 33 and the driver 3 coupled to the drum 2 are arranged side by side to rotate the drum 2 stably.

In some cases, where the rear panel 11 of the cabinet is made of a thin steel plate, which may be easily deformed or vibrated even with a fairly small external force. Moreover, the rear panel 11 receives a load of the driver 3 as well as a load of the drum 2 via the rotation shaft 33 and thus may not maintain its shape.

Further, when the laundry is eccentric inside the drum 2 or repeatedly falls into the drum 2 during rotation, a repeated external force is transmitted to the rear panel 11 and the rear panel 11 may vibrate.

When vibration or external force is transmitted to the rear panel 11 and thus the rear panel 11 is bent or deformed even temporarily, the rotation shaft 33 connecting the driver 3 and the drum 2 may be distorted. Accordingly, vibration or noise may occur in the driver 3, and in severe cases, the rotation shaft 33 may be damaged. Further, there is a problem that noise is generated while the rear panel 11 is bent or deformed.

Further, while the rear panel 11 vibrates, a distance between the rotor 32 and the stator 31 may temporarily change, causing the rotor 32 to collide with the stator 31 or generate vibration and noise.

In some cases, when the driver 3 further includes the speed reducer 37, the rotation shaft 33 coupled to the speed reducer 37 and a speed reducing shaft 33a extending from the speed reducer 37 to the drum 2 are separated from each other. In some examples, the speed reducer 37 is supported on the rear panel 11 via the stator 31 or the rotation shaft 33. Thus, when the rear panel 11 is deformed even a little, the speed reducing shaft 33a and the rotation shaft 33 may be distorted or misaligned with each other.

In other words, the speed reducing shaft 33a connected to the drum 2 may have a smaller displacement amount due to the load of the drum 2 than that of the rotation shaft 33 connected to the driver 3. Therefore, when the rear panel 11 is temporarily bent or deformed, inclinations of the rotation shaft 33 and the speed reducing shaft 33a are different from each other, and thus, the rotation shaft 33 and the speed reducing shaft 33a are misaligned with each other.

Therefore, in the conventional laundry treating apparatus, when the driver 3 operates, the rotation shaft 33 and the speed reducing shaft 33a may be misaligned with each other. Thus, reliability of the speed reducer 37 may not be ensured, and the speed reducer 37 may be damaged.

Further, the conventional dryer may not include a channel through which the air of the drum flows in the base located below the drum or a structure to treat condensate condensed in the channel.

SUMMARY

The present disclosure describes a laundry treating apparatus including a channel switching valve disposed on a side

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surface of a duct cover such that it is easy to repair and maintain the channel switching valve.

The present disclosure describes a laundry treating apparatus including a channel switching valve that is integrally formed with the duct cover such that an extension length of the channel switching portion can be reduced.

The present disclosure describes a laundry treating apparatus capable of increasing a pressure of water discharged to an evaporator by reducing a distance between the channel switching valve and a water collector body.

The present disclosure describes a laundry treating apparatus that can simplify an assembly process of an air circulating channel that defines a passage configured to carry air discharged from the drum.

The present disclosure describes a laundry treating apparatus including a cleaning water channel that is defined on a top of the duct cover, thereby simplifying production and assembly processes of the apparatus.

According to one aspect of the subject matter described in this application, a laundry treating apparatus includes a cabinet having an opening defined at a front surface thereof, a drum rotatably that is disposed in the cabinet and has a laundry inlet defined at a front surface thereof and configured to introduce laundry into the drum, a base that is disposed below the drum and defines a space configured to receive air discharged from the drum and to guide the air to the drum, and a motor configured to rotate the drum. The base includes an air circulating channel configured to fluidly communicate with the drum, where the air circulating channel is configured to receive the air discharged from the drum and to guide the air to the drum, a heat exchanger including (i) a first heat exchanger disposed inside the air circulating channel and configured to cool the air discharged from the drum and (ii) a second heat exchanger spaced apart from the first heat exchanger and configured to heat the air cooled by the first heat exchanger, a water collector body that is disposed outside the air circulating channel and in fluid communication with the air circulating channel, where the water collector body is configured to collect water condensed from the air in the first heat exchanger, a cleaning water channel disposed above the air circulating channel and configured to receive the water from the water collector body and to discharge the received water to the first heat exchanger, a pump configured to move the water collected in the water collector body to the cleaning water channel, and a channel switching valve connected to the pump and configured to selectively supply the water from the pump to the cleaning water channel. The channel switching valve includes a water receiving portion connected to the pump and configured to receive the water from the pump, and a connective portion connected to the water receiving portion and disposed at a lateral side of the air circulating channel, where the connective portion is configured to deliver the water from the water receiving portion to the cleaning water channel, and at least portion of the connective portion is located lower than a top surface of the air circulating channel.

Implementations according to this aspect can include one or more of the following features. For example, the air circulating channel can include an air flow duct that extends upward from the base and accommodates the first heat exchanger and the second heat exchanger, and a duct cover that is coupled to the air flow duct and covers the first heat exchanger and the second heat exchanger, the cleaning water channel being disposed at a top surface of the duct cover. The connective portion can be disposed at a side surface of the duct cover, the at least portion of the connective portion

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being located lower than the top surface of the duct cover. In some examples, the duct cover can define a cover through-hole that passes through the top surface of the duct cover and faces at least a portion of the first heat exchanger; and a valve communication hole that passes through a surface of the cleaning water channel and fluidly communicates the cleaning water channel with the connective portion. The cleaning water channel can extend from the valve communication hole to the cover through-hole and be configured discharge the water toward the first heat exchanger through the cover through-hole.

In some examples, the connective portion can be integrally formed with the duct cover such that the water is delivered to the cleaning water channel without leakage at the connective portion. In some examples, the connective portion can define a supply hole connected to the water receiving portion and configured to receive the water from the water receiving portion, and a valve communication hole that passes through a bottom surface of the cleaning water channel and is configured to deliver the water supplied from the supply hole to the cleaning water channel. The supply hole and the valve communication hole can be spaced apart from each other and face different directions from each other.

In some implementations, the connective portion can further include a receiving channel configured to guide the water from the water receiving portion to the cleaning water channel, where the receiving channel has a first end that defines the supply hole and a second end that defines the valve communication hole, and the receiving channel extends in an inclined direction with respect to the top surface of the duct cover.

In some implementations, the laundry treating apparatus can include a water storage tank spaced from the base and connected to the connective portion, where the water storage tank is configured to receive the water from the water collector body. The connective portion can further define a water receiving hole connected to the water receiving portion and configured to receive the water from the water receiving portion, and a water discharge hole connected to the water storage tank and configured to discharge the water received through the water receiving hole toward the water storage tank. The water discharge hole can be spaced apart from the water receiving hole and facing away from the water receiving hole. In some examples, the connective portion can further define a connective transfer channel that has a first end defining the water receiving hole and a second end defining the water discharge hole, where the connective transfer channel is configured to guide the water from the water receiving portion to the water storage tank, and the connective transfer channel and the receiving channel are defined in a single body.

In some implementations, the channel switching valve can further include a water delivering portion disposed between the water receiving portion and the connective portion and configured to guide the water supplied from the water receiving portion to the connective portion, where the connective portion is coupled to the water delivering portion and configured to receive the water from the water receiving portion through the water delivering portion. In some examples, the channel switching valve can further include a sealing member disposed between the connective portion and the water delivering portion and configured to block the water guided from the water delivering portion from leaking out between the connective portion and the water delivering portion.

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In some implementations, the laundry treating apparatus can include a nozzle cover that is coupled to the top surface of the duct cover and covers the cleaning water channel, where the nozzle cover can include a fastening part that protrudes from a top surface of the nozzle cover and is coupled to the water delivering portion. The water delivering portion can include a first fastening portion that extends from one side of the water delivering portion and is coupled to the fastening part of the nozzle cover, and a second fastening portion that extends from another side of the water delivering portion and is coupled to the connective portion. In some examples, the connective portion can include a first fixing protrusion that protrudes upward from a top of the connective portion and a second fixing protrusion that protrudes downward from a bottom of the connective portion. The water delivering portion can include a first fastening portion that extends from one side of the water delivering portion and receives the first fixing protrusion, and a second fastening portion that extends from another side of the water delivering portion and receives the second fixing protrusion.

In some implementations, the cabinet can include a first side panel that defines a first side of the cabinet and faces one side of the drum, and a second side panel that defines a second side of the cabinet and faces another side of the drum, where the air flow duct and the duct cover are located closer to the second side panel than to the first side panel, and the connective portion extends from the duct cover toward the first side panel. In some examples, the connective portion can extend toward the first side panel along an inclined direction relative to an extension direction of the duct cover. In some examples, the water collector body can be located between the first side panel and the air flow duct, and the channel switching valve can be located between the drum and the water collector body and overlap with the water collector body along a vertical direction.

In some implementations, a top of the water receiving portion can be located lower than a bottom of the drum and is spaced apart from the bottom of the drum. In some examples, the cleaning water channel can include a plurality of cleaning water channels, and the receiving channel can include a plurality of receiving channels that are connected to the plurality of cleaning water channels, respectively, where a number of the plurality of receiving channels is equal to a number of the plurality of cleaning water channels.

In some implementations, the channel switching valve can further include a scroll accommodated in the water receiving portion and configured to selectively supply the water in the water receiving portion to the receiving channel, where the scroll is configured to, based on rotating about an axis, selectively communicate the water receiving portion with one of the plurality of receiving channels. In some examples, the water receiving portion can include a valve rotatable shaft coupled to the scroll and configured to rotate the scroll, and a valve driver coupled to the valve rotatable shaft and configured to rotate the valve rotatable shaft.

The scroll can define a scroll communication hole, where a width of the scroll communication hole is equal to a diameter of one of the plurality of receiving channels, and the scroll communication hole is configured to selectively communicate with one of the plurality of receiving channels based on a rotation angle of the valve rotatable shaft. In some implementations, the scroll can include a scroll plate that defines the scroll communication hole and a scroll

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coupling groove, the scroll coupling groove being defined at a rotation center of the scroll plate and coupled to the valve rotatable shaft.

In some implementations, the channel switching valve can be disposed at a side face of an air circulating channel so that the channel switching valve is disposed closer to a water collector body.

In some implementations, the duct cover and the nozzle cover can be coupled to each other to define a cleaning water channels such that a separate washing pipe is omitted.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows a structure of a conventional dryer in related art.

FIGS. 2A and 2B show a dryer in related art.

FIG. 3 shows an outer shape of an example of a laundry treating apparatus according to the present disclosure.

FIG. 4 briefly shows an inside of the laundry treating apparatus.

FIG. 5 is an exploded perspective view showing example of internal components of the laundry treating apparatus.

FIGS. 6A and 6B show an outer shape of an example of a speed reducer.

FIG. 7 is an enlarged and detailed cross-sectional view of examples of the motor and the speed reducer.

FIG. 8 shows an example of a base and a rear plate.

FIG. 9 shows an example of the rear plate, the speed reducer, and the motor.

FIG. 10 shows an example of the speed reducer and the stator.

FIG. 11 shows an example of a speed reducer and a motor.

FIG. 12 is a perspective view showing an example of the base.

FIG. 13 is an exploded perspective view showing an example of a duct cover and a water collector cover coupled to an open top surface of a water collector body that are separated from the base of the FIG. 12.

FIG. 14 is a cross-sectional view showing an example of an arrangement relationship of a drum and an air circulating channel.

FIG. 15 is a perspective view showing an example of a cleaning water channel disposed on a top surface of a duct cover.

FIG. 16 is a top view showing an example of a duct cover having a cleaning water channel.

FIG. 17 is a perspective view showing a bottom surface of an example of a duct cover.

FIG. 18 is an exploded perspective view showing an example of a channel switching valve.

FIG. 19 is a perspective view showing an example of a duct cover to which a nozzle cover is coupled.

FIG. 20 is a cross-sectional view showing an example of a nozzle cover.

FIG. 21 is a cross-sectional view showing an example of a nozzle cover.

FIG. 22 is a cross-sectional view showing an example of a nozzle cover.

FIGS. 23A and 23B are a side view and a bottom view of the nozzle cover shown in FIG. 22.

FIG. 24 is a cross-sectional view showing one example in which a nozzle cover and a channel defining portion are coupled to each other.

FIG. 25 is a cross-sectional view showing another example in which a nozzle cover and a channel defining portion are coupled to each other.

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FIGS. 26A and 26B are a perspective view showing an example of a connective portion and a water delivering portion are coupled to each other.

FIG. 27 shows an example of an internal cross-sectional view of the connective portion and the water delivering portion.

FIGS. 28A and 28B are perspective views of an example of a connective portion, a water delivering portion, and a nozzle cover are coupled to each other.

DETAILED DESCRIPTIONS

FIG. 3 shows an outer shape of an example of a laundry treating apparatus according to the present disclosure.

In some implementations, the laundry treating apparatus can include a cabinet 100 that defines an outer shape of the laundry treating apparatus.

For instance, the cabinet 100 can include a front panel 110 that defines a front surface of the laundry treating apparatus, a top panel 150 that defines a top surface thereof, and a side panel 140 that defines a side surface thereof. The side panel 140 can include a first side panel 141 defining a left side face. The front panel 110 can include an opening 111 communicating with an interior of the cabinet 100 and a door 130 pivotably coupled to the cabinet 100 to open and close the opening 111.

In some implementations, the front panel 110 can be equipped with a manipulating panel 117. The manipulating panel 117 can include an input unit 118 that receives a control command from a user, and a display 119 that outputs information such as a control command selectable by the user. The control commands can include a drying course or a drying option that can perform a series of drying cycles. Inside the cabinet 100, a control box (see FIG. 12) that controls internal components to execute the control commands input through the input unit 118 can be installed. The control box can be connected to components inside the laundry treating apparatus to control the components to perform an input command.

The input unit 118 includes a power supply request unit that requests a power supply of the laundry treating apparatus, a course input unit that enables a user to select a course among a plurality of courses, and an execution request unit that requests start of a course selected by the user.

The display 119 can be configured to include at least one of a display panel capable of outputting texts and figures, and a speaker capable of outputting a voice signal and sound.

In some implementations, the laundry treating apparatus can include a water storage tank 120 configured to separately store moisture generated in a process of drying the laundry. The water storage tank 120 can include a handle which the user can grip to withdraw the water storage tank 120 from one side of the front panel 110 to an outside. The water storage tank 120 can be configured to collect the condensate generated during a drying cycle. Thus, the user can withdraw the water storage tank 120 from the cabinet 100, remove the condensate therefrom, and put the water storage tank 120 back into the cabinet 100. Accordingly, the laundry treating apparatus can be disposed in a place where a sewer or the like is not installed.

In some examples, the water storage tank 120 can be disposed on top of the door 130. Accordingly, when the user withdraws the water storage tank 120 from the front panel 110, the user can bend a waist in a relatively smaller amount, thereby increasing the user's convenience.

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FIG. 4 briefly shows an inside of the laundry treating apparatus.

The laundry treating apparatus includes a drum 200 accommodated inside the cabinet 100 to accommodate the laundry, a driver for rotating the drum 200, and a heat exchanger 900 configured to supply hot air to the drum 200, and a base 800 having an air circulating channel 820. The air circulating channel 820 is communicating with the drum 200. Air discharged from the drum 200 can be supplied to the air circulating channel 820. Further, the air discharged from the air circulating channel 820 can be supplied to the drum 200 again.

The driver can include a motor 500 that provides power to rotate the drum 200. The driver can be in direct connection with the drum 200 to rotate the drum 200. For example, the driver can be implemented as a Direct Drive (DD) type driver. Accordingly, the driver can control a rotation direction of the drum 200 or a rotation speed of the drum 200 by directly rotating the drum 200 while the driver is free of a belt and a pulley.

The motor 500 can rotate at high revolutions per minute (RPM). For example, the laundry inside the drum 200 can rotate at a much higher RPM than RPM at which it can rotate while being coupled to an inner wall of the drum 200.

However, when the laundry inside the drum 200 rotates while being continuously coupled to the inner wall of the drum 200, there is a problem in that drying efficiency decreases because a portion of the laundry coupled to the inner wall of the drum is not exposed to hot air.

When the motor 500 is rotated at a low RPM in order that the laundry roll or are mixed with each other inside the drum 200 without being coupled to the inner wall of the drum 200, there can be a problem that an output or torque that can be generated by the driver may not be properly utilized.

Therefore, the driver of the laundry treating apparatus can further include a speed reducer 600 that can reduce the RPM to increase the torque while taking advantage of a maximum output of the motor 500.

Further, the driver can include a drum rotation shaft 6341 connected to the drum 200 to rotate the drum 200.

The drum 200 can be formed in a cylindrical shape to accommodate the laundry therein. Further, unlike the drum used for washing, water may not be put inside the drum 200 used only for drying, and liquid water condensed inside the drum 200 may not be discharged out of the drum 200. Therefore, through-holes defined in a circumferential face of the drum 200 can be omitted. That is, the drum 200 used only for drying can be different from the drum 200 used for washing.

The drum 200 can be formed in an integral cylindrical shape, or can be manufactured in a structure in which a drum body 210 including a circumferential face and a drum rear face 220 constituting a rear face are coupled to each other.

A laundry inlet 211 through which laundry enters and exits can be defined in a front face of the drum body 210. The driver that rotates the drum can be connected to the drum rear face 220. The drum body 210 and the drum rear face 220 can be coupled to each other via a fastener such as a bolt. The disclosure is not limited thereto. As long as the drum body 210 and the drum rear face 220 are coupled to each other while both rotate together, they can be coupled to each other using various methods.

The drum body 210 can have a lift 213 for lifting the laundry up so that the laundry accommodated therein can be mixed with each other under the rotation. When the drum 200 rotates, the laundry accommodated therein can repeatedly rise up and fall due to the lift 213. The laundry

accommodated inside the drum 200 can be in contact with hot air while the laundry repeatedly rise up and fall. Therefore, the drying efficiency increases, and the drying time is shortened.

A reinforcing bead 212 can be formed on a circumferential face of the drum body 210. The reinforcing bead 212 can be configured to be recessed into or protrude from the circumferential face of the drum 200. The reinforcing bead can include a plurality of beads which can be configured to be spaced apart from each other. The reinforcing beads can form a certain pattern and can be recessed into or protrude from the circumferential face.

Rigidity of the drum body 210 can increase due to the reinforcing bead 212. Accordingly, even when a large amount of laundry is accommodated in the drum body 210 or a sudden rotation force is transmitted via the driver, the drum body 210 can be prevented from being distorted. Further, when the reinforcing bead 212 is provided, a spacing between the laundry and an inner circumferential face of the drum body can increase, compared to a case where the circumferential face of the drum body 210 is a flat face, so that the hot air supplied to the drum 200 is more effectively introduced between the laundry and the drum 200. Durability of the drum increases due to the reinforcing bead, and the drying efficiency of the laundry treating apparatus increases due to the bead.

In general, in a DD-type washing machine, the driver can be fixed to a tub that accommodates the drum 200, and the drum 200 can be coupled to the driver and supported on the tub. However, because a laundry treating apparatus can be configured to intensively perform a drying cycle, a tub fixed to the cabinet 100 to accommodate the drum 200 is omitted.

Accordingly, the laundry treating apparatus can further include a support 400 configured to fix or support the drum 200 or the driver inside the cabinet 100.

The support 400 can include a front plate 410 disposed in front of the drum 200 and a rear plate 420 disposed in rear of the drum 200. The front plate 410 and the rear plate 420 can have a plate shape and can be disposed to respectively face front and rear faces of the drum 200. A distance between the front plate 410 and the rear plate 420 can be set to be equal to a length of the drum 200 or be larger than the length of the drum 200. The front plate 410 and the rear plate 420 can be fixedly supported on the bottom face of the cabinet 100 or the base 800.

The front plate 410 can be disposed between the front panel constituting the front face of the cabinet and the drum 200. Further, the front plate 410 can have an inlet-communication hole 412 communicating with the laundry inlet 211. Because the front plate 410 has the inlet-communication hole 412, the front face of the drum 200 is supported thereon, laundry can be put into or taken out from the drum 200.

The front plate 410 can include a duct connector 416 disposed below the inlet-communication hole 412. The duct connector 416 can constitute a lower portion of the front plate 410.

The front plate 410 can include a duct communication hole 417 extending through the duct connector 416. The duct communication hole 417 can have a hollow shape to guide the air discharged through the laundry inlet 211 of the drum to a bottom of the drum 200. Further, the air discharged through the laundry inlet 211 can be guided to the air circulating channel 820 positioned under the drum 200.

In some implementations, a filter can be installed in the duct communication hole 417 to filter foreign substances such as large lint or large particles generated from the

laundry. The filter filters the air discharged from the drum 200 to prevent foreign substances from accumulating inside the laundry treating apparatus, and to prevent foreign substances from accumulating and thus interfering with circulation of the air.

Because the laundry inlet 211 can be disposed in a front face, the driver can be installed on the rear plate 420 rather than the front plate 410. The driver can be configured to be supported and mounted on the rear plate 420. This allows the driver to rotate the drum 200 while the position of the driver is stably fixed due to the rear plate 420.

At least one of the front plate 410 and the rear plate 420 can support the drum 200 such that the drum can rotate. At least one of the front plate 410 and the rear plate 420 can accommodate a front or rear end of the drum 200 such that the drum can rotate.

For example, a front portion of the drum 200 can be rotatably supported on the front plate 410, and a rear portion of the drum 200 can be spaced apart from the rear plate 420 and can be connected to the motor 500 mounted on the rear plate 420 and thus can be indirectly supported on the rear plate 420. In this way, an area where the drum 200 contacts or rubs against the support 400 can be minimized and noise or vibration can be prevented from occurring.

In another example, the drum 200 can be configured to be rotatably supported on both the front plate 410 and the rear plate 420.

One or more support wheels 415 supporting the front portion of the drum 200 can be disposed at a lower portion of the front plate 410. The support wheel 415 can be rotatably disposed on a rear face of the front plate 410. The support wheel 415 can be rotated while in contact with a lower portion of the drum 200.

When the drum 200 is rotated by the driver, the drum 200 can be supported on the drum rotation shaft 6341 connected to the rear portion of the drum. When the laundry is accommodated in the drum 200, a load imposed to the drum rotation shaft 6341 due to the laundry can increase. Therefore, there is a risk of the drum rotation shaft 6341 being bent by the load.

When the support wheel 415 supports the front and lower portion of the drum 200, the load on the drum rotation shaft 6341 can be reduced. This can prevent the drum rotation shaft 6341 from being bent and prevent noise from being generated due to the vibration.

The support wheels 415 can be disposed at positions symmetrical to each other around a center of rotation of the drum 200 so as to support the load of the drum 200. The support wheels 415 can be disposed at left and right sides of the lower portion of the drum 200 to support the drum 200 thereon. However, the present disclosure is not limited thereto, and a larger number of support wheels 415 can be included according to an operating environment of the drum 200.

The air circulating channel 820 disposed in the base 800 can circulate the air inside the drum 200 such that the air is input back into the drum 200.

The air circulating channel 820 can include an inflow duct 821 into which the air discharged from the drum 200 flows, an air discharge duct 823 that supplies the air to the drum 200, and an air flow duct 822 connecting the inflow duct 821 and the air discharge duct 823 to each other.

When air is discharged from the front face of the drum 200, the air flow duct 822 can be located at a front side of the air circulating channel 820. The air discharge duct 823 can be located at a rear side of the air circulating channel 820.

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The air discharge duct **823** can further include a blower **8231** that discharges air out of the air circulating channel **820**. The blower **8231** can be disposed at the rear side of the air discharge duct **823**. Air exhausted through the blower **8231** can flow to the drum **200**.

A duct cover **830** can be coupled to a top of the air circulating channel **820**, so that an open top face of the air circulating channel **820** can be partially shielded therewith. The duct cover **830** can prevent air from leaking out of the air circulating channel **820**. In other words, the duct cover **830** can constitute one face of a channel through which air is circulated.

Further, a heat exchanger **900** disposed in the base **800** can include a first heat exchanger **910** disposed inside the air circulating channel **820** to cool the air, and a second heat exchanger **920** disposed inside the air circulating channel **820** to heat the air cooled in the first heat exchanger **910**.

The first heat exchanger **910** dehumidifies the air discharged from the drum **200**, and the second heat exchanger **920** can heat the dehumidified air. The heated air is supplied to the drum **200** again to dry the laundry accommodated in the drum **200**.

Each of the first heat exchanger **910** and the second heat exchanger **920** can be implemented as a heat exchanger through which refrigerant flows. When being implemented as a heat exchanger through which the refrigerant flows, the first heat exchanger **910** can be implemented as an evaporator, and the second heat exchanger **920** can be implemented as a condenser. The refrigerant flowing along the first heat exchanger **910** and the second heat exchanger **920** can exchange heat with air discharged from the drum **200**.

The heat exchanger **900** can include an air circulating channel fan **950** that is installed in the air circulating channel **820** to generate air flow inside the air circulating channel **820**. Further, the heat exchanger **900** can further include an air circulating channel fan motor **951** that rotates the air circulating channel fan **950**. The air circulating channel fan **950** can be rotated upon receiving rotation power from the air circulating channel fan motor **951**. When the air circulating channel fan **950** operates, the air dehumidified by the first heat exchanger **910** and then heated by the second heat exchanger **920** can flow to the rear portion of the drum **200**.

The air circulating channel fan **950** can be installed in one of the inflow duct **821**, the air flow duct **822**, and the air discharge duct **823**. Because the air circulating channel fan **950** can be configured to rotate, noise can be generated when the air circulating channel fan **950** operates. Therefore, the air circulating channel fan **950** can be disposed in rear of the air circulating channel **820**.

The air circulating channel fan **950** can be installed at the blower **8231**. Further, the air circulating channel fan motor **951** can be located in rear of the blower **8231**. When the air circulating channel fan **950** is rotated by the air circulating channel fan motor **951**, air inside the air circulating channel **820** can be discharged out of the air circulating channel **820** via the blower **8231**.

Because the laundry inlet **211** of the drum **200** can be disposed at a relatively higher position in order for the user to easily withdraw the laundry located inside the drum **200**, the air circulating channel **820** and the heat exchanger **900** can be disposed below the drum **200**.

The rear plate **420** can be disposed in rear of the drum **200** to guide the air discharged from the air circulating channel **820** to the drum **200**. The rear plate **420** can be configured to be spaced apart from the drum rear face **220**. The air circulating channel **820** can receive air inside the drum **200** through the front plate **410** and supply air to the drum **200**

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through the rear plate **420**. Air discharged from the air circulating channel **820** can be guided to the drum **200** through the rear plate **420**.

The base **800** can further include a connector **850** that guides the air discharged from the air circulating channel **820** to the rear plate **420**. The connector **850** can guide the exhaust air to spread evenly throughout the rear plate **420**.

The connector **850** can be installed at the blower **8231**. That is, the connector **850** can guide the air discharged from the blower **8231** to the rear plate **420**. The hot air supplied to the rear plate **420** can flow into the drum **200** through the drum rear face **220**.

The drum **200** of the laundry treating apparatus can be rotated while being directly connected to the driver positioned in rear of the drum **200**, rather than being indirectly rotated while being coupled to a belt. Therefore, unlike a drum of a conventional dryer that has a cylindrical shape in which front and rear faces are open, a rear face of the drum of the laundry treating apparatus can be shielded and can be directly coupled to the driver.

As described above, the drum **200** can include the drum body **210** having a cylindrical shape to accommodate laundry and the drum rear face **220** coupled to the rear portion of the drum body **210** to define a rear face of the drum.

The drum rear face **220** can be configured to shield the rear face of the drum body **210**, and provide a coupling face for direct engagement with the driver. That is, the drum rear face **220** can be connected to the driver and receive the rotation power to rotate an entirety of the drum **200**. As a result, the front face of the drum body **210** can have the laundry inlet **211** into which laundry is put, and the rear face thereof can be shielded with the drum rear face **220**.

The drum rear face **220** can be equipped with a bushing **300** connecting the driver and the drum rear face **220** to each other. The bushing **300** can be disposed at the drum rear face **220** to define a center of rotation of the drum **200**. The bushing **300** can be formed integrally with the drum rear face **220**, or can be made of a material with greater rigidity and durability than that of the drum rear face **220** in order to be firmly coupled to the rotation shaft that transmits power. The bushing **300** can be seated on and coupled to the drum rear face **220** so as to be coaxial with the center of rotation of the drum rear face **220**.

The drum rear face **220** can include a circumferential portion **221** coupled to an outer circumferential face of the drum body **210**, and a mount plate **222** that can be disposed inwardly of the circumferential portion **221** and can be coupled to the driver. The bushing **300** can be seated on the mount plate **222** and can be coupled thereto. The rotation shaft that rotates the drum can be coupled to the mount plate **222** via the bushing **300**, and thus can be more firmly coupled thereto. Further, this can prevent deformation of the drum rear face **220** from occurring.

The drum rear face **220** can include an intake hole **224** extending through a portion between the circumferential portion **221** and the mount plate **222**, and air-communicating in a front and rear direction of the drum rear face **220**. The hot air supplied through the air circulating channel **820** can be introduced into the drum body **210** through the intake hole **224**. The intake hole **224** can include a plurality of holes extending through the drum rear face **220** or as a mesh.

The driver that rotates the drum **200** can be located in rear of the rear plate **420**. The driver can include a motor **500** that generates rotation power and a speed reducer **600** that reduces the rotation force of the motor **500** and transmits the reduced force to the drum **200**.

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The motor **500** can be disposed in rear of the rear plate **420**. The motor **500** can be coupled to the rear face of the rear plate **420** via the speed reducer **600**.

The speed reducer **600** can be fixed to the rear face of the rear plate **420**, and the motor **500** can be coupled to the rear face of the speed reducer **600**. That is, the rear plate **420** can provide a support face on which the speed reducer **600** or the motor **500** is supported. However, the present disclosure is not limited thereto, and the motor **500** can be coupled to the rear plate **420**.

FIG. 5 is an exploded perspective view showing internal components constituting the laundry treating apparatus in a separated state from each other.

The laundry treating apparatus can include the drum **200** for accommodating the laundry, the front plate **410** for supporting the front face of the drum, the rear plate **420** located in rear of the drum, and the base **800** disposed below the drum to provide a space in which the air inside the drum is circulating or moisture contained in the air is condensed, and the motor **500**, which is located in rear of the drum and provides the rotation power to the drum, the speed reducer **600** to reduce the rotation speed of the motor and deliver the rotation power to the drum, and a rear cover **430** that can be coupled to the rear plate **420** to prevent the motor from being exposed to the outside.

The base **800** can include the air circulating channel **820** which communicates with the drum **200**, and receives the air from the drum or discharges the air to the drum.

The front plate **410** can include a front panel **411** constituting a front face thereof, and the inlet-communication hole **412** that is formed to extend through the front panel **411** and communicates with the drum **200**. The front plate **410** can have a front gasket **413** which can be disposed on the rear face of the front panel **411** and can be configured to surround a radially outer side of the inlet-communication hole **412** and can accommodate a portion of the drum body **210**.

The front gasket **413** can support the drum body **210** such that the drum body can rotate, and can be in contact with the outer circumferential face or an inner circumferential face of the laundry inlet **211**. The front gasket **413** can prevent the hot air inside the drum **200** from leaking into a space between the drum body **210** and the front plate **410**. The front gasket **413** can be made of a plastic resin or an elastic body. A separate sealing member can be additionally coupled to the front gasket **413** to prevent laundry or the hot air from escaping from the drum body **210** to the front plate **410**.

In some examples, the front plate **410** can include a duct communication hole **417** extending through an inner circumferential face of the inlet-communication hole **412**. Further, the front plate **410** can include a duct connector **416** extending downwardly of the duct communication hole **417** to define a channel communicating the drum body **210** and the air circulating channel **820** to each other.

The duct connector **416** can communicate with the drum body **210** through the duct communication hole **417**, and the air discharged from the drum body **210** can flow into the duct connector **416** through the duct communication hole **417** and can be guided to the air circulating channel **820**. Because the air discharged from the drum body **210** is guided to the air circulating channel **820** via the duct connector **416**, this can prevent the air inside the drum from leaking out.

A filter member that filters foreign substances or lint from the air discharged from the drum **200** and prevents foreign substances from entering the air circulating channel **820** can be installed in the duct connector **416**.

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The support wheels **415** supporting the lower portion of the drum **200** and being rotatably installed on the rear face of the front panel **411** can be installed on the front plate **410**. The support wheel **415** supports the front face of the drum **200** and thus prevents the rotation shaft connected to the drum from being bent.

The front plate **410** can have a water storage tank support hole **414** which can be configured to extend through the front panel **411**, and which the water storage tank **120** (see FIG. 3) in which the condensate generated in the drying process is stored can be withdrawn through or supported on. When the water storage tank support hole **414** can be disposed at a top level, the user may not have to bend his back when withdrawing the water storage tank, so that the user's convenience increases.

The drum **200** for accommodating the laundry therein can include the drum body **210** having the laundry inlet **211** defined in a front portion thereof through which the laundry is input or output, and the drum rear face **220** constituting a rear face thereof.

The drum rear face **220** can include the circumferential portion **221** connected to the drum body **210**, the intake hole **224** defined inwardly of the circumferential portion **221** and extending through the drum rear face **220**, and the mount plate **222** disposed at the center of rotation of the drum rear face **220**, and coupled to the rotation shaft. Air can be introduced to the rear face of the drum through the intake hole **224**.

The drum rear face **220** can further include a reinforcing rib **225** extending from the circumferential portion **221** toward the center of rotation. The reinforcing rib **225** can extend while bypassing the intake hole **224**. The reinforcing rib **225** has the effect of preventing the rigidity of the drum rear face **220** from being reduced due to the intake hole **224**. The reinforcing rib **225** can be configured to extend radially from the outer circumferential face of the mount plate **222** toward an inner circumferential face of the circumferential portion **221**.

Further, the drum rear face **220** can further include a circumferential rib **227** extending in the circumferential direction of the drum rear face **220** to connect the reinforcing ribs **225** to each other. The intake holes **224** can be respectively disposed between adjacent ones of the reinforcing rib **225**, the circumferential rib **227**, and the circumferential portion **221**. The reinforcing rib **225** and the circumferential rib **227** have the effect of preventing the drum rear face **220** from being deformed upon receiving the rotation force from the motor **500**.

The inflow duct **821** can communicate with the duct communication hole **417** of the front plate **410** to communicate with a channel installed inside the front plate **410**. The air flow duct **822** can extend from a distal end of the inflow duct **821** toward the rear face of the drum **200**, and the air discharge duct **823** can be disposed at a distal end of the air flow duct **822** to direct the air to the drum **200**.

The blower **8231** can be located downstream of the air discharge duct **823**, and the blower **8231** can provide a space where the air circulating channel fan is installed. When the circulation fan channel fan operates, the air introduced into the inflow duct **821** can be discharged upwardly of the blower **8231**.

In some examples, the base **800** can be equipped with the heat exchanger **900** that can cool and heat the air circulating inside the drum **200**. The heat exchanger **900** can include a compressor **930** connected to the first heat exchanger and the second heat exchanger to supply compressed refrigerant. The compressor **930** can be configured so as not to directly

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exchange heat with the circulating air, and thus can be located out of the air circulating channel **820**.

Further, the heat exchanger can include the air circulating channel fan motor **951** supported on a rear face of the blower **8231** to rotate the air circulating channel fan. The air circulating channel fan motor **951** can be coupled to the rear face of the blower **8231**.

In some examples, the laundry treating apparatus can further include the connector **850** which can be coupled to the air circulating channel **820** for guiding the hot air discharged from the air circulating channel **820** to the rear portion of the drum **200** or the rear plate **420**.

The connector **850** can be disposed on top of the air discharge duct **823** and be configured to guide the hot air heated through the second heat exchanger **920** upwards beyond the air discharge duct **823**. Further, the connector **850** can be coupled to an opening disposed above the blower **8231**.

The connector **850** can be configured to have a channel defined therein. The connector **850** can be configured to evenly guide the flow of air generated by the air circulating channel fan to the rear plate **420**. That is, the connector **850** can be configured so that an area of the channel therein increases as a distance thereof from the blower **8231** increases.

The rear plate **420** can be coupled to the base **800** or supported on the base **800** and be positioned in rear of the drum **200**. The rear plate **420** can be configured to include the rear panel **421** positioned to face toward the front plate **410**, and a duct portion **423** recessed in the rear panel **421** to define a channel through which air flows and to guide the air discharged from the air circulating channel **820** to the drum.

The rear plate **420** can include a mount **425** to or on which the driver is coupled or supported. The mount **425** can be configured to extend through the rear panel **421** and disposed on an inner circumferential face of the duct portion **423**. The mount **425** can be configured to be spaced apart from an inner circumferential face of the duct portion **423** inwardly in a radial direction.

In some implementations, the driver can include a combination of the speed reducer **600** and the motor **500** as described above. In some examples, the driver can include only the motor **500**. That is, a component that generates power and transmits the rotation power to the drum can be referred to as a driver.

The driver can be mounted on the mount **425**. The mount **425** can support the driver's load. The driver can be connected to the drum **200** while supported on the mount **425**.

The duct portion **423** can be configured to receive a portion of the drum rear face **220**. The duct portion **423** can have the channel defined therein through which air flows together with the drum rear face **220**.

The driver can be installed on the mount **425** so as not to interfere with the duct portion **423**. In other words, the driver can be radially inwardly spaced away from an inner circumferential face of the duct portion **423**. The driver can be installed on the mount **425**, while a rear portion thereof can be exposed to the outside so that it can be cooled by external air.

The driver can include the motor **500** that provides power to rotate the drum **200**. The motor **500** can include a stator **510** that generates a rotating magnetic field, and a rotor **520** that can be configured to rotate by the stator **510**.

The rotor **520** accommodates the stator **510** and can be equipped with an outer rotor type configured to rotate along the circumference of the stator **510**. In some examples, the

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rotor **520** can be coupled to a drive shaft and can be directly connected to the drum **200** through the stator **510** and the mount **425**. In this case, the rotor **520** can directly transmit the power to rotate the drum **200**.

The rotor **520** can be coupled to the drive shaft via a washer **540**. The washer **540** can perform a function of connecting the drive shaft and the rotor **520** to each other. Because a contact area between the rotor **520** and the drive shaft can increase due to the washer **540**, the rotation of the rotor **520** can be transmitted more effectively.

The speed reducer **600** can be configured to connect the motor **500** and the drum **200** to each other. The speed reducer **600** can convert the power of the motor **500** to rotate the drum **200**. The speed reducer **600** can be disposed between the motor **500** and the drum **200** to receive the power of the motor **500**, convert the same, and transmit the same to the drum **200**. The speed reducer **600** can be configured to convert the RPM of the rotor to a small RPM but increase a torque value thereof and transfer the converted RPM to the drum **200**.

Specifically, the speed reducer **600** can be coupled to the rotor **520** and the drive shaft that rotates with the rotor **520**. The speed reducer **600** can include a gear assembly that can be engaged with the drive shaft and rotates therewith to reduce the RPM of the drive shaft but increase the torque thereof. The gear assembly can be coupled to the drum **200** and can be connected to the drum rotation shaft to rotate the drum. Thus, when the drive shaft **530** rotates, the drum rotation shaft rotates at a slower RPM than that of the drive shaft, but with the increased torque.

The performance of this speed reducer **600** depends on whether the drive shaft and the drum rotation shaft can be kept coaxial with each other. That is, when the drive shaft and the drum rotation shaft are misaligned with each other, there is a risk that coupling between the parts constituting the gear assembly inside the speed reducer **600** and at least one of the drive shaft and the drum rotation shaft can loosen or can be disengaged. Therefore, the power of the drive shaft may not be properly transmitted to the drum rotation shaft, or the drive shaft can rotate in vain.

Further, when the drive shaft and the drum rotation shaft are misaligned with each other even temporarily, the gears inside the speed reducer **600** can be misaligned with each other and collide with each other, resulting in vibration or noise.

Further, there is a risk that the speed reducer **600** can completely deviate from its correct position or be damaged when an angle by which the drive shaft and the drum rotation shaft are misaligned with each other increases even temporarily.

In some implementations, in the laundry treating apparatuses having the speed reducer, the speed reducer **600** and the motor **500** can be fixed to a support that maintains its original state without deformation even when an external force is applied thereto.

For example, in a washing machine, the tub accommodating the drum can be first fixed to the cabinet, and then the motor and the speed reducer can be second fixed to a bearing housing made of a rigid body built into the tub by injection molding. This allows the speed reducer and the driver to tilt or vibrate together with the bearing housing or the fixing steel plate, even when significant vibrations occur in the tub. As a result, the speed reducer and the driver themselves can maintain a combined state therewith, and the drive shaft and the rotation shaft can be kept coaxial.

In some implementations, the laundry treating apparatus is a dryer, the tub fixed inside the cabinet can be omitted.

Further, because the rear panel of the cabinet is made of a relatively thin plate, even when the stator **510** is fixed thereto, the rear panel can easily vibrate or be bent due to a repulsive force when the rotor **520** rotates. When the rear panel vibrates or bends even temporarily, the rotation centers of the speed reducer **600** and the motor **500** which are in combination with the drum **200** can be misaligned with each other.

Further, because the rear panel is made of a thin steel plate, the rear panel may not support both the speed reducer **600** and the motor **500**. For example, when the speed reducer **600** and the motor **500** are coupled to the rear panel and are arranged side by side, a rotational moment can be generated due to a total length and weight of the speed reducer **600** and the motor **500**, such that the speed reducer **600** sags downwards. As a result, the drum rotation shaft itself coupled to the drum can be misaligned with the speed reducer **600**. Thus, the drum rotation shaft and the drive shaft may not be maintained at the coaxial state.

In some examples, a configuration can be considered that the stator **510** can be coupled to the rear plate **420** to support the motor **500**. When a large amount of laundry is accommodated inside the drum **200**, or when eccentricity occurs, the drum rotation shaft can be distorted according to the displacement of the laundry whenever the drum **200** rotates. In some examples, the stator **510** can be separated from the drum **200** and fixed to the rear plate **420**, so that the drum rotation shaft can vibrate at a different dimension or tilt at a different angle than the stator **510** can do. Therefore, the coaxiality of the drum rotation shaft and the drive shaft may not be maintained.

From another point of view, the drum **200** can be supported on the front plate **410** and the rear plate **420** so that an installed position thereof can be fixed to a certain degree. Therefore, the position of the drum rotation shaft coupled to the drum **200** can be fixed to a certain degree. Therefore, even when vibration occurs in the drum **200**, the vibration can be buffered by at least one of the front plate **410** or the rear plate **420**.

However, when the vibration generated in the drum **200** is transmitted to the motor **500**, and even when the speed reducer **600** and the motor **500** are fixed to the rear plate **420**, the vibration amplitude of the drum rotation shaft is larger than the vibration amplitude of the motor **500** and the rear plate **420**. Thus, there can be a problem that the drive shaft and the drum rotation shaft cannot be maintained in a coaxial relationship with each other.

In order to solve this problem, in the laundry treating apparatus, the motor **500** can be fixedly coupled to the speed reducer **600**. In other words, the speed reducer **600** itself can serve as a reference point of an entirety of the driver. In other words, the speed reducer **600** can serve as a reference of the overall vibration amplitude and tilting angle of the driver.

The motor **500** is not fixed to other components of the laundry treating apparatus, but is fixed only to the speed reducer **600**. Thus, when vibration is transmitted to the driver or external force is transmitted thereto, the motor **500** can tilt or vibrate simultaneously together with the speed reducer **600** when the speed reducer **600** tilts or vibrates.

As a result, the speed reducer **600** and the motor **500** can constitute one vibration system, and the speed reducer **600** and the motor **500** can be maintained in a fixed state with each other while not performing relative motion with respect to each other.

The stator **510** of the motor **500** can be directly coupled to the speed reducer **600** and fixed thereto. In this way, the installed position of the drive shaft **530** relative to the speed

reducer **600** may not be changed. A center of the drive shaft **530** and a center of the speed reducer **600** can coincide with each other, and thus the drive shaft **530** can rotate in the coaxial state with the center of the speed reducer **600**.

A first axis **M1** can refer to an imaginary line extending in a front-rear direction along the center of rotation of the drum **200**. That is, the first axis **M1** can extend in parallel to an X axis.

Each of a second axis **M2** and a third axis **M3** can refer to an imaginary line extending in a left and right direction of the laundry treating apparatus. That is, each of the second axis **M2** and the third axis **M3** can be orthogonal to an XZ plane and parallel to a Y axis.

The first axis **M1** and the second axis **M2** can intersect each other at the speed reducer **600**. Further, the first axis **M1** and the third axis **M3** can intersect with each other at the mount **425**.

The speed reducer **600** and the motor **500** can be designed to be arranged along the first axis **M1** parallel to a ground when there is no load on the drum **200** or when the motor **500** is not running.

However, when vibration occurs in the drum **200** or the motor **500**, the vibration is transmitted to the speed reducer **600** and thus the speed reducer **600** is tilted, thereby causing the speed reducer **600** to temporarily tilt along the second axis **M2**.

In some examples, the motor **500** can be coupled to the speed reducer **600**, and thus can vibrate or tilt together with the speed reducer **600**. Thus, the motor **500** and the speed reducer **600** can be arranged side by side along the second axis **M2**. Thus, the drive shaft and the drum rotation shaft can be arranged side by side along the second axis **M2**.

As a result, even when the speed reducer **600** is tilted, the motor **500** can move integrally with the speed reducer **600**, and thus the drive shaft and the drum rotation shaft can be maintained in a coaxial state with each other.

The speed reducer **600** can be fixedly coupled to the rear plate **420**. In this case, the speed reducer **600** will tilt or vibrate while being coupled to the rear plate **420**, so that the rear plate **420** plays the role of the center of the vibration system including the speed reducer **600**, the motor **500**, and the drum **200**. Even in this case, the motor **500** can be not directly coupled to the rear plate **420**, but can be only coupled to the speed reducer **600** and fixed thereto.

The speed reducer **600** and the motor **500** and the drum **200** can be arranged side by side along the first axis **M1**. However, the vibration of the drum **200** or the motor **500** causes the speed reducer **600** to be inclined in parallel to the third axis **M3**. The third axis **M3** can extend through the speed reducer **600** coupled to the rear plate **420**. In some examples, the speed reducer **600** and the motor **500** are coupled to each other, so that the motor **500** can be tilted in parallel to the third axis **M3**, just like the speed reducer **600**.

Eventually, the motor **500** and the drum **200** can be coupled to the speed reducer **600**, so that the motor **500** and the drum **200** can be tilted in parallel manner with respect to the speed reducer **600** or vibrate at the same time with the vibration of the speed reducer.

The coaxiality and the coincidence as above-mentioned may not mean physically perfect coaxiality and coincidence, but can allow an error range acceptable in mechanical engineering or as recognized as coaxiality or coincidence by a person skilled in the art. For example, a state in which the drive shaft **530** and the drum rotation shaft **6341** are misaligned with each other by a range within 5 degrees can be

defined as being coaxial or coincident. However, the angle value is only an example, and the allowable error in design can be changed.

Because the drive shaft **530** rotates relative to the speed reducer **600** but is fixed thereto to prevent tilting of the drive shaft **530**, and the stator **510** is fixed to the speed reducer **600**, a distance between the stator **510** and the rotor **520** can be maintained to be constant. As a result, the collision between the stator **510** and the rotor **520** can be prevented. The noise or vibration that can occur due to the change of the rotation center as the rotor **520** rotates the stator **510** can be fundamentally blocked.

The drum rotation shaft **6341** can be configured to extend from the inside of the speed reducer **600** toward the drum **200**, and can vibrate together with the speed reducer **600** and tilt together with the speed reducer **600**. That is, the drum rotation shaft **6341** can be only configured to be rotatably coupled to the speed reducer **600**, but the installed position thereof can be fixed. As a result, the drum rotation shaft **6341** and the drive shaft **530** can be arranged side by side and coaxial with each other. In other words, the center of the drum rotation shaft **6341** and the center of the drive shaft **530** can be maintained in a coinciding manner with each other.

In some examples, a sealing portion **450** can be disposed between the drum rear face **220** and the rear plate **420**. The sealing portion **450** can seal between the drum rear face **220** and the rear plate **420** so that the air introduced into the duct portion **423** of the rear plate **420** may not flow out thereof and flows into the intake hole **224**.

The sealing portion **450** can be disposed on each of an outer side face and an inner side face of the duct portion **423**. A first sealing **451** can be disposed at a radially outer side of the duct portion **423**, and a second sealing **452** can be disposed at a radially inner side. The first sealing **451** can prevent hot air between the drum rear face **220** and the duct portion **423** from leaking radially outwardly. The second sealing **452** can prevent hot air between the drum rear face **220** and the duct portion **423** from leaking radially inwardly.

In other words, the sealing portions **450** can be disposed at the radially outer and inner sides of the intake hole **224**, respectively. The first sealing **451** can be disposed at the radially outer side of the intake hole **224**, and the second sealing **452** can be disposed at the radially inner side of the intake hole **224**.

The sealing portion **450** can be configured to be in contact with both the drum rear face **220** and the rear plate **420** in order to prevent the hot air from leaking out. Because the drum **200** rotates during the operation of the laundry treating apparatus, continuous friction from the drum rear face **220** is applied to the sealing portion **450**. Therefore, the sealing portion **450** can be made of a material that can seal between the drum rear face **220** and the duct portion **423** without deterioration in performance even due to the frictional force and frictional heat generated according to rotation.

In some examples, the motor **500** or the speed reducer **600** can be coupled to the rear face of the rear plate **420**, and the rear plate **420** can be made of a thin sheet metal, so that the rear plate **420** can be bent or deformed due to the load transmitted to the speed reducer **600** via the speed reducer **600** and the drum **200**. That is, the rigidity of the rear plate **420** can be secured to install the speed reducer **600** and the motor **500** thereon.

In some implementations, the rear plate **420** can further include a bracket **700** to reinforce coupling rigidity. The rear plate **420** can additionally be coupled to the bracket **700** and

the speed reducer **600** and the motor **500** can be coupled to the rear plate **420** via the bracket **700**.

The speed reducer **600** can be coupled simultaneously to the bracket **700** and the rear plate **420**. The fastener can simultaneously extend through and couple the speed reducer **600**, the rear plate **420**, and the bracket **700** to each other. The rear plate **420** can be coupled to the bracket **700** to ensure rigidity thereof. The speed reducer **600**, the motor **500**, etc. can be coupled to the rear plate **420** with the secured rigidity.

The fastening can be made in such a way that the speed reducer **600** is first coupled to the bracket **700** and the bracket **700** is then coupled to the rear plate **420**. That is, the speed reducer may not be directly coupled to the rear plate **420**, but can be fixed to the rear plate **420** via the bracket **700**.

In some examples, when the motor **500** or the speed reducer **600** can be coupled to the rear face of the rear plate **420**, the motor **500** and the speed reducer **600** can be exposed to the outside. In some examples, the motor **500** can be avoided from being exposed to the outside while being coupled to the rear face of the rear plate **420**. Further, the duct portion **423** can be heated by the hot air. Therefore, the rear face of the duct portion **423** can be thermally insulated.

The rear cover **430** can be coupled to the rear face of the rear plate **420** to prevent the duct portion **423** and the motor **500** or the speed reducer **600** from being exposed to the outside. The rear cover **430** can be spaced apart from the duct portion **423** and the driver.

The rear cover **430** has the effect of preventing the motor **500** from being damaged due to external interference, or preventing the drying efficiency from being lowered due to heat loss through the duct portion **423**.

FIGS. **6A** and **6B** show examples of an outer shape of the speed reducer.

The speed reducer **600** can include a speed reducer housing **610** and **620** constituting an outer shape thereof. The speed reducer housing can include a first housing **610** configured to face toward the drum and a second housing **620** to face toward the motor.

The speed reducer **600** can include a gearbox. The gearbox can be configured to receive power from the motor and convert the motor's RPM to a small RPM but increase the torque value and transmit the converted rotation force to the drum. A significant portion of the gearbox can be housed inside the second housing **620**, and the first housing **610** can be configured to shield the inside of the speed reducer **600**. In this way, an overall thickness of the speed reducer **600** can be reduced. The detailed configuration of the gearbox will be described later.

The first housing **610** can include a first housing shielding body **611** configured to shield the second housing **620** and a first housing shaft receiving portion **612** extending from the first housing shielding body **611** in a direction away from the second housing **620**. The first housing shaft receiving portion **612** can receive the drum rotation shaft **6341** and can support the drum rotation shaft **6341** such that the drum rotation shaft **6341** can rotate.

The first housing **610** can include a stator coupling portion **613** configured to support the motor. The stator coupling portion **613** can extend from a circumferential face of the first housing shielding body **611** in a direction away from the first housing shaft receiving portion **612**.

The stator coupling portion **613** can include a stator fastening hole **615** to which the motor can be fastened. The stator fastening hole **615** can be recessed in the stator coupling portion **613**. A separate fastener can be inserted

into the stator fastening hole **615**. The stator coupling portion **613** and the motor can be coupled to each other using the fastener.

The first housing **610** can further include a coupling guide **614** to guide the coupling of the motor. The coupling guide **614** can extend from the circumferential face of the first housing shielding body **611** in a direction away from the first housing shaft receiving portion **612**. The coupling guide **614** can extend from the first housing shielding body **611** so as to be connected to the stator coupling portion **613**. The coupling guide **614** can guide a position of the stator **510** when the stator **510** can be coupled to the stator coupling portion **613**. Thus, the assembly can be improved.

Referring to FIGS. **6A** and **6B**, the second housing **620** can house the gear assembly therein. In general, the gearbox **600** can include a sun gear, a planetary gear orbiting the sun gear, and a ring gear that accommodates the planetary gear and allows the planetary gear to rotate. The second housing **620** can include a second housing coupling body **621** coupled to the first housing **610**, a second housing shielding body **622** extending from the second housing coupling body **621** in a direction away from the first housing **610** and defining a space in which the gearbox is accommodated, and a second housing shaft receiving portion extending from an inner circumferential face of the second housing shielding body **622** in a direction away from the first housing **610** to support the drive shaft **530**.

In some implementations, a center of the first housing **610** and a center of the second housing **620** can be designed to be coaxial with each other. When the drive shaft **530** and the drum rotation shaft **6341** are coaxial with each other, this is advantageous for power transmission. Accordingly, the first housing shaft receiving portion **612**, which rotatably supports the drum rotation shaft **6341**, and the second housing shaft receiving portion, which rotatably supports the drive shaft **530**, can be coupled to each other so as to be coaxial with each other.

The drive shaft **530** can be inserted into the second housing **620** and rotatably supported within the second housing **620**. The drive shaft **530** can be coupled to the washer **540** that rotatably supports the rotor **520**. The washer **540** can include a receiving body **542** having a shaft support hole **543** defined in a center thereof for receiving the drive shaft **530**, and a washer coupling body **541** extending radially from an outer circumferential face of the receiving body **542** to define a face to which the rotor is coupled. The shaft support hole **543** can be formed in a groove shape corresponding to a protrusion formed on an outer circumferential face of the drive shaft **530** such that the protrusion can be received in the groove.

The washer **540** can include at least one washer coupling protrusion **5411** configured to protrude from the washer coupling body **541** in a direction away from the speed reducer. Further, the washer **540** can include one or more washer coupling holes **5412** extending through the washer coupling body **541**.

The washer coupling protrusion **5411** can be coupled to a receiving groove formed in the rotor. A fastener passing through the rotor can be inserted into the washer coupling hole **5412** to couple the rotor and the washer **540** to each other.

A plurality of washer coupling protrusions **5411** and a plurality of washer coupling holes **5412** can be alternately arranged along a circumferential direction and can be disposed on a surface of the washer coupling body **541**.

FIG. **7** is an enlarged cross-sectional view of the driver.

The driver can include the motor **500** that generates rotation power and the speed reducer that reduces the rotation speed of the motor **500** and delivers the rotation power having the reduced speed to the drum. The speed reducer **600** can include the drum rotation shaft **6341** that rotates the drum.

The motor **500** can include the stator **510** that generates a rotating magnetic field upon receiving external power and the rotor **520** that surrounds an outer circumferential face of the stator **510**. Permanent magnets can be disposed on an inner circumferential face of the rotor **520**.

The permanent magnets located on an inner circumferential face of the rotor **520** can move in a specific direction via rotating magnetism generated by the stator **510**, and the permanent magnet can be fixed to an inner circumferential face of the rotor **520**. Therefore, the rotor **520** can be rotated under the rotating magnetic field of the stator **510**.

The drive shaft **530** that rotates together with the rotor **520** and transmits the rotation power of the rotor **520** can be coupled to a center of rotation of the rotor **520**. The drive shaft **530** can be configured to rotate together with the rotor **520**. The drive shaft **530** can be coupled to the rotor **520** via the washer **540**.

The drive shaft **530** can be directly connected to the rotor **520**. Alternatively, when the drive shaft **530** is connected to the rotor via the washer **540**, the rotor **520** can be coupled thereto more firmly and thus can transmit the rotation force of the rotor **520** more effectively. Further, this can prevent the load from being concentrated on the drive shaft **530**, thereby increasing the durability of the drive shaft **530**.

The drive shaft **530** can be directly connected to the drum. In some cases, the drive shaft **530** can rotate at the same speed as that of the rotor **520**. In some cases, the speed of the drive shaft **530** can be decelerated to rotate the drum. For example, the drive shaft **530** can be connected to the speed reducer, and the speed reducer can be connected to the drum. That is, the speed reducer can decelerate the rotation of the drive shaft **530** to rotate the drum in the decelerated manner.

The speed reducer **600** can include a first housing **610** and a second housing **620** constituting an outer shape, and the gearbox **630** for reducing the power of the drive shaft **530**. The second housing **620** can provide a space to accommodate the gearbox **630** therein, and the first housing **610** can shield the accommodating space defined in the second housing **620**.

The second housing **620** can include a second housing coupling body **621** coupled to the first housing **610**, a second housing shielding body **622** extending rearwards from an inner circumferential face of the second housing coupling body **621** to define the receiving space for receiving the gearbox **630**, and a second housing shaft receiving portion **623** extending rearwardly from the second housing shielding body **622** and configured to receive the drive shaft **530**.

The gearbox **630** can include the ring gear **633** installed along an inner circumferential face of the second housing shielding body **622**. One or more planetary gear **632** meshed with the ring gear **633** can be disposed on an inner circumferential face of the ring gear **633**. The planetary gear **632** can be meshed with the ring gear **633**, and the sun gear **631** can rotate together with the drive shaft **530**.

The sun gear **631** can be configured to rotate while being coupled to the drive shaft **530**. The sun gear **631** can be implemented as a separate member from the drive shaft **530**. The disclosure is not limited thereto, and the sun gear **631** can be formed integrally with the drive shaft **530**.

Each of the sun gear **631**, the planetary gear **632** and the ring gear **633** can be implemented as a helical gear. When

each gear is implemented as the helical gear, noise can be reduced and power transmission efficiency can increase. However, the present disclosure is not limited thereto, and each of the sun gear **631**, the planetary gear **632**, and the ring gear **633** can be implemented as a spur gear.

In an operation example of the gearbox **630**, as the rotor rotates, the drive shaft **530** and the sun gear **631** connected to the drive shaft **530** rotate. Thus, the planetary gear **632** meshed with an outer circumferential face of the sun gear **631** can rotate while being disposed between the ring gear **633** and the sun gear **631**.

The planetary gear **632** can include a planetary gear shaft **6323** that is inserted into the center of rotation. The planetary gear shaft **6323** can rotatably support the planetary gear **632**.

The speed reducer can further include a first carrier **6342** and a second carrier **6343** supporting the planetary gear shaft **6323**. A front face of the planetary gear shaft **6323** can be supported on the second carrier **6343**, while a rear face thereof can be supported on the first carrier **6342**.

The drum rotation shaft **6341** can extend from the rotation center of the second carrier **6343** in a direction away from the motor. The drum rotation shaft **6341** can be implemented as a separate component from the second carrier **6343** and can be coupled thereto such that both rotate together. To the contrary, the drum rotation shaft **6341** can extend from the second carrier **6343** and be integrally formed with the second carrier **6343**.

The drum rotation shaft **6341** can be coupled to the drum to rotate the drum. As described above, the drum rotation shaft **6341** can be coupled to the drum via a connecting body such as a bushing, or can be directly coupled to the drum without a separate connecting body.

The drum rotation shaft **6341** can be supported on the first housing **610**. The first housing **610** can include a first housing shielding body **611** shielding the receiving space of the second housing **620**, and a first housing shaft receiving portion **612** extending from the first housing shielding body **611** in a direction away from the second housing **620** to accommodate the drum rotation shaft **6341** therein. A first bearing **660** and a second bearing **670** can be press-fitted to an inner circumferential face of the first housing shaft receiving portion **612** to rotatably support the drum rotation shaft **6341**.

The first housing **610** and the second housing **620** can be coupled to each other via a speed reducer fastener **681**. Further, the speed reducer fastener **681** passes through the first housing **610** and the second housing **620** at the same time and can couple both to each other. Further, the speed reducer fastener **681** passes through the first housing **610**, the second housing **620** and the rear plate **420** simultaneously to couple the first housing **610** and the second housing **620** to each other and at the same time to fix the speed reducer **600** to the rear plate **420**.

The rear plate **420** can be made of a thin steel plate. Therefore, the rear plate **420** may not secure the rigidity thereof to support all of the speed reducer **600**, the motor **500** connected to the speed reducer **600**, and the drum **200** connected to the speed reducer **600**. Therefore, the bracket **700** can be used to secure the rigidity of the rear plate **420** when coupling the speed reducer **600** to the rear plate **420**. The bracket **700** can be made of a material with higher rigidity than that of the rear plate **420** and can be coupled to the front face or rear of the rear plate **420**.

The bracket **700** can be coupled to the front face of the rear plate **420** to secure the rigidity such that the speed reducer **600** can be coupled thereto, and the speed reducer **600** can be coupled to the rear plate **420** and the bracket **700**

at the same time. A fastener such as a bolt can be used to couple the rear plate **420** to the bracket **700** and the speed reducer.

Further, in order to secure the speed reducer **600** to the rear plate **420**, the speed reducer fastener **681** that is used to couple the first housing **610** and the second housing **620** to each other can be used. That is, the speed reducer fastener **681** can extend through the second housing **620**, the first housing, the rear plate **420** and the bracket **700** at the same time to couple all thereof to each other. Thus, a front face of the rear plate **420** can be supported on the bracket **700** and a rear face thereof can be supported on the first housing **610**. Thus, when the speed reducer **600** can be coupled to the rear plate **420**, the rigidity thereof can be secured. However, the present disclosure is not limited thereto. First, only the first housing **610** and the second housing **620** can be coupled to each other using the speed reducer fastener **681**, and then the speed reducer **600** can be coupled to the rear plate **420** using a separate fastener.

Further, the stator coupling portion **613** to which the motor **500** can be coupled can be formed at a radially outer side of the first housing **610**. The stator coupling portion **613** can include a coupling groove formed by recessing which the stator coupling portion **613**.

The stator **510** can be directly coupled to the rear plate **420**, or can be coupled to the stator coupling portion **613**. The stator **510** can include a fixing rib **512** that can be disposed on an inner circumferential face thereof to support the stator. The fixing rib **512** can be coupled to the stator coupling portion **613**. The fixing rib **512** and the stator coupling portion **613** can be coupled to each other via a stator coupling pin **617**.

The motor **500** can be coupled to the speed reducer **600** while being spaced apart from the rear plate **420**, so that the motor **500** and the speed reducer **600** can constitute a single vibrating body. Therefore, even when external vibration is applied, the drive shaft **530** coupled to the rotor **520** and the drum rotation shaft **6341** connected to the speed reducer **600** can easily maintain the coaxial relationship with each other.

There is a risk that an axial direction of the drum rotation shaft **6341** can tilt due to the vibration of the drum **200**. However, the motor **500** can be coupled to the first housing **610** supporting the drum rotation shaft **6341**, such that even when the axial direction of the drum rotation shaft **6341** tilts, an axial direction of the drive shaft **530** can tilt by the same degree via the first housing **610**. That is, the motor **500** can move integrally with the speed reducer **600** so that the drum rotation shaft **6341** and the drive shaft **530** can be maintained in a coaxial relationship with each other even when the external force is applied thereto.

Under the above coupling structure, the efficiency and reliability at which the power generated from the motor **500** is transmitted to the drum **200** can increase, and wear, decrease in power transmission efficiency and durability and reliability of the gearbox **630** as caused by the axial misalignment between the drum rotation shaft **6341** and the drive shaft **530** can be prevented.

FIG. **8** shows the base and the rear plate.

Referring to FIG. **8**, the rear plate **420** can be located in rear of the drum. The rear plate **420** can guide the hot air discharged from the air circulating channel **820** to the drum. That is, the rear plate **420** can be located in rear of the drum to define a channel so that the hot air is uniformly supplied to an entirety of the drum.

The rear plate **420** can include the rear panel **421** facing toward the drum rear face, and the duct portion **423** that can be configured to be recessed rearwardly in the rear panel **421**

to define a channel. The duct portion **423** can be formed by pressing backwards the rear panel **421**. The duct portion **423** can be configured to receive a portion of the drum rear face.

The duct portion **423** can include an air inlet **4233** positioned in rear of the air circulating channel and an air flow portion **4231** positioned in rear of the drum. The air flow portion **4231** can be configured to receive a portion of the drum. The air flow portion **4231** can accommodate a portion of the drum, and can define a channel located in rear of the drum.

The air flow portion **4231** can be formed in an annular shape so as to face toward the intake hole defined in the rear face of the drum. The air flow portion **4231** can be configured to be recessed in the rear panel **421**. That is, the air flow portion **4231** can be configured so that a front face thereof is open, and can define a channel together with the rear face of the drum.

When the front face of the air flow portion **4231** can be configured to be open, the hot air flowing to the air flow portion **4231** can directly flow to the drum without passing through a separate component. Accordingly, this can prevent heat loss from occurring while hot air passes through the separate component. That is, there is an effect that can increase the drying efficiency by reducing the heat loss of the hot air.

The rear plate **420** can include the mount **425** disposed at the radially inner side of the air flow portion **4231**. The mount **425** can provide a location to which the speed reducer **600** or the motor **500** is coupled. That is, the rear plate **420** can include the mount **425** disposed at an inner side thereof, and the air flow portion **4231** formed in an annular shape and disposed at a radially outer side of the mount **425**.

Specifically, the air flow portion **4231** can include an outer circumferential portion **4231a** disposed outwardly of and surrounding an inner space in which hot air flows. Further, the air flow portion **4231** can include an inner circumferential portion **4231b** disposed inward of and surrounding the inner space in which hot air flows. That is, the outer circumferential portion **4231a** can constitute an outer circumference of the air flow portion **4231**, and the inner circumferential portion **4231b** can constitute an inner circumference of the air flow portion **4231**.

Further, the air flow portion **4231** can include a recessed face **4232** that forms a rear face of the channel through which the hot air flows. The recessed face **4232** can be configured to connect the outer circumferential portion **4231a** and the inner circumferential portion **4231b** to each other. That is, a space in which the hot air discharged from the air circulating channel **820** flows can be defined by the inner circumferential portion **4231b**, the outer circumferential portion **4231a**, and the recessed face **4232**.

Further, the recessed face **4232** prevents the hot air from leaking rearwardly and guides the hot air toward the drum. That is, the recessed face **4232** can be recessed in the air flow portion **4231**.

The air inlet **4233** can be positioned to face toward the air circulating channel **820**. The inlet can be positioned to face toward the blower **8231**. The air inlet **4233** can be configured to be recessed backwards in the rear panel **421** to prevent interference with the blower **8231**. A top of the air inlet **4233** can be connected to the air flow portion **4231**.

The laundry treating apparatus can include the connector **850** connected to the blower **8231**. The connector **850** can guide the hot air discharged from the blower **8231** to the air flow portion **4231**. The connector **850** can have a channel defined therein to guide the hot air discharged from the blower **8231** to the air flow portion **4231**. That is, the

connector **850** can define the channel for connecting the blower **8231** and the air flow portion **4231** to each other. A cross-sectional area of the channel defined inside the connector **850** can be configured to increase as the channel extends away from the blower **8231**.

The connector **850** can be positioned to face toward the air inlet **4233**. The air inlet **4233** can be formed to be recessed backwards to prevent interference with the connector **850**. Further, a top of the connector **850** can be configured to partition the air flow portion **4231** and the air inlet **4233** from each other. That is, the hot air discharged from the connector **850** can be introduced into the air flow portion **4231**, but can be prevented from flowing into the air inlet **4233**.

The connector **850** can be configured to evenly supply the hot air to the air flow portion **4231**. The connector **850** can be configured so that a width thereof increases as a distance thereof from the blower **8231** increases. The top of the connector **850** can be positioned along a circumferential extension line of the outer circumferential portion **4231a**.

Accordingly, the hot air discharged from the connector **850** can be supplied to an entirety of the air flow portion **4231** without flowing to the air inlet **4233**. The connector **850** prevents the hot air from being concentrated on one side of the air flow portion **4231**, so that the hot air can be evenly supplied to the inside of the drum. Therefore, there is an effect of increasing the drying efficiency of laundry.

The connector **850** can be configured to increase in a width thereof as it extends toward an upstream side, so that a velocity of hot air flowing along the connector **850** can be reduced according to a flow direction. That is, the connector **850** can perform a function of a diffuser that adjusts a speed of the hot air. The connector **850** can reduce the speed of the hot air to prevent the hot air from being concentrated on a specific portion of the drum.

Due to the shape of the connector **850** as described above, the air inlet **4233** configured to face toward the connector **850**, and configured to prevent interference with the connector **850** can be configured to increase in a width thereof as a distance thereof from the blower **8231** increases. Due to the shape of the air inlet **4233**, an overall shape of the duct portion **423** can have a character '9' in a front view.

Because the drum can be configured to rotate during the drying cycle, the drum can be configured to be spaced apart from the air flow portion **4231** by a predetermined distance. Hot air can escape through a separation space.

Accordingly, the laundry treating apparatus can further include the sealing portion **450** that prevents the hot air from leaking into the separation space between the drum and the air flow portion **4231**. The sealing portion **450** can be positioned along a perimeter of the air flow portion **4231**.

The sealing portion **450** can include the first sealing **451** extending along the outer circumference of the air flow portion **4231**. The first sealing **451** can be disposed between the drum and the outer circumference of the air flow portion **4231**. Further, the first sealing **451** can be configured to contact both the drum rear face **220** and the rear plate **420** to prevent the leakage more effectively.

In some examples, the first sealing **451** can be configured to be in contact with the front face of the connector **850**. Further, the first sealing **451** can be configured to be in contact with the top of the connector **850**. The connector **850** can define a channel through which hot air flows together with the air flow portion **4231**. Therefore, the first sealing **451** can be configured to be in contact with connector **850** to prevent the hot air from leaking into a space between the drum and the connector **850**.

The sealing portion **450** can include the second sealing **452** extending along an inner circumference of the air flow portion **4231**. The second sealing **452** can be disposed between the drum and an inner circumference of the air flow portion **4231**. Further, the second sealing **452** can be configured to contact both the drum rear face **220** and the rear plate **420**. The second sealing **452** can prevent the hot air flowing along the air flow portion **4231** from leaking toward the mount **425**.

Because the drum **200** rotates during the operation of the laundry treating apparatus, continuous friction from the drum rear face **220** is applied to the sealing portion **450**. Therefore, the sealing portion **450** can be made of a material capable of sealing between the drum rear face **220** and the air flow portion **4231** without deterioration in performance even with frictional force and frictional heat generated according to the rotation.

FIG. 9 shows a combined structure of the rear plate and the speed reducer, and the motor.

Referring to FIG. 9, the speed reducer **600** can be supported on the rear plate **420**, and the motor **500** can be coupled to the speed reducer **600**. That is, the rear plate **420** can be configured to support both the speed reducer **600** and the motor **500**.

The motor **500** that provides the rotation power and a speed reducer **600** that decelerates the power of the motor and transmits the same to the drum can be located in rear of the rear plate **420**.

The speed reducer **600** can be installed on the rear plate **420** so as to be located inside the duct portion **423**. The speed reducer **600** can be positioned radially inwardly of the air flow portion **4231** to prevent interference with the air flow portion **4231**.

A gear unit inside the speed reducer **600** can be damaged by the heat of the hot air flowing along the air flow portion **4231**. Accordingly, the air flow portion **4231** and the speed reducer **600** can be configured to be spaced apart from each other by a predetermined distance.

The speed reducer **600** can be coupled to and extend through the rear plate **420**. Therefore, the speed reducer **600** can be connected to the drum located in front of the rear plate **420**.

The stator **510** can be coupled to the speed reducer **600**. The stator **510** can be coupled to the speed reducer **600** and can be installed to be spaced apart from the rear plate **420**. In some examples, the speed reducer **600** can be located between the drum and the motor and can support the drum and the motor such that the drum and the motor are spaced apart from the rear plate **420**. That is, the speed reducer **600** can act as a center supporting the drum and the motor.

In some examples, the stator **510** can include the main body **511** formed in a ring shape, the fixing rib **512** that extends from an inner circumferential face of the main body **511** and can be coupled to the stator coupling portion **613** of the speed reducer, teeth **514** extending from and along an outer circumferential face of the main body **511** so that a coil is wound around the teeth, and a pole shoe **515** disposed at a free end of the teeth **514** to prevent the coil from being removed.

The rotor **520** can include the rotor body **521** that can be formed in a cylindrical hollow shape. Further, the rotor **520** can include an installation body **522** that is recessed frontwards in a rear face of the rotor body **521**. The rotor **520** can have permanent magnets disposed along an inner circumferential face of the rotor body **521**.

The rotor **520** can be coupled to the drive shaft **530** to transmit the rotation power of the rotor **520** to an external

component via the drive shaft **530**. The drive shaft **530** can be connected to the rotor **520** via the washer **540**.

Further, the motor **500** can include the washer **540** that supports the drive shaft **530**. The washer **540** can include the washer coupling body **541** that is coupled to the rotor. The washer coupling body **541** can be formed in a disk shape.

The washer **540** can include the receiving body **542** that is housed in the rotor. The receiving body **542** can be configured to protrude rearward from the washer coupling body **541**. The washer **540** can include the shaft support hole **543** extending through the center of the receiving body **542**. The drive shaft **530** can be inserted into the shaft support hole **543** and supported on the washer **540**.

Further, the washer **540** can include the washer coupling hole **5412** extending through the washer coupling body **541**. Further, the installation body **522** can include a rotor coupling hole **526** disposed at a position corresponding to that of the washer coupling hole **5412**. That is, the washer **540** and the rotor **520** can be coupled to each other via a coupling member that passes through the washer coupling hole **5412** and the rotor coupling hole **526** at the same time. That is, the washer **540** and the rotor **520** can be coupled to each other so as to rotate together.

Further, the washer **540** can include the washer coupling protrusion **5411** that projects rearward from the washer coupling body **541**. Further, the installation body **522** can include a washer protrusion receiving hole **525** configured to correspond to the washer coupling protrusion **5411**. The washer coupling protrusion **5411** can be inserted into the washer protrusion receiving hole **525** to support the coupling between the washer **540** and the rotor **520**.

Further, the rotor **520** can include a rotor installation hole **524** that extends through a center of the installation body **522**. The rotor installation hole **524** can accommodate the receiving body **542** therein. Accordingly, the washer **540** can rotate together with the drive shaft **530** via the rotor **520** and can firmly support the coupling between the drive shaft **530** and the rotor **520**. Therefore, this can secure the durability and reliability of an entirety of the motor **500**.

FIG. 10 shows a coupling structure of the speed reducer and the stator from the rear.

The stator **510** can include the main body **511** formed in a ring shape and fixed to the speed reducer **600**, the fixing rib **512** extending from an inner circumferential face of the main body **511** and coupled to the stator fastening hole **615** of the speed reducer, the teeth **514** extending from and along the outer circumferential face of the main body **511** and configured so that the coil is wound around the teeth, the pole shoe **515** disposed at the free end of the teeth **514** to prevent the coil from being removed, and a terminal that controls supply of the current to the coil.

The stator **510** can include a receiving space **513** defined inside the main body **511** and extending through the main body **511**. A plurality of fixing ribs **512** can be arranged to be spaced apart by a certain angular spacing around the receiving space **513** and can be disposed inside the main body **511**. A fixed rib hole **5121** where a fixing member is installed can be defined inside the fixing rib **512** so that the fixed rib hole **5121** and the stator fastening hole **615** of the speed reducer can be coupled to each other using the fixing member such as a pin.

When the stator **510** is directly coupled to the speed reducer **600**, a portion of the speed reducer **600** can be configured to be accommodated in the stator **510**. In particular, when the speed reducer **600** is accommodated in the stator **510**, an overall thickness of the driver including both

the speed reducer and the motor can be reduced, so that to volume of the drum can be further expanded.

In some examples, the speed reducer **600** can have a diameter smaller than a diameter of the main body **511**. That is, each of the first housing **610** and the second housing **620** can have a largest diameter smaller than the diameter of the main body **511**. Accordingly, the speed reducer **600** can be configured such that at least a portion thereof is accommodated in the main body **511**. However, the stator coupling portion **613** can extend from the housing of the speed reducer so as to overlap the fixing rib **512**. Accordingly, the stator coupling portion **613** can be coupled to the fixing rib **512** and portions of the first housing and the second housing **620** can be positioned inside the main body **511**.

FIG. **11** shows combination of the speed reducer and the motor.

The stator **510** can be coupled to the speed reducer **600**. The stator can be coupled to the stator coupling portion **613** protruding outwardly from the housing of the speed reducer **600** so that at least a portion of the speed reducer can be accommodated inside the main body **511**. Thus, the center of the main body **511** and the centers of the drive shaft **530** and the speed reducer **600** can be kept in a coaxial relationship with each other.

In some examples, the rotor **520** can be positioned to accommodate the stator **510** therein while being spaced apart from the pole shoe **515** by a certain distance. Because the drive shaft **530** is fixed to the speed reducer **600** housed in the main body **511**, a gap **G1** between the rotor **520** and the stator **510** can be maintained.

Therefore, the rotor **520** and the stator **510** can be prevented from colliding with each other or the rotor can be prevented from rotating while the rotor is temporarily misaligned with the stator, so that noise or vibrations can be prevented.

In some examples, all of an imaginary first diameter line **K1** passing through the center of the speed reducer **600** and the center of the drive shaft **530**, and an imaginary second diameter line **K2** passing through the center of the main body **511**, and an imaginary third diameter line **K3** passing through the center of the rotor **520** can meet each other at the rotation center of the speed reducer **600**.

In this way, the speed reducer **600** itself can act as the center of rotation of the drive shaft **530**, and the stator **510** can be fixed directly to the speed reducer **600**, so that the drive shaft **530** can be prevented from being misaligned with the speed reducer **600**. As a result, the reliability of the speed reducer **600** can be guaranteed.

FIG. **12** is a perspective view showing the base **800** of the laundry treating apparatus.

Referring to FIG. **12**, the base **800** can include the air circulating channel **820** which can be disposed at one side of the base **800**, and circulate the air in the drum. Further, at the other side of the base **800**, a component mount **810** that provides a space in which components for the operation of the dryer are installed can be provided. The component mount **810** can be disposed out of the air circulating channel **820**.

In the conventional dryer, the air circulating channel **820** may be disposed on the base **800**, and the driver for rotating the drum **200** may be installed on the base **800**. Because the driver occupies a large portion of an installation space of the base **800**, the component mount **810** formed in a space of the base **800** except for the air circulating channel **820** has a small space. Thus, it may not be easy to install other components of the laundry treating apparatus on the component mount.

In some implementations, in the laundry treating apparatus, the motor **500** rotating the drum **200** can be spaced apart from the base **800** and can be disposed in rear of the drum **200**. Thus, without the motor **500** in the base, a space of the base **800** can be utilized in various ways.

A compressor **930** for compressing refrigerant for heat exchange can be installed at the component mount **810**. Further, the base **800** can include a water collector **860** which can be configured to be spaced apart from the compressor **930**, and into which the condensate generated in the air circulating channel **820** is collected. A control box **190** for controlling the compressor **930** and the motor can be installed on the component mount **810**.

The control box **190** can be installed on the base and supported thereon firmly. Further, the control box **190** and a connection line for connecting components controlled by the control box to each other can be firmly supported on the base **800**.

In another example, the water collector **860** may not be disposed between the compressor **930** and the air circulating channel **820**, but can be disposed to overlap the compressor **930** in the front-rear direction. Because the water collector **860** may be located in a space where the motor is conventionally disposed, a volume of the water collector **860** can be expanded. When the volume of the water collector **860** increases, a frequency of emptying the collected condensate can be reduced, so that the user's convenience can be improved.

A side face of the base **800** can be coupled to the side panel constituting the side face of the cabinet. The side panel can include the first side panel **141** and the second side panel **142**. The control box **190** can be installed on the component mount **810** and can be installed closer to one of the side panels.

The control box **190** can control all operations of the laundry treating apparatus. Therefore, there can be many cases of checking or repairing the control box **190**.

When the control box **190** is adjacent to the first side panel **141**, the user can access the control box **190** by removing only the first side panel **141**. Accordingly, there is an effect that easiness of maintenance increases.

When the first side panel **141** is removed, various components such as the compressor **930** and the control box **190** can be easily accessed by the user, so that the first side panel **141** can be referred to as a service panel.

FIG. **12** shows a state in which the component mount **810** is located at a left side of the base **800** and the control box **190** can be accessed by the user when the first side panel **141** is removed. However, the present disclosure is not limited thereto. When the air circulating channel **820** is formed on at the left side and the component mount **810** is formed at the right side, the control box or the compressor can be repaired and checked by removing the right panel.

In some examples, the air circulating channel **820** can further include the duct cover **830** positioned at a top of the air circulating channel **820** to define a channel through which air discharged from the drum flows. The duct cover **830** can be coupled to an open top face of the air circulating channel **820**.

The top faces of the inflow duct **821** and the air flow duct **822** are open so that air can flow in and out through the open top faces. The duct cover **830** can shield an open top face of the air flow duct **822**. Therefore, the duct cover **830** allows the air of the drum to flow into the channel through the inflow duct **821**, and prevents the air flowing into the inflow duct **821** from flowing out of the channel through the open top face of the air flow duct **822**. That is, the duct cover **830**

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can constitute one face of the channel that guides the air introduced through the inflow duct **821** to the air discharge duct **823**.

The air discharge duct **823** can include the blower **8231** that discharges air out of the air discharge duct **823**. The blower **8231** can discharge the air that has passed through the inflow duct **821** and the air flow duct **822** out of the air discharge duct **823**.

The blower **8231** can provide a space where the air circulating channel fan **950** that circulates the air inside the drum is installed. The air circulating channel fan **950** can increase a circulating speed of air by forcibly flowing the air, and thus has the effect of shortening a drying time by increasing a drying speed of laundry.

When the air circulating channel fan **950** rotates, air can flow in such a way that the air is discharged through an opening formed above the blower **8231**. The air discharged from the blower **8231** can flow back into the drum and can be used to dry the laundry.

The air circulating channel fan **950** can employ various types of fans. For example, a sirocco fan can be applied so that air is introduced in a direction of the rotation shaft and is discharged in a radial direction. However, the present disclosure is not limited thereto, and various fans can be used to generate the air flow according to design purposes.

The duct cover **830** can include a communication cover body **8312** coupled to a top of the inflow duct **821** and a shielding cover body **8311** coupled to the top of the air flow duct **822**. The shielding cover body **8311** can extend from the communication cover body **8311**, and the shielding cover body **8311** can be formed integrally with the communication cover body **8312**.

The communication cover body **8312** can include an inflow communication hole **8314** that communicates the drum and the inflow duct **821** with each other. Even when the communication cover body **8312** can be coupled to the inflow duct **821**, the inflow communication hole **8314** can guide the air discharged from the drum to the inflow duct **821**.

Further, the shielding cover body **8311** can shield the top face of the air flow duct **82**. Thus, the air introduced into the inflow duct **821** can be guided to the air discharge duct **823** while not flowing out of the air circulating channel **820** via the air flow duct **822**.

The shielding cover body **8311** can include a cleaning water channel **833** through which water can flow and which can be disposed in a top face of the shielding cover body **8311**. The cleaning water channel **833** can receive water and spray the water toward the first heat exchanger located below the duct cover **830**.

A cover through-hole **8313** vertically extending through the shielding cover body **8311** can be disposed downstream of the cleaning water channel **833**. Water flowing along the cleaning water channel **833** can be sprayed downwardly of the shielding cover body **8311** through the cover through-hole **8313**.

The first heat exchanger to dehumidify the air discharged from the drum can be disposed below the cover through-hole **8313**. Therefore, the water passing through the cover through-hole **8313** can be sprayed towards the first heat exchanger to wash the first heat exchanger.

A nozzle cover can be coupled to a top of the cleaning water channel **833**. The nozzle cover can shield an open top face of the cleaning water channel **833**. The nozzle cover can prevent the air flowing along the air flow duct **822** from leaking through the cover through-hole **8313**. Further, the nozzle cover shields the top face of the cleaning water

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channel **833** to prevent the water flowing along the cleaning water channel **833** from scattering to the outside.

Alternatively, the air circulating channel **820** can further include a duct filter that can be disposed in front of the first heat exchanger to filter foreign substances of air that has passed through the inflow duct **821**. The duct filter can be disposed between the inflow duct **821** and the first heat exchanger to prevent foreign substances from being deposited on a front face of the first heat exchanger, thereby improving the drying efficiency and heat exchange efficiency of the first heat exchanger.

When the foreign substances are deposited on the duct filter, the circulation of air passing through the inflow duct **821** and the air flow duct **822** can be disturbed. In order to solve the above problem, the cleaning water channel **833** can remove the foreign substances deposited on the duct filter using water pressure via spraying water toward the duct filter.

However, for convenience of description, the following description will be based on the laundry treating apparatus in which the duct filter is omitted.

A channel switching valve **870** that can be coupled to the cleaning water channel **833** and supplies water for cleaning to the cleaning water channel **833** can be further included. The channel switching valve **870** can be connected to a water supply source to selectively supply water to the cleaning water channel **833**. The water supply source can include the water collector **860**.

The channel switching valve **870** can be connected to the water collector **860** via a hose to guide the water collected in the water collector **860** to the cleaning water channel **833**. The channel switching valve **870** can guide the water collected in the water collector **860** to the water storage tank **120** (refer to FIG. 1).

FIG. 13 is an exploded perspective view showing the duct cover and the water collector cover in a separated state from the base in FIG. 12.

Referring to FIG. 13, below the duct cover **830**, the first heat exchanger **910** and the second heat exchanger **920** which sequentially exchange heat with the air inside the drum **200** can be installed so as to be spaced apart from each other in the front and rear direction. The air inside the drum **200** introduced into the inflow duct **821** can be heat-exchanged in the first heat exchanger **910** such that the moisture is removed therefrom, and the air from which the moisture has been removed can be heat-exchanged in the second heat exchanger **920** and thus can be heated. The heated air can be supplied back into the drum **200** through the air discharge duct **823**.

The air circulating channel **820** can further include a water cover **826** disposed between the first heat exchanger **910** and a bottom face of the air flow duct **822**. The water cover **826** can be configured to be supported on the air flow duct **822**.

The water cover **826** can be configured to be positioned under the first heat exchanger **910** to support the bottom face of the first heat exchanger **910**. The water cover **826** can support the first heat exchanger **910** so as to be spaced away from the bottom face of the air flow duct **822**.

In the first heat exchanger **910**, condensate can be generated by condensing the wet steam discharged from the drum **200**. When the condensate is not discharged from the inside of the laundry treating apparatus and remains, there is a problem that an odor is generated or the drying efficiency is reduced. Thus, the condensate can be collected while being spaced away from the first heat exchanger **910** or the second heat exchanger **920**, and then discharged.

The water cover **826** can support the first heat exchanger **910** so as to be spaced apart from the bottom face of the air flow duct **822** to define a space between the bottom face of the air flow duct **822** and the water cover **826**. The condensate can flow into the water collector **860** along the space defined by the water cover **826**.

The air dehumidified through the first heat exchanger **910** can be heated in the second heat exchanger **920**. The air passing through the second heat exchanger **920** has a low moisture content. As the air is heated, an amount of saturated steam increases, so that it is difficult to generate condensate. Accordingly, the water cover **826** can be positioned on a bottom face adjacent to the first heat exchanger **910**, and the water cover **826** can be configured to be spaced apart from the second heat exchanger **920**.

Because only a portion of a top face of the water cover **826** is shown in FIG. 13, a shape of the channel formed by the water cover **826** and a detailed structure of the water cover **826** will be described later.

In some examples, the base **800** can include the water collector **860** that can be configured to be spaced apart from the air circulating channel **820** and configured to collect the condensate generated in the air circulating channel **820**. The water collector **860** can include the water collector body **862** that has a space defined therein where condensate is collected.

The water collector **860** can further include a water collector cover **863** shielding an open top face of the water collector body **862**. Moisture-sensitive components can be installed around the water collector **860**. In some examples, the condensate collected in the water collector body **862** can be blocked from scattering to the outside. For example, the water collector cover **863** can be coupled to the water collector body **862** to block the condensate from leaking to the top face of the water collector body **862**.

Further, the water collector **860** can include a pump that allows the condensate collected inside the water collector body **862** to flow to the outside. In order for the pump to function properly, the inside of the water collector body **862** can be sufficiently sealed. The water collector cover **863** seals the inside of the water collector body **862** to increase the reliability of the pump.

The water collector cover **863** can include a water collector cover body **8631** that constitutes a shielding face of the water collector body **862**. Further, the water collector cover **863** can include at least one of a support body **8635** configured to support the water collector cover body **8631** and a fastening hook **8636** configured to couple the water collector cover body **8631** to the water collector body **862**.

The support body **8635** can protrude from a circumference of the water collector cover body **8631** and be seated on the base. The fastening hook **8636** can be formed to protrude from the water collector cover body **8631**. The fastening hook **8636** can firmly fix the water collector cover body **8631** to the water collector body **862**. The fastening hook **8636** can be fixedly inserted into a hook hole to be described later.

The condensate generated in the air circulating channel **820** is collected inside the water collector body **862**. The top face of the water collector body **862** can be open, such that the condensate can be scattered to the outside. However, the water collector body **862** is located adjacent to the control box **190**, the compressor **930**, and the like. Thus, when the condensate scatters out of the water collector body **862**, a failure of the mechanical devices can occur.

The water collector cover **863** can shield the open top face of the water collector body **862** using the water collector

cover body **8631** to prevent the condensate from scattering. The support body **8635** and the fastening hook **8636** can firmly fix the water collector cover body **9631** to the water collector body **862**. Therefore, this can prevent the condensate from scattering and thus a failure of the device from occurring.

Further, the water collector cover **863** can include a pump receiving portion **8634** configured to extend through the water collector cover body **8631** and to receive the pump. Further, the water collector cover **863** can include a drain channel **8637** that protrudes upwardly from the water collector cover body **8631** and is formed in a pipe shape communicating an inside and an outside of the water collector body **862** to each other.

The pump receiving portion **8634** can receive therein the pump configured to move the condensate collected inside the water collector body **862** out of the water collector body **862**. When the pump is activated, the condensate stored in the water collector body **862** can be discharged through the drain channel **8637**.

The hose can be connected to the drain channel **8637** to guide the discharged condensate out of the water collector body **862**. One end of the hose can be connected to the drain channel **8637**, and the other end thereof can be connected to the channel switching valve **870**. However, the disclosure is not limited thereto, and the other end of the hose can be located out of the cabinet to drain the condensate directly out of the cabinet. The other end of the hose can be connected to the water storage tank **120** (refer to FIG. 1) located on a top of the cabinet, so that the condensate collected in the water collector body **862** can be guided to the water storage tank **120**.

The water collector cover **863** can further include a return channel **8638** which can be spaced apart from the drain channel **8637** and communicate an inside and an outside of the water collector body **862** with each other. The return channel **8638** can communicate the water collector body **862** and the water storage tank with each other. The return channel **8638** can guide water from the water storage tank back to the water collector body **862**.

The return channel **8638** can be connected via the hose to the water storage tank **120** disposed on the top of the cabinet (see FIG. 3). To prevent water from overflowing the water storage tank, when the water storage tank is full of water, the water stored in the water storage tank can flow back to the water collector body **862** via the hose connecting the return channel **8638** and the water storage tank to each other. There is an effect that the user's convenience can be improved by reducing the frequency at which the user directly drains the water.

In some examples, the channel switching valve **870** for switching the channel along which the condensate collected in the water collector **860** flow can be further included. The pump can be connected to the channel switching valve **870** via the hose. The water stored in the water collector body **862** can flow, under the operation of the pump, to the channel switching valve **870**. The channel switching valve **870** can guide the flowing water to various paths.

The channel switching valve **870** can be connected to the cleaning water channel **833** to move the water to the cleaning water channel **833**. Water directed to the cleaning water channel **833** can be used to clean the first heat exchanger.

Further, the channel switching valve **870** can be connected to the water storage tank **120** via the hose to guide the condensate flowing from the water collector body **862** to the

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water storage tank **120**. The user can directly drain water from the water storage tank where the condensate is stored.

The channel switching valve **870** can be controlled by the control box **190**, and can operate in a different manner depending on an operation timing of the laundry treating apparatus. For example, when an operation of the first heat exchanger **910** has been completed in the drying cycle, the control box **190** can control the channel switching valve **870** to direct the condensate to the cleaning water channel **833**. Further, when washing of the first heat exchanger **910** has been completed, the control box **190** can control the channel switching valve **870** to guide the condensate to the water storage tank **120**.

In some examples, as described above, in order for the pump to operate normally, it is desirable to seal an inside of the space to which the pump drains water. Because the water collector cover **863** can be firmly coupled to the water collector body **862** using the support body **8635** and the fastening hook **8636**, this can easily seal the space where the condensate is stored. Thus, operational reliability of pump **861** can be improved. A sealing can be added to a portion where the water collector cover **863** and the water collector body **862** are joined to each other, thereby improving water-tightness of the space.

In some examples, the water collector cover **863** can be configured to seal the inside of the water collector body **862**, and can be detachably coupled to the water collector body **862**. Foreign substances such as lint included in the condensate generated by the first heat exchanger **910** can flow into the water collector body **862**. When the foreign substances with large particles are introduced thereto, there can be a problem that the substances can interfere with the operation of the pump.

In some examples, the water collector cover **863** can be removed to remove the foreign substances introduced into the water collector body **862**. Accordingly, the water collector cover **863** can be detachably coupled to the water collector body **862**. In some examples, there is an effect that the water collector cover **863** can be easily removed from the water collector body **862** using the fastening hook **8636**.

That is, in a general use environment, the support body **8635** and the fastening hook **8636** can securely shield the open top face of the water collector body **862** to prevent the condensate from scattering to the outside.

In some cases, when the water collector cover **863** is removed in order to remove foreign substances deposited on the water collector body **862**, the fastening hook **8636** can be used to easily remove the water collector cover.

In some examples, the duct cover **830** can include a cover mount hook **8391** formed along a perimeter thereof, and a duct protrusion **824** protruding from and along a periphery of the air circulating channel **820** and coupled to the cover mount hook **8391**.

The cover mount hook **8391** can be coupled to the duct protrusion **824** to couple the duct cover **830** to the air circulating channel **820**. That is, the duct cover **830** can be securely fastened to the duct protrusion **824** using the cover mount hook **8391** in a state seated around the inflow duct **821** and the air flow duct **822**.

A sealing can be added to a contact face of the duct cover **830** and the air circulating channel **820** to prevent air from leaking from the inside of the air circulating channel **820** to the outside.

FIG. **14** is a cross-sectional view showing an arrangement relationship of the drum and the air circulating channel in the laundry treating apparatus. Descriptions of those duplicate with the configurations as described in FIG. **13** are omitted.

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The cabinet **100** can include the first side panel **141** positioned on one side of the drum **200** to constitute one side face thereof, and the second side panel **142** positioned on the other side of the drum **200** to constitute the other side face thereof.

In this case, the air circulating channel **820** can be disposed closer to one of the first side panel **141** and the second side panel **142** than to the other thereof. The water collector **860** can be disposed closer to the other of the first side panel **141** and the second side panel **142**.

In some examples, the air circulating channel **820** can be disposed closer to the second side panel **142** than to the first side panel **141**. The air flow duct **822** and the duct cover **830** can be disposed closer to the second side panel **142** than to the first side panel **141**. The first side panel **141** can define a left side face with respect to the drum **200**, and the second side panel **142** can define a right side face with respect to the drum **200**.

Accordingly, the water collector **860** can be spaced apart from the air circulating channel **820** and disposed out of the air circulating channel **820**. The water collector **860** can be installed between the second side panel **142** and the air circulating channel **820**.

In some examples, the channel switching valve **870** can be coupled to the air circulating channel **820** so as to communicate with the cleaning water channel **833** and can be configured to deliver the condensate to the cleaning water channel **833**. In some examples, the channel switching valve **870** can be coupled to the air circulating channel **820** and extend by a predetermined length **L9**. Thus, depending on an arrangement of the channel switching valve **870**, the channel switching valve **870** can interfere with the drum **200**.

To solve this situation, the channel switching valve **870** can be positioned at a lower level than that of the top face of the duct cover **830** and can be configured to face toward a side face of the air flow duct **822**. In some examples, the channel switching valve **870** can be disposed between the air circulating channel **820** and the first side panel **141** and can face toward the water collector **860**. A vertical level of a top of the channel switching valve **870** can be lower than that of the top face of the duct cover **830**.

Thus, the channel switching valve **870** may not interfere with the drum **200**. Thus, the user can remove the first side panel **141** without removing the drum **200** to easily repair and maintain the channel switching valve **870**.

Further, the duct cover **830** can include a valve connector **838** extending toward the water collector **860** and facing toward the water collector **860**. The valve connector **838** can be configured to be disposed above the water collector **860**, and can be configured to be disposed side by side with the water collector **860**.

The channel switching valve **870** can be coupled to a bottom face of the valve connector **838** and can extend toward the water collector **860**. The cleaning water channel **833** can be configured such that one end thereof is formed on a top face of the valve connector **838** and communicates with the channel switching valve **870**.

Coupling the channel switching valve **870** to the bottom face of the valve connector **838** can allow the channel switching valve **870** to be further prevented from interfering with the drum **200**. Further, in the laundry treating apparatus, a radius **R** of the drum **200** can be further expanded within a range in which interference with the channel switching valve **870** is prevented. The channel switching valve **870** can be freely positioned according to the position of the valve connector **838**.

In some examples, regarding a detailed structure of the channel switching valve **870**, the channel switching valve **870** can include a water receiving portion **871** communicating with the pump **861** and receiving the water from the pump **861**, and a connective portion **879** communicating with the water receiving portion **871** and coupled to the duct cover **830** to deliver the water to the cleaning water channel **833**.

Further, the channel switching valve **870** can further include a water delivering portion **872** disposed between the water receiving portion **871** and the connective portion **879**. The water delivering portion **872** can be coupled to each of the water receiving portion **871** and the connective portion **879** and guide the water supplied from the water receiving portion **871** to the connective portion **879**. In other words, the water receiving portion **871**, the water delivering portion **872**, and the connective portion **879** can be arranged in this order along a direction in which the condensate flows.

In some examples, the connective portion **879** can be coupled to the valve connector **838** and extend toward the water collector **860**, and the connective portion **879** can be disposed to face toward the water collector body **862**. Further, the connective portion **879** can be coupled to the bottom face of the valve connector **838** and communicate with the cleaning water channel **833** to deliver the condensate to the cleaning water channel **833**. The connective portion **879** can be disposed at a side in a longitudinal direction of the air circulating channel **820**. At least a portion of the connective portion **879** is positioned at a lower level than that of the top face of the air circulating channel **820**.

In some examples, the water collector **860** can include the drain channel **8637** protruding upward from the water collector cover **863** and communicating the water collector body **862** with the outside of the water collector cover **863**, and a first water collector drain pipe **8911a** for connecting the drain channel **8637** and the channel switching valve **870** to each other such that the condensate flows from the pump **861** to the channel switching valve **870**. The first water collector drain pipe **8911a** can act as a passage through which the condensate flows from the pump **861** to the channel switching valve **870**.

In some examples, the water receiving portion **871** can be connected to the first water collector drain pipe **8911a** and can receive condensate from the pump **861** through the first water collector drain pipe **8911a**. The condensate supplied to the water receiving portion **871** can be delivered to the water delivering portion **872** and the connective portion **879**.

As the channel switching valve **870** extends from the valve connector **838** toward the water collector **860**, the water receiving portion **871** can be disposed on top of the water collector **860** and face toward the water collector **860**.

Thus, a distance between the water receiving portion **871** and the pump **861** can be reduced. Thus, an extension length of the first water collector drain pipe **8911a** for connecting the pump **861** and the water receiving portion **871** to each other can be reduced to prevent the condensate from remaining in the first water collector drain pipe **8911a**.

FIG. 15 is a perspective view showing the cleaning water channel disposed on the top face of the duct cover in the laundry treating apparatus.

The duct cover **830** can include a shielding cover body **8311** coupled to a top of the air flow duct **822** for shielding the first heat exchanger **910** and the second heat exchanger **920**, and a communication cover body **8312** extending forward from the shielding cover body **8311** and coupled to a top of the inflow duct **821**.

The shielding cover body **8311** can be configured to shield the open top face of the air flow duct **822**, and the communication cover body **8312** can be configured to be seated on a top face of the inflow duct **821**.

In this case, the shielding cover body **8311** and the communication cover body **8312** can be formed integrally with each other. Accordingly, an assembly process of the duct cover **830** can be simplified, and the air inside the air flow duct **822** and the inflow duct **821** can be prevented from leaking to a space between the shielding cover body **8311** and the communication cover body **8312**.

Further, the communication cover body **8312** can include an inflow communication hole **8314** that passes through one face thereof and communicates the drum **200** and the inflow duct **821** with each other. The inflow communication hole **8314** can communicate with the duct communication hole **417**. Thus, the air discharged from the drum **200** can be introduced through the inflow communication hole **8314**.

The inflow duct **821** can have a width larger than a width of the air flow duct **822**. Accordingly, the communication cover body **8312** seated on the top face of the inflow duct **821** can have a greater width than that of the shielding cover body **8311**.

Further, the inflow communication hole **8314** formed in the communication cover body **8312** has a larger width than that of the shielding cover body **8311**. Accordingly, the air inside the drum **200** can be smoothly introduced into the inflow communication hole **8314** that communicates with the drum **200**.

The inflow communication hole **8314** has a larger diameter than that of the shielding cover body **8311**, and one portion of the inflow communication hole **8314** extends in a parallel manner to the shielding cover body **8311** and the other portion of the inflow communication hole **8314** can be configured to protrude toward the connective portion **879**.

In some examples, the water supplied to the cleaning water channel **833** through the connective portion **879** flows along the top face of the shielding cover body **8311** and is discharged to the first heat exchanger **910**. In this way, the foreign substances attached to the front face of the first heat exchanger **910** can be removed.

In some implementations, the shielding cover body **8311** can include a cover through-hole **8313** extending through a top face thereof and facing toward at least a portion of the first heat exchanger **910**. The cover through-hole **8313** can be disposed at an end of the cleaning water channel **833** so as to communicate the cleaning water channel **833** and the first heat exchanger **910** with each other.

The cover through-hole **8313** can act as an outlet of the cleaning water channel **833**, and the water flowing along the cleaning water channel **833** can be sprayed to the first heat exchanger **910** via the cover through-hole **8313**.

Accordingly, the foreign substances attached to the first heat exchanger **910** can be removed by the water discharged from the cleaning water channel **833** through the cover through-hole **8313** without the user having to separate the first heat exchanger **910** to clean the same.

The cover through-hole **8313** can correspond to a width direction of the shielding cover body **8311**, and can extend parallel to the extension direction of the valve connector **838**. A width **W5** of the cover through-hole **8313** can be smaller than a width of the shielding cover body **8311**, and can correspond to a width of the first heat exchanger **910** shown in FIG. 14.

In some examples, the connective portion **879** can be connected to the water delivering portion **872** shown in FIG. 14 and configured to deliver water to the cleaning water

channel **833**. For instance, the connective portion **879** can include receiving channels **8791a**, **8791b**, and **8791c** that communicate with the water delivering portion **872** and receive water from the water delivering portion **872**. The receiving channels **8791a**, **8791b**, and **8791c** can extend through the valve connector **838** and communicate with the cleaning water channel **833** and deliver the condensate supplied from the water delivering portion **872** to the cleaning water channel **833**.

Further, the cleaning water channel **833** can include a valve communication hole **8382** passing through the bottom face thereof and communicating with the receiving channels **8791a**, **8791b**, and **8791c**. The condensate supplied from the receiving channels **8791a**, **8791b**, and **8791c** can flow into the cleaning water channel **833** through the valve communication hole **8382**. The valve communication hole **8382** can be disposed on a top face of the valve connector **838**, and can be disposed on a top face of the shielding cover body **8311** and along an extension direction of the receiving channels **8791a**, **8791b**, and **8791c**.

In some examples, the cleaning water channel **833** can be disposed on a top face of the shielding cover body **8311** to guide the water flowing therein from the valve communication hole **8382** to the cover through-hole **8313**. That is, the cleaning water channel **833** can extend from the valve communication hole **8382** to the cover through-hole. The valve communication hole **8382** can act as a starting point of the cleaning water channel **833**, and the cover through-hole **8313** can act as an ending point of the cleaning water channel **833**.

For example, one end of the cleaning water channel **833** can be disposed on a top face of the valve connector **838**, and the other end thereof can be connected to the cover through-hole **8313**. Further, one end of the cleaning water channel **833** can extend toward the valve connector **838**, and the other end thereof can extend toward the cover through-hole **8313**.

In some examples, the condensate supplied to the cleaning water channel **833** through the valve communication hole **8382** can friction with an inner face of the cleaning water channel **833** while flowing along the cleaning water channel **833**, and thus a flow speed can gradually decrease. Accordingly, the condensate inside the cleaning water channel **833** may not be discharged from the cleaning water channel **833** but can remain therein.

In some implementations, the shielding cover body **8311** can include an inclined face **8316** configured such that a portion of a top face extends in a downwardly inclined manner and frontwards. At least a portion of the cleaning water channel **833** can be disposed in the inclined face **8316**.

This can minimize an amount of residual water that is not discharged from the cleaning water channel **833**. Further, as the water flowing through the cleaning water channel **833** flows along the inclined face **8316**, the flow speed thereof naturally increases to remove the foreign substances remaining in the first heat exchanger **910**.

In some examples, the inclined face **8316** can include a first inclined face **8316a** which extends from the top face of the shielding cover body **8311** in a downwardly inclined manner and in a frontward direction, and a second inclined face **8316b** extending from the first inclined face **8316a** toward the communication cover body **8312** in an inclined manner. An inclination of the first inclined face **8316a** can be greater than that of the second inclined face **8316b**.

The cleaning water channel **833** can include a guide channel **8331** that communicates with the valve communication hole **8382** and receives water from the valve com-

munication hole **8382**, and a discharge channel **8332** that is connected to the guide channel **8331** and extends to the cover through-hole **8313**.

The guide channel **8331** can be disposed on the top face of the shielding cover body **8311** which is positioned at a higher level than that of the inclined face **8316**, and the discharge channel **8332** can be disposed on the inclined face **8316**.

Further, the discharge channel **8332** can include a first discharge channel **8332a** connected to the guide channel **8331** and disposed on the first inclined face **8316a**, and a second discharge channel **8332b** connected to the first discharge channel **8332a** and disposed on the second inclined face **8316b**.

The guide channel **8331** can have one end disposed on the top face of the valve connector **838** and extending toward the first inclined face **8316a**. One end of the first discharge channel **8332a** can communicate with the guide channel **8331**, while the other end thereof can communicate with the second discharge channel **8332b**, so that water flowing from the guide channel **8331** can be guided to the second discharge channel **8332b**.

One end of the second discharge channel **8332b** can communicate with the first discharge channel **8332a** while the other end thereof can be connected to the cover through-hole **8313**, so that water flowing from the first discharge channel **8332a** can be guided to the cover through-hole **8313**.

Accordingly, the flow speed of the water supplied to the guide channel **8331** from the valve communication hole **8382** can increase naturally as it passes through the first discharge channel **8332a** and the second discharge channel **8332b**. In other words, as the first discharge channel **8332a** and the second discharge channel **8332b** extend in a downwardly inclined manner, the flow speed of the water flowing from the guide channel **8331** to the cover through-hole **8313** can increase naturally.

Further, the water inside the cleaning water channel **833** flows along the first discharge channel **8332a** and the second discharge channel **8332b** to the cover through-hole **8313**. Thus, the situation can be prevented in which the water inside the cleaning water channel **833** is not be discharged to the cover through-hole **8313** and remains inside the cleaning water channel **833**.

In some examples, as the flow speed of the water increases as the water flow along the channel, a diameter thereof becomes narrower. Thus, the water inside the cleaning water channel **833** may not be uniformly dispersed at a distal end of the cleaning water channel **833**. This can result in concentrated discharge to only a specific area of the cover through-hole **8313**, and thus can result in water not being evenly supplied to the surface of the first heat exchanger **910**.

Accordingly, the cleaning water channel **833** can include a plurality of channels and disposed on the top face of the shielding cover body **8311**. Each of the distal ends of the plurality of cleaning water channels **833** can be connected to the cover through-hole **8313**. Accordingly, a width of the distal end of one of the plurality of cleaning water channels **833** can be smaller than that of a single cleaning water channel **833** when the cleaning water channel **833** only includes the single cleaning water channel **833**.

The cleaning water channel **833** can include the first cleaning water channel **833a** disposed closest to one end of the shielding cover body **8311** among the plurality of cleaning water channels **833**, the second cleaning water channel **833b** disposed closest to the other end of the

shielding cover body **8311** among the plurality of cleaning water channels **833**, and the third cleaning water channel **833c** disposed between the first cleaning water channel **833a** and the second cleaning water channel **833b**.

The distal end of the first cleaning water channel **833a** can be connected to one end of the cover through-hole **8313**, and the distal end of the second cleaning water channel **833b** can be connected to the other end of the cover through-hole **8313**.

The cover through-hole **8313** can be configured to be in connection with distal ends of the first cleaning water channel **833a**, the second cleaning water channel **833b**, and the third cleaning water channel **833c**.

Further, widths of the first cleaning water channel **833a**, the second cleaning water channel **833b**, and the third cleaning water channel **833c** can be equal to each other. When the water may not be dispersed to a specific area due to a structure of the cleaning water channel **833**, the widths of the first cleaning water channel **833a**, the second cleaning water channel **833b**, and the third cleaning water channel **833c** can be different from each other.

Further, one end of each of the first cleaning water channel **833a**, the second cleaning water channel **833b**, and the third cleaning water channel **833c** can be referred to as a first end. In some examples, the first ends of the first cleaning water channel **833a**, the second cleaning water channel **833b**, and the third cleaning water channel **833c** can be configured to be in contact with each other and to be disposed on a top face of the valve connector **838**. The first cleaning water channel **833a**, the second cleaning water channel **833b**, and the third cleaning water channel **833c** can extend in a separate manner from each other and along a flowing direction of the condensate. The other end of each of the first cleaning water channel **833a**, the second cleaning water channel **833b** and the third cleaning water channel **833c** can extend to the cover through-hole **8313**.

Further, the channel switching valve **870** shown in FIG. **14** can be configured to communicate with the first cleaning water channel **833a**, the second cleaning water channel **833b** and the third cleaning water channel **833c** and to selectively supply water to the first cleaning water channel **833a**, the second cleaning water channel **833b** and the third cleaning water channel **833c**.

Specifically, the valve communication hole **8382** can include the number of holes corresponding to the number of the plurality of cleaning water channels **833**. The number of the receiving channels **8791**, for example, **8791a**, **8791b**, and **8791c** can correspond to the number of the cleaning water channels **833**.

The receiving channel **8791** can include the first receiving channel **8791a** communicating with the first cleaning water channel **833a**, the second receiving channel **8791b** communicating with the second cleaning water channel **833b**, and the third receiving channel **8791c** communicating with the third cleaning water channel **833c**.

The condensate can be selectively supplied to the first receiving channel **8791a**, the second receiving channel **8791b** and the third receiving channel **8791c** through the water receiving portion **871** based on an operation of the channel switching valve **870** shown in FIG. **14**. Accordingly, the water can be selectively supplied to one of the first receiving channel **8791a**, the second receiving channel **8791b** and the third receiving channel **8791c**. The water can be then supplied to one of the plurality of cleaning water channels **833** and discharged to the cover through-hole **8313**.

Accordingly, a water pressure of water discharged from one of the plurality of cleaning water channels **833** can be

greater than that compared to a case in which the condensate from the channel switching valve **870** is supplied to all of the plurality of cleaning water channels **833**. As the pressure of water discharged from the cleaning water channel **833** increases, the foreign substances generated in the first heat exchanger **910** can be completely removed.

In some examples, the cleaning water channel **833** can include a channel defining portion **834** defining a channel through which water flowing into the valve communication hole **8382** can flow to the cover through-hole **8313**. The channel defining portion **834** can protrude from the top face of the shielding cover body **8311** and be formed integrally with the shielding cover body **8311**.

Accordingly, the cleaning water channel **833** may not be separately coupled to the shielding cover body **8311**, so that a manufacturing cost of the duct cover **830** can be reduced, and an assembly process thereof can be simplified.

The channel defining portion **834** can extend from the valve communication hole **8382** towards the cover through-hole **8313**.

That is, the channel defining portion **834** can constitute an inner circumferential face of the cleaning water channel **833**. Specifically, the channel defining portion **834** can be configured to constitute an inner circumferential face of the guide channel **8331** and an inner circumferential face of the discharge channel **8332**. Further, the channel defining portion **834** can be configured to constitute an inner circumferential face of each of the first discharge channel **8332a** and the second discharge channel **8332b**.

In some examples, the cleaning water channel **833** can include a discharge rib **835** configured to guide the water discharged from the cleaning water channel **833** to the first heat exchanger **910**.

The discharge rib **835** can extend frontwards from the distal end of the second discharge channel **8332b**. The discharge rib **835** can extend downward so that the distal end of the discharge rib **835** can be positioned in the cover through-hole **8313** and can further extend toward the first heat exchanger **910**. Thus, the water discharged from the cleaning water channel **833** can flow uniformly along the discharge rib **835** towards the first heat exchanger **910**.

FIG. **16** is a top view of the duct cover having the cleaning water channel in a laundry treating apparatus.

The flow speed of the condensate flowing into the guide channel **8331** through the valve communication hole **8382** can increase naturally as it passes through the first discharge channel **8332a** and the second discharge channel **8332b**. As the flow speed of the water increases as the water flow along the channel, a diameter thereof becomes narrower. Thus, the cleaning water channel **833** can be configured so that a width thereof increases in the direction in which the condensate flows, so that the condensate can be spread widely at a distal end thereof.

Specifically, the guide channel **8331** can be configured such that a width **t1** thereof increases as it extends from the valve communication hole **8382** toward the first discharge channel **8332a**.

Further, the first discharge channel **8332a** has a larger width than that of the guide channel **8331**, so that water flowing from the guide channel **8331** to the first discharge channel **8332a** can be uniformly discharged. A width **t2** of the first discharge channel **8332a** can be greater than the width **t1** of the guide channel **8331**.

Further, the second discharge channel **8332b** has a larger width than that of the first discharge channel **8332a**, so that water flowing from the first discharge channel **8332a** to the second discharge channel **8332b** can be uniformly dis-

charged. A width t_3 of the second discharge channel **8332b** can be greater than the width t_2 of the first discharge channel **8332a**.

Further, a width of each of the first discharge channel **8332a** and the second discharge channel **8332b** can increase as it extends along the flowing direction of the water.

Accordingly, the cleaning water channel **833** can evenly spray the water on the front face of the first heat exchanger **910**, and as a result, an entirety of water can be uniformly supplied to the first heat exchanger **910**.

In some examples, the pressure of water as discharged from the valve communication hole **8382** can be lowered as the water flows toward the cover through-hole **8313**. A thickness of the channel defining portion **834** can decrease as it extends along the direction of movement of the water. That is, a thickness t_5 of the channel defining portion **834** can decrease as a distance thereof from the valve communication hole **8382** increases. Alternatively, the thickness t_5 of the channel defining portion **834** can be uniform in order to facilitate molding of an entirety of the duct cover **830**.

In some examples, the channel defining portion **834** can include a first channel defining portion **834** constituting an inner circumferential face of the first cleaning water channel **833a**, a second channel defining portion **834** constituting an inner circumferential face of the second cleaning water channel **833b**, and a third channel defining portion **834** constituting an inner circumferential face of the third cleaning water channel **833c**.

A distal end of the first channel defining portion **834** and a distal end of the third channel defining portion **834** can be configured to be in contact with each other. A distal end of the first channel defining portion **834** and a distal end of the second channel defining portion **834** can be configured to be in contact with each other.

Further, the cleaning water channel **833** can include a partitioning rib **836** configured to partition the first cleaning water channel **833a**, the second cleaning water channel **833b**, and the third cleaning water channel **833c** from each other.

The partitioning rib **836** can extend from the distal end of the first channel defining portion **834** and the distal end of the third channel defining portion **834** toward the cover through-hole **8313**. That is, the partitioning rib **836** can extend from a point at which the distal end of the first channel defining portion **834** and the distal end of the third channel defining portion **834** contact each other toward the cover through-hole **8313**.

Accordingly, the water discharged from the second discharge channel **8332b** can be uniformly discharged to the cover through-hole **8313** along the partitioning rib **836**. The partitioning rib **836** can extend from the channel defining portion **834** toward the discharge rib **835** and be disposed on a top face of the discharge rib **835**.

Further, the cleaning water channel **833** can include a communication channel **8333** that communicates the discharge channel **8332** and the cover through-hole **8313** with each other.

The communication channel **8333** can be disposed on a top of the cover through-hole **8313** and can be configured to face toward the cover through-hole **8313**. The communication channel **8333** can be configured so that water to be discharged from the discharge channel **8332** flows to the cover through-hole **8313**.

Further, the channel defining portion **834** can be configured to constitute an inner circumferential face of the communication channel **8333**, so that water discharged from

the discharge channel **8332** can be prevented from flowing out of the cover through-hole **8313**.

In some examples, the cleaning water channel **833** can include a support **837** supporting the channel defining portion **834**.

The channel defining portion **834** can include the support **837** extending outwardly from an outer circumferential face thereof. The support **837** can be configured to protrude from a top face of the shielding cover body **8311**, and can be coupled to an outer circumferential face of the channel defining portion **834** to support the channel defining portion **834**. The support **837** can include a plurality of supports arranged along the outer circumferential face of the channel defining portion **834**.

Thus, the support **837** can support the channel defining portion **834** so that the channel defining portion **834** can withstand the water pressure of water therein. Thus, durability and reliability of the channel defining portion **834** can be improved.

FIG. 17 is a perspective view showing a bottom face of the duct cover in the laundry treating apparatus.

The duct cover **830** can include a first heat-blocking rib **8315a** and a second heat-blocking rib **8315b** that can prevent the heat from the first heat exchanger **910** from being transferred to the cleaning water channel **833**.

The first heat-blocking rib **8315a** can protrude from a bottom face of the shielding cover body **8311** and extend in an away direction from the cover through-hole **8313**. The second heat-blocking rib **8315b** can protrude from the bottom face of the shielding cover body **8311** and extend in parallel to the cover through-hole **8313**.

Each of the first heat-blocking rib **8315a** and the second heat-blocking rib **8315b** can include a plurality of heat-blocking ribs. The second heat-blocking rib **8315b** can extend in a perpendicular manner to the first heat-blocking rib **8315a** and be connected to the plurality of first heat-blocking ribs **8315a**.

The first heat-blocking rib **8315a** and the second heat-blocking rib **8315b** can be configured to face toward the first heat exchanger **910**. Thus, an amount of the heat transfer from the first heat exchanger **910** to the cleaning water channel **833** via the first heat-blocking rib **8315a** and the second heat-blocking rib **8315b** can be reduced.

Further, the shielding cover body **8311** can include an evaporator cover body **83111** facing toward the first heat exchanger **910** and a condenser cover body **83112** extending rearwards from the evaporator cover body **83111** and facing toward the second heat exchanger **920**. The first heat-blocking rib **8315a** and the second heat-blocking rib **8315b** can be disposed on a bottom face of the evaporator cover body **83111**, and the cover through-hole **8313** can extend through the bottom face of the evaporator cover body **83111**.

In some examples, the duct cover **830** can include a channel inserted groove **8349** that is recessed in a bottom face of the duct cover and constitutes the channel defining portion **834**. The channel inserted groove **8349** can be recessed in the bottom face of the shielding cover body **8311** and extend to the channel defining portion **834**.

The channel inserted groove **8349** can extend along an extending direction of the channel defining portion **834**. The channel inserted groove **8349** can be formed in a process of injection molding the channel defining portion **834**, and a load applied to the channel defining portion **834** can be distributed, thereby reinforcing structural rigidity of the channel defining portion **834**.

In some examples, the duct cover **830** can include a duct cover extension **832** extending in a thickness direction from

an outer face of each of the shielding cover body **8311** and the communication cover body **8312** and along a circumference of each of the shielding cover body **8311** and the communication cover body **8312**. The air flow duct **822** and the inflow duct **821** shown in FIG. 13 can be coupled to the duct cover extension **832**.

The duct cover extension **832** can protrude in the thickness direction from at least one of both side faces, a front face, and a rear face of each of the shielding cover body **8311** and the communication cover body **8312** to improve durability of each of the shielding cover body **8311** and the communication cover body **8312** and to provide a space in which a separate component can be seated on top of each of the shielding cover body **8311** and the communication cover body **8312**.

In some examples, the duct cover extension **832** can include an inserted portion **8322** extending in the thickness direction and inserted into an inner face of each of the inflow duct and the air flow duct **822**, and a step portion **8223** spaced outwardly from an outer circumferential face of the inserted portion **8322** and extending in the thickness direction **Z** and coupled to an outer face of each of the air flow duct **822** and the inflow duct **821**.

Between an inner circumferential face of the step portion **8223** and an outer circumferential face of the inserted portion **8322**, a sealing seat portion **8324** into which a top each of the air flow duct **822** and the inflow duct **821** is inserted can be disposed. The air flow duct **822** and the inflow duct **821** of the air circulating channel **820** can be inserted into the sealing seat portion **8324** and be coupled to a portion between the step portion **8223** and the inserted portion **8322**. Accordingly, the air flow duct **822** and the inflow duct **821** can be coupled to the shielding cover body **8311** and the communication cover body **8312**, respectively, so that open top faces thereof can be shielded.

FIG. 18 is an exploded perspective view of the channel switching valve in a laundry treating apparatus.

FIG. 18 illustrates a detailed structure of the channel switching valve **870** that selectively supplies the water to the plurality of cleaning water channels **833**. FIG. 18 is a view of the channel switching valve **870** viewed in a direction from a bottom to a top (**Z** direction).

The channel switching valve **870** can include the water receiving portion **871** communicating with the pump **861** and receiving the water from the pump **861**, and the connective portion **879** communicating with the water receiving portion **871** and connected to the valve connector **838** to deliver the water to the cleaning water channel **833**. The channel switching valve **870** can include the water delivering portion **872** disposed between the water receiving portion **871** and the connective portion **879** and coupled to the water receiving portion **871** and the connective portion **879**.

In some examples, the connective portion **879** can include a connective transfer channel **8792** that communicates with the water delivering portion **872** and receives water from the water delivering portion **872**. The connective transfer channel **8792** can act as a passage which can be in communication with the water storage tank **120** and along which the water supplied from the water delivering portion **872** can flow to the water storage tank **120**.

Accordingly, the water storage tank **120** can receive the water transferred to the channel switching valve **870** through the channel switching valve **870** from the pump **861** via the connective transfer channel **8792** and can temporarily store therein the water.

In this case, the connective transfer channel **8792** can be configured such that one end thereof faces toward the water delivering portion, and the other end thereof faces toward the water storage tank **120**.

Further, one end and the other end of the connective transfer channel **8792** can be spaced apart from each other so as to be prevented from facing toward each other. The connective transfer channel **8792** can be configured such that one end and the other end thereof can be prevented from facing toward each other in a straight line manner.

In some examples, the water receiving portion **871** can include a scroll receiving portion **8712** coupled to the water delivering portion **872**, and a water inlet portion **8711** extending from the scroll receiving portion **8712** toward the water collector **860** (see FIG. 14) and connected to the first water collector drain pipe **8911a**.

The water inlet portion **8711** can communicate with an inside of the scroll receiving portion **8712** and receive water from the first water collector drain pipe **8911a** and can move the water into the inside of the scroll receiving portion **8712**.

Further, the water receiving portion **871** can include a driver receiving portion **8713** extending from the scroll receiving portion **8712** in a direction away from the water delivering portion **872**, a valve driver **873** installed in the driver receiving portion **8713** to provide rotation power, and a valve rotatable portion **874** disposed within the scroll receiving portion **8712** and coupled to the valve driver **873** and configured to rotate. The water receiving portion **871** can include a driver fixing member **8716** that secures the valve driver **873** to the driver receiving portion **8713**.

Further, the water receiving portion **871** can include a scroll **875** accommodated in the scroll receiving portion **8712** and coupled to the valve rotatable portion **874** and configured to rotate.

The valve rotatable portion **874** can include a second valve rotation shaft **8742** coupled to the valve driver **873** so as to rotate, and a first valve rotation shaft **8741** coupled to the second valve rotation shaft **8742** and the scroll **875** so as to rotate.

In some examples, the water delivering portion **872** can include a delivering body **8721** to which the scroll receiving portion **8712** is coupled, and a contact portion **8726** extending from the delivering body **8721** toward the connective portion **879** and coupled to the connective portion **879**.

Further, the water delivering portion **872** can include a supply channel **8722** that passes through the delivering body **8721** and the contact portion **8726** and communicates with the connective transfer channel **8792** and the receiving channels **8791a**, **8791b**, and **8791c**.

The supply channel **8722** can include a plurality of supply channels arranged along a circumference of the contact portion **8726**. The plurality of supply channels can communicate with the plurality of receiving channels **8791a**, **8791b**, and **8791c** and the connective transfer channel **8792**, respectively.

The scroll **875** can include a scroll plate **8751** that is accommodated in the scroll receiving portion **8712** and rotates, a scroll communication hole **8752** that passes through the scroll plate **8751** and selectively communicates with the plurality of supply channels **8722**, and a scroll coupling groove **8753** passing through the scroll plate **8751** and coupled to the first valve rotation shaft **8741**.

The scroll plate **8751** can rotate while being in contact with one end of the supply channel **8722**. The scroll communication hole **8752** can be configured to selectively communicate with one of the supply channels **8722** according to the rotation of the scroll plate **8751**.

Accordingly, water flowing into the water inlet portion **8711** according to the rotation of the scroll plate **8751** can be selectively guided to the connective transfer channel **8792** and the receiving channels **8791a**, **8791b**, and **8791c**.

When water is supplied to the connective transfer channel **8792**, the water stored in the water collector **860** can flow to the water storage tank **120**. Further, when water is supplied to one of the receiving channels **8791a**, **8791b**, and **8791c**, water can be supplied to one of the cleaning water channels **833**.

Accordingly, according to the operation of the channel switching valve **870**, water can be selectively supplied to one of the water storage tank **120** or the cleaning water channel **833**. Further, when water is supplied to one of the plurality of cleaning water channels **833**, the pressure of water discharged to the first heat exchanger **910** can be greater than that in a case when water is continuously supplied to all of the plurality of cleaning water channels **833**.

In some examples, when the water supplied to the channel switching valve **870** flows into a location between the connective portion **879** and the valve connector **838**, various devices for the operation of the laundry treating apparatus can come into contact with the water.

In some implementations, the receiving channels **8791a**, **8791b**, and **8791c** can be formed integrally with the valve connector **838**. This prevents water from leaking to a location between the connective portion **879** and the valve connector **838**.

The receiving channels **8791a**, **8791b**, and **8791c** can pass through the bottom face of the valve connector **838** and communicate with the cleaning water channel **833**. The receiving channels **8791a**, **8791b**, and **8791c** can extend first downwards from the valve connector **838** and then extend in a direction away from the valve connector **838**.

Each of the receiving channels **8791a**, **8791b**, and **8791c** can be formed to be positioned at a lower level than that of the top face of the valve connector **838**. Each of the receiving channels **8791a**, **8791b**, and **8791c** can extend through the valve connector **838** so that one end thereof can be inserted into the cleaning water channel **833**.

In some examples, the connective portion **879** can include a connective extension **8793** extending from the outer circumferential face of the connective transfer channel **8792** and the outer circumferential face of each of the receiving channels **8791a**, **8791b**, and **8791c**.

The connective extension **8793** can couple the connective transfer channel **8792** to the receiving channels **8791a**, **8791b**, and **8791c**. The connective extension **8793** can be integrally formed with the connective transfer channel **8792** and the receiving channels **8791a**, **8791b**, and **8791c**, and can serve to fix the connective transfer channel **8792** and the receiving channels **8791a**, **8791b**, and **8791c**.

In some examples, the water delivering portion **872** can include a fastening portion **8725** that extends from an outer circumferential face of the contact portion **8726** and can be coupled to the connective extension **8793**. The connective portion **879** can include fixing protrusion **8794** extending from the connective extension **8793** to the fastening portion **8725** and coupled to the fastening portion **8725**.

The fixing protrusion **8794** and the fastening portion **8725** can be configured to face toward each other. One end of the fastening portion **8725** can be coupled to and accommodated in the fixing protrusion **8794**. As shown in the figure, the fixing protrusion **8794** can be disposed at each of one side and the other side of the connective extension **8793**. The fastening portion **8725** can be disposed at each of one side

and the other side of the contact portion **8726** and can face toward the fixing protrusion **8794**.

Further, the connective portion **879** can include a connective protrusion **8795** protruding from the outer circumferential face of the connective extension **8793** and spaced apart from the fixing protrusion **8794**. Further, the water delivering portion **872** can include a mount hook **8724** which extends from the outer circumferential face of the contact portion **8726** and into which the connective protrusion **8795** is inserted.

The mount hook **8724** can be disposed at a position corresponding to that of the connective protrusion **8795** and can be coupled to the connective protrusion **8795**. In an example, as shown in the figure, the connective protrusion **8795** can be formed to protrude from each of one side and the other side in a vertical direction (the Z-direction) of the connective extension **8793**. The mount hook **8724** can be disposed at each of one side and the other side in a vertical direction (the Z-direction) of the contact portion **8726**.

Accordingly, the water delivering portion **872** can be coupled to the connective extension **8793** through the connective protrusion **8795** and the fixing protrusion **8794**, and can prevent the water delivering portion **872** from being spaced from the connective extension **8793**.

Further, the water delivering portion **872** can include a protrusion **8727** that protrudes from a center of the contact portion **8726** toward the connective extension **8793** and is inserted into the connective extension **8793**. The protrusion **8727** can be inserted into the connective extension **8793** to prevent the water delivering portion **872** from being removed from the connective portion **879**.

In some examples, the water delivering portion **872** can include a fixing member **8723** for fixing the scroll receiving portion **8712** to the delivering body **8721**. The scroll receiving portion **8712** can include a fixing groove **8715** into which the fixing member **8723** is inserted. Further, the water receiving portion **871** can have a protruding hook **8717** that extends from an outer circumferential face of the scroll receiving portion **8712** and is coupled to the delivering body **8721**.

In some examples, the channel switching valve **870** can include a sealing member **8773** disposed between the connective portion **879** and the water delivering portion **872**. The sealing member **8773** can be disposed between the connective extension **8793** and the contact portion **8726** to prevent water from leaking to a location between each of the receiving channels **8791a**, **8791b**, and **8791c** and the supply channel **8722**.

The sealing member **8773** can be accommodated in one of the contact portion **8726** or the connective extension **8793**. The sealing member **8773** can be configured to surround the receiving channels **8791a**, **8791b**, and **8791c**.

A sealing portion **877** can include a shaft sealing member **8772** disposed between the second valve rotation shaft **8742** and the first valve rotation shaft **8741** to prevent water from leaking to the valve driver **873**, and a scroll sealing member **8771** that surrounds an outer circumferential face of the scroll plate **8751** and prevents water from leaking to a location between the scroll receiving portion **8712** and the delivering body **8721**.

An elastic member **876** for pressing the scroll **875** in a direction away from the first valve rotation shaft **8741** can be disposed between the scroll **875** and the first valve rotation shaft **8741**.

FIG. 19 is a perspective view showing the duct cover to which the nozzle cover is coupled in the laundry treating apparatus.

The air circulating channel **820** can further include the nozzle cover **840** that shields the cleaning water channel **833** and prevents water flowing through the cleaning water channel **833** from scattering to the outside.

The nozzle cover **840** can be coupled to the top of the cleaning water channel **833** and can be disposed above the shielding cover body **8311**. When the shielding cover body **8311** is viewed from above the nozzle cover **840**, the nozzle cover **840** can accommodate the cleaning water channel **833** and can be coupled to the top of the cleaning water channel **833** so that the cleaning water channel **833** can be screened with the nozzle cover **840**.

The nozzle cover **840** can extend along an extension direction of the cleaning water channel **833**. That is, the nozzle cover **840** can extend from one side thereof at which the channel switching valve **870** can extend to the other side at which the inflow communication hole **8314** is disposed. For example, a direction toward one side can be a direction in which the channel switching valve **870** extends from the valve connector **838**, while a direction toward the other side can be a direction toward the inflow communication hole **8314**, that is, a frontward direction (X direction).

Further, a length **L4** by which the nozzle cover **840** extends frontwards and rearwards can be smaller than or equal to a length **L2** by which the shielding cover body **8311** extends. The length **L4** by which the nozzle cover **840** extends forwards and backwards can be larger than or equal to a length by which the cleaning water channel **833** extends, which can be appropriately designed according to an amount of water to wash the first heat exchanger **910**.

The nozzle cover **840** can be coupled to a top of the channel defining portion **834** shown in FIG. **15** and can be configured to shield the cleaning water channel **833**. As shown, the nozzle cover **840** can be coupled to a top of each of the first cleaning water channel **833a**, the second cleaning water channel **833b**, and the third cleaning water channel **833c** and can be configured to shield the first cleaning water channel **833a**, the second cleaning water channel **833b** and the third cleaning water channel **833c**.

Accordingly, the nozzle cover **840** can prevent the water flowing through the cleaning water channel **833** from scattering to the outside.

FIG. **20** is a cross-sectional view showing an example of the nozzle cover in the laundry treating apparatus. FIG. **20** is a cross-sectional view in a longitudinal direction B-B' showing an inside of the duct cover **830** and the nozzle cover **840** shown in FIG. **19**.

The nozzle cover **840** can include a nozzle cover body **841** shielding the cleaning water channel **833**.

The nozzle cover body **841** can be coupled to a top **8341** of the channel defining portion **834** shown in FIG. **15** and extend along the extension direction of the cleaning water channel **833**. The nozzle cover body **841** can extend in parallel with the guide channel **8331**, and a distance between the nozzle cover body **841** and the cleaning water channel **833** can gradually increase along a direction in which the water flows.

That is, a distance between a bottom face of each of the first discharge channel **8332a** and the second discharge channel **8332b** and the nozzle cover body **841** can gradually increase along the direction in which the water flows.

Further, the nozzle cover **840** can further include a shielding rib **843** that moves water flowing along the cleaning water channel **833** to the cover through-hole **8313**.

The shielding rib **843** can extend from a distal end of the nozzle cover body **841** to the shielding cover body **8311**. The shielding rib **843** together with the nozzle cover body **841**

can serve to shield the cover through-hole **8313**, and can be disposed at the distal end of the cover through-hole **8313**.

That is, one end of the cover through-hole **8313** can be connected to the second discharge channel **8332b** and the other end thereof can be connected to the shielding rib **843**. Alternatively, the shielding rib **843** can be spaced apart from the cover through-hole **8313** and positioned in front of the cover through-hole **8313**.

The shielding rib **843** can serve to temporarily store the water discharged from the cleaning water channel **833** inside the cleaning water channel **833**. Water flowing along the cleaning water channel **833** can collide with the shielding rib **843** such that the water can flow to the cover through-hole **8313**.

In some examples, the condensate discharged from the second discharge channel **8332** can be discharged through the cover through-hole **8313** and along the discharge rib **835**. In some examples, the condensate may not be discharged to the first heat exchanger **910**, but can be discharged to a location in front of the first heat exchanger **910** along the extension direction of the discharge rib **835**. In particular, as a speed of the condensate passing through the discharge rib **835** increases, the number of times the condensate comes into contact with an inlet face of the first heat exchanger **910** can be reduced.

In some implementations, the nozzle cover **840** can further include a switching rib **846** for guiding the water passing through the discharge rib **835** toward the first heat exchanger **910**.

The switching rib **846** can be configured to extend from the shielding rib **843** toward the cover through-hole **8313** and to face toward the discharge rib **835**. The switching rib **846** can extend toward the first heat exchanger **910** so that a distal end of the switching rib **846** can be configured to protrude downwardly beyond the cover through-hole **8313**. The switching rib **846** can extend in an inclined manner relative to the discharge rib **835**, and a distal end of the switching rib **846** and a distal end of the discharge rib **835** can be configured to be spaced apart from each other.

The distal end of the switching rib **846** can be disposed in front of a front face of the first heat exchanger **910**, and the distal end of the discharge rib **835** can be disposed in rear of the front face of the first heat exchanger **910**. Accordingly, the water passing through the discharge rib **835** can collide with the switching rib **846** and thus be discharged to a location between the distal end of the switching rib **846** and the distal end of the discharge rib **835**.

In some examples, an inclination angle $\theta 1$ of the first discharge channel **8332a**, that is, the inclination angle $\theta 1$ of the first inclined face **8316a** can be greater than or equal to an inclination angle $\theta 2$ of the second discharge channel **8332b**, that is, the inclination angle $\theta 2$ of the second inclined face **8316b**.

Accordingly, the water flowing into the cleaning water channel **833** can flow to the cover through-hole **8313** due to gravity while passing through the first discharge channel **8332a** and the second discharge channel **8332b**. Thus, the water can be completely discharged. Further, a thickness of each of the first inclined face **8316a** and the second inclined face **8316b** can be uniform.

FIG. **21** is a cross-sectional view showing another example of the nozzle cover in the laundry treating apparatus. Hereinafter, the description will be based on a different configuration from that of the nozzle cover **840** in FIG. **20**.

The nozzle cover **840** can further include an inserted portion **849** that reduces a distance between the cleaning water channel **833** and the nozzle cover body **841**.

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The inserted portion **849** can be configured to protrude from the nozzle cover body **841** toward the inside of the cleaning water channel **833**. The inserted portion **849** can be configured to protrude from a top face of the nozzle cover body **841** toward the first discharge channel **8332a** and the second discharge channel **8332b**.

As the inserted portion **849** protrudes from the nozzle cover body **841** toward the cleaning water channel **833**, a thickness of the nozzle cover body **841** can increase. The inserted portion **849** can be configured such that a length by which the inserted portion **849** protrudes from the nozzle cover body **841** gradually increases along a flowing direction of the condensate.

The inserted portion **849** can be configured such that one face thereof facing toward the cleaning water channel **833** has an inclination angle corresponding to the inclined face **8316**.

In some examples, an inclination angle $\theta 4$ of one face of the inserted portion **849** facing toward the first inclined face **8316** can correspond to the inclination angle $\theta 1$ of the first inclined face. An inclination angle $\theta 3$ of one face of the inserted portion **849** facing toward the second inclined face **8316b** can correspond to the inclination angle $\theta 2$ of the second inclined face.

A distance between one face of the inserted portion **849** facing toward the first discharge channel **8332a** and the first inclined face **8316a** can correspond to a vertical dimension between a bottom face and a top face of the guide channel **8331**.

Further, a distance between one face of the inserted portion **849** facing toward the second inclined face **8316b** and the second inclined face **8316b** can correspond to a vertical dimension between the bottom face and the top face of the guide channel **8331**.

The inserted portion **849** can serve to reduce an internal space of the cleaning water channel **833**. As a result, a vertical dimension of the cleaning water channel **833** can be reduced so that the speed of the water reaching the shielding rib **843** can increase and thus the water can flow quickly to the cover through-hole **8313**.

Further, as the inserted portion **849** is formed, a vertical dimension of the cleaning water channel **833** can be uniform. Accordingly, when the water flows in the cleaning water channel **833**, a volume occupied by air inside the cleaning water channel **833** can be reduced. Further, the noise and vibration generated when the water inside the cleaning water channel **833** collides with an inner circumferential face of the cleaning water channel **833** can be reduced.

Further, even when the water first reaches a specific area of the inserted portion **849**, the water can be uniformly discharged along an entire area of the inserted portion **849** and through the cover through-hole **8313**.

FIG. 22 is a cross-sectional view showing another example of the nozzle cover in the laundry treating apparatus. FIG. 22 is a cross-sectional view of an inside of the duct cover **830** and the nozzle cover **840** (B-B').

The nozzle cover body **841** can include a welded plate **8411** coupled to the channel defining portion **834** and shielding the guide channel **8331**, a first inclined plate **8412** extending from the welded plate **8411** and coupled to the channel defining portion **834** and shielding the first discharge channel **8332a**, and a second inclined plate **8413** extending from the first inclined plate **8412** and coupled to the channel defining portion **834** and shielding the second discharge channel.

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The shielding rib **843** can extend downward from the distal end of the second inclined plate **8413** and can be coupled to the top face of the duct cover body **831**. The switching rib **846** can extend from the second inclined plate **8413** or the shielding rib **843** toward the cover through-hole **8313**.

The first inclined plate **8412** can extend from the welded plate **8411** in an inclined manner along the flowing direction of water, and the second inclined plate **8413** can extend from the first inclined plate **8412** in an inclined manner along the flowing direction of water.

An inclination angle $\theta 3$ of the first inclined plate with respect to the welded plate can correspond to the inclination angle $\theta 1$ of the first inclined face. An inclination angle $\theta 4$ of the second inclined plate with respect to the welded plate **8411** can correspond to the inclination angle $\theta 2$ of the second inclined face. Accordingly, an internal vertical dimension of the cleaning water channel **833** can be constant.

A thickness of each of the welded plate **8411**, the first inclined plate **8412** and the second inclined plate **8413** can be uniform, which has the effect of lowering a manufacturing cost of the nozzle cover **840**.

FIGS. 23A and 23B are a side view and a bottom view of the nozzle cover shown in FIG. 22. FIG. 23A is a side view of the nozzle cover **840**, and FIG. 23B is a bottom view of the nozzle cover **840**.

The switching rib **846** can extend from the second inclined plate **8413** or the shielding rib **843** toward the cover through-hole **8313**. An angle $\theta 5$ between the switching rib **846** and the shielding rib **843** can be in a range of 10 degrees to 80 degrees. The angle $\theta 5$ between the switching rib **846** and the shielding rib **843** can be designed in various manners depending on an arrangement of the shielding rib **843** and the first heat exchanger **910** or an arrangement of the shielding rib **843** and the cover through-hole **8313**.

A vertical dimension H7 of the shielding rib **843** can be smaller than a vertical dimension of the second inclined face **8316b**. A vertical dimension of the second inclined plate **8413** can be smaller than a vertical dimension of the first inclined face **8316a** of H6 and be larger than a vertical dimension of the second inclined face **8316b**.

Accordingly, the channel defining portion **834** can protrude by a certain vertical dimension and can be coupled to the nozzle cover **840**. The nozzle cover **840** can face toward the first inclined face **8316a**, the second inclined face **8316b**, and the cleaning water channel **833** and can have a certain vertical dimension.

In some examples, the nozzle cover **840** can include a partitioning rib **848**. The rib **848** together with the partitioning rib **836** can partitioning the water discharged from the plurality of cleaning water channels **833**.

The partitioning rib **848** can extend from the switching rib **846** towards the partitioning rib **836**. The partitioning rib **848** can overlap the partitioning rib **836**. For example, the partitioning rib **848** can be coupled to the partitioning rib **836**.

The partitioning rib **848** together with the partitioning rib **836** can partition the water discharged from the plurality of cleaning water channels **833**. Accordingly, the partitioning rib **848** prevents the water discharged from one cleaning water channel **833** from flowing to another cleaning water channel **833**, so that water is uniformly sprayed to the first heat exchanger **910**.

As shown in FIG. 23B, the nozzle cover **840** can include a coupling portion **844** coupled to the channel defining portion **834**.

The coupling portion **844** can extend from the nozzle cover body **841** toward the channel defining portion **834**, and can be configured to be coupled to the top of the channel defining portion **834**.

The coupling portion **844** can be welded onto the top of the channel defining portion **834** so as to be integrally formed with the channel defining portion **834**. The welded plate **8411** can be configured to be in contact with the top of the channel defining portion **834**. The coupling portion **844** can face toward the channel defining portion **834** and extend along an extension direction of the channel defining portion **834**.

Further, the nozzle cover **840** can include an extension rib **842** that prevents the nozzle cover body **841** from being removed from the cleaning water channel **833**.

The extension rib **842** can be configured to extend from an outer circumferential face of the nozzle cover body **841** in the thickness direction and to accommodate therein the channel defining portion **834**. The extension rib **842** can be configured to have a larger width than a width of the channel defining portion **834** and to accommodate therein the outer circumferential face of the channel defining portion **834**.

Alternatively, when the support **837** can be disposed on the outer circumferential face of the channel defining portion **834**, the extension rib **842** can be configured to accommodate therein a top of the support **837**.

In some examples, a distance **t1** between both opposing inner faces of the coupling portion **844** extending from the welded plate **8411** can correspond to a width **t1** of the guide channel **8331**.

A distance **t2** between both opposing inner faces of the coupling portion **844** extending from the first inclined plate **8412** can correspond to a width **t2** of the first discharge channel **8332a**. A distance **t3** between both opposing inner faces of the coupling portion **844** extending from the second inclined plate **8413** can correspond to a width **t3** of the second discharge channel **8332b**.

Accordingly, the welded plate **8411** can shield the cleaning water channel **833**, such that the water inside the cleaning water channel **833** can be prevented from leaking to the outside.

FIG. **24** is a cross-sectional view showing an example in which the nozzle cover and the channel defining portion are coupled to each other in the laundry treating apparatus.

The support **837** can include a curved portion **8371** for easy coupling of the extension rib **842** thereto.

The support **837** can include the curved portion **8371** configured to be spaced apart from the at least a portion of the extension rib **842**. The curved portion **8371** can be formed at a distal end of the support coupled to the extension rib **842**.

The extension rib **842** can extend from the outer circumferential face of the nozzle cover body **841** in the thickness direction and can be coupled to the support **837** at the curved portion **8371** thereof. Thus, this can prevent burr from occurring in a process where a lower end **8422** of the extension rib **842** is coupled to the support **837**.

Further, a vertical dimension **H7** by which the channel defining portion **834** protrudes from the top face of the duct cover body **831** can be larger than or equal to a vertical dimension **H8** by which the support **837** protrudes. Accordingly, the nozzle cover body **841** can be configured to be spaced apart from the support **837**.

In some examples, a thickness **t5** of the channel defining portion **834** can be smaller than or equal to a width **t1** of the cleaning water channel **833**. A vertical dimension of the

cleaning water channel **833** can correspond to a vertical dimension **H7** of the channel defining portion **834**.

Further, the nozzle cover body **841** can be coupled to a top **8341** of the channel defining portion **834**, and the channel defining portion **834** can be integrally coupled to the nozzle cover body **841** through a thermal welding process. For example, the channel defining portion **834** can be coupled to the nozzle cover body **841** by welding.

In some examples, the thermal welding process can refer to a process of bonding surfaces of two thermoplastic members to each other by applying heat and pressure thereto. In other words, heat can be applied to the coupling portion **844** and then the coupling portion **844** can be brought into contact with the channel defining portion **834** so that the coupling portion **844** is integrally formed with the channel defining portion **834**.

Alternatively, the channel defining portion **834** can be coupled to the nozzle cover body **841** through a vibrating welding process.

In some examples, the vibration welding process can refer to a process in which two thermoplastic members are melted with frictional heat generated at a contact area therebetween via vertical or left and right vibrations while pressing the two thermoplastic members against each other, and then the melted solidified members are joined to each other and are solidified.

In other words, the vibration welding process vibrates the nozzle cover body **841** or the channel defining portion **834** to generate the frictional heat between the coupling portion **844** and the channel defining portion **834** and the couples the coupling portion **844** and the channel defining portion **834** to each other using the frictional heat.

As a result, the nozzle cover body **841** can shield the cleaning water channel **833** more efficiently than in an approach in which the channel defining portion **834** and the body **841** are coupled to each other in a hook or bolt-nut coupling manner. Thus, a lifespan of a final product can extend as a modification and repair period can extend.

Further, the nozzle cover body **841** is integrally coupled to the channel defining portion **834**, thereby reducing a material cost and simplifying an assembly process thereof.

Further, even when a separate cleaning water pipe is not configured, the cleaning water channel **833** can be formed via the combination of the nozzle cover **840** and the duct cover **830**, so that a manufacturing process thereof can be easy.

FIG. **25** is a cross-sectional view showing another example in which the nozzle cover and the channel defining portion are coupled to each other in the laundry treating apparatus.

In FIG. **24**, the nozzle cover **840** is shown to be coupled to the channel defining portion **834**. FIG. **25** is a view showing a state in which the nozzle cover **840** is spaced apart from the channel defining portion **834** by a predetermined distance before being coupled to the channel defining portion **834**.

The channel defining portion **834** can further include a first coupling rib **8342** constituting an inner side face of the cleaning water channel **833** and a second coupling rib **8343** constituting an outer side face of the cleaning water channel **833**.

The first coupling rib **8342** can protrude from the channel defining portion **834** and be coupled to the nozzle cover body **841**. The second coupling rib **8343** can be coupled to the coupling portion **844** while protruding from the channel defining portion **834** so as to be spaced apart from the first coupling rib **8342**. A vertical dimension **H11** by which the

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second coupling rib **8343** can protrude from the channel defining portion **834** can correspond to the first coupling rib **8342**.

In some examples, a lower end of the coupling portion **844** can be configured to be in contact with the top of the second coupling rib **8343**, and the welded plate **8411** and the first coupling rib **8342** can be configured to be in contact with each other.

The second coupling rib **8343** can be coupled to the coupling portion **844** via a thermal welding process, or via a vibration welding process. In this process, the coupling portion **844** can be melted and coupled to the second coupling rib **8343**. In some examples, the channel defining portion **834** can further include a sealing groove **8344** disposed between the first coupling rib **8342** and the second coupling rib **8343** and extending in the extension direction of the cleaning water channel **833**, and a sealing member **8345** seated in the sealing groove **8344** for shielding a space between the nozzle cover body **841** and the cleaning water channel **833**.

A vertical dimension **H11** by which the second coupling rib **8343** and the first coupling rib **8342** protrude can correspond to a diameter of the sealing member **8345**.

The sealing member **8345** can be configured to be in contact with the nozzle cover body **841** and to shield a space between the nozzle cover body **841** and the sealing groove **8344**, and to prevent the water inside the cleaning water channel **833** from leaking out through the nozzle cover body **841**. That is, the sealing member **8345** can prevent water leakage from the inside of the cleaning water channel **833** to the outside.

Further, a plurality of sealing grooves **8344** and a plurality of sealing members **8345** can be defined in the channel defining portion **834** and can overlap each other along the width direction.

When the plurality of sealing grooves **8344** and the plurality of sealing members are provided, a shielding force of the nozzle cover body **841** can further increase compared to a case when a single sealing groove **8344** and a single sealing member **8345** are provided.

In some examples, a distance **t6** between both opposing inner faces of the channel inserted groove **8349** can be smaller than a thickness **t5** of the channel defining portion **834**, so that the channel inserted groove **8349** can be accommodated in the channel defining portion **834**.

FIGS. **26A** and **26B** are perspective views showing a state in which the connective portion and the water delivering portion are coupled to each other in the laundry treating apparatus. Hereinafter, descriptions of those with the above-described structures will be omitted.

FIG. **26A** is a perspective view showing a state in which the water receiving portion **871** is omitted from the channel switching valve **870** and the connective portion **879** and the water delivering portion **872** are present. FIG. **26B** is a perspective view of a state in which the connective portion **879** and the water delivering portion **872** in FIG. **26A** are coupled to each other when viewed in a different direction.

The connective portion **879** can be coupled to the valve connector **838** and extend toward the water delivering portion **872**. The water delivering portion **872** can be connected to the connective portion **879** and guide the condensate supplied from the water receiving portion **871** to the connective portion **879**. Further, the water receiving portion **871** can be connected to the water delivering portion **872** to supply the condensate to the water delivering portion **872**.

The water receiving portion **871** can be located on top of the water collector **860** and can be connected to the water

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delivering portion **872**. Accordingly, the water delivering portion **872** can receive a load due to a weight of the water receiving portion **871**.

Further, the water delivering portion **872** can be located on top of the water collector **860** and can be connected to the connective portion **879**. Accordingly, the connective portion **879** can receive a load due to the weight of the water receiving portion **871** and the weight of the water delivering portion **872**.

Accordingly, the connective portion **879** can include the fixing protrusion **8794** to which the fastening portions **8725a** and **8725b** of the water delivering portion **872** are coupled, in order to support the water delivering portion **872** and the water receiving portion **871**. The fixing protrusion **8794** can be configured to protrude from an outer circumferential face of the connective extension **8793**.

The connective extension **8793** can have a distal end facing toward the water delivering portion **872** and protruding beyond the valve connector **838** in order to prevent the fixing protrusion **8794** from coming into contact with the valve connector **838**. A top of the fixing protrusion **8794** can be positioned above the valve connector **838** and the first coupling rib **8342**.

Further, the fixing protrusion **8794** can include first fixing protrusion **8794a** extending from the connective extension **8793** to one side and second fixing protrusion **8794b** extending from the connective extension **8793** to the other side.

For example, one side to which the first fixing protrusion **8794a** can extend can refer to a side above the connective extension **8793**, and the other side to which the second fixing protrusion **8794b** can extend can refer to a side below the connective extension **8793**.

In some examples, the water delivering portion **872** can include a receiving portion **8728** extending toward the connective portion **879** and an outer circumferential face of the plurality of supply channels **8722**.

The receiving portion **8728** can be integrally formed with the plurality of supply channels **8722**, and can be disposed closer to the valve connector **838** than the distal end of the supply channel **8722** can be.

Further, the receiving portion **8728** can be connected to the connective extension **8793** and accommodate the plurality of receiving channels **8791** therein. The sealing member **8773** can be seated on an inner circumferential face of the receiving portion **8728** and can shield a space between the supply channel **8722** and the receiving channel **8791** and the connective transfer channel **8792**.

Further, the water delivering portion **872** can include a fastening portion **8725** that extends from the outer circumferential face of the receiving portion **8728** and is coupled to the connective extension **8793**.

In some examples, the fastening portion **8725** can include a first fastening portion **8725a** coupled to the first fixing protrusion **8794a** and a second fastening portion **8725b** coupled to the second fixing protrusion **8794b**.

The first fastening portion **8725a** can extend to one side from the outer circumferential face of the receiving portion **8728** and be positioned in a corresponding manner to that of the first fixing protrusion **8794a**. The second fastening portion **8725b** can extend from the outer circumferential face of the receiving portion **8728** to the other side and be disposed at a position corresponding to that of the second fixing protrusion **8794b**.

In some examples, one side to which the first fastening portion **8725a** extends from the outer circumferential face of the receiving portion **8728** can refer to a side above the receiving portion **8728**. The other side to which the second

fastening portion **8725b** extends from the outer circumferential face of the receiving portion **8728** can refer to a side below the receiving portion **8728**.

In some examples, the fastening portion **8725** can include a fastening rib **87251** that accommodates the fixing protrusion **8794** therein. The fastening rib **87251** has a diameter larger than a diameter of the fixing protrusion **8794** and can be configured to accommodate the fixing protrusion **8794** therein.

The fastening rib **87251** can include a first fastening rib **87251a** protruding from the first fastening portion **8725a** and accommodating the first fixing protrusion **8794a** therein, and a second fastening rib **87251b** protruding from the second fastening portion **8725b** and accommodating the second fixing protrusion **8794b** therein.

Accordingly, the fastening portions **8725a** and **8725b** can be prevented from moving in the vertical direction from the fixing protrusion **8794**, so that the coupling force between the fastening portions **8725a** and **8725b** and the fixing protrusion **8794** can increase.

In some examples, the water delivering portion **872** can include a protrusion **8729** extending from an outer circumferential face of the receiving portion **8728**. The connective portion **879** can include the mount hook **8797** that is coupled to the protrusion **8729**.

The mount hook **8797** can be configured to protrude from an outer circumferential face of the connective extension **8793** and extend toward the protrusion **8729**.

In some examples, the supply channel **8722** can communicate with one of the receiving channel **8791** or the connective transfer channel **8792**. A plurality of supply channels **8722** can be provided and can be arranged along a circumference of the receiving portion **8728**.

In some examples, the plurality of supply channels **8722** can include the first supply channel **8722a** in communication with the first receiving channel **8791a**, the second supply channel **8722b** in communication with the second receiving channel **8791b**, the third supply channel **8722c** that communicates with the third receiving channel **8791c**, and the fourth supply channel **8722d** that communicates with the connective transfer channel **8792**.

Further, the water delivering portion **872** can include a protrusion **8727** that can be disposed between the first supply channel **8722a**, the second supply channel **8722b**, the third supply channel **8722c**, and the fourth supply channel **8722d**.

The connective portion **879** can include the inserted portion **8799** which is disposed at a position corresponding to that of the protrusion **8727** and into which the protrusion **8727** is inserted.

The inserted portion **8799** can be disposed between the first receiving channel **8791a**, the second receiving channel **8791b**, the third receiving channel **8791c**, and the connective transfer channel **8792** and face toward the protrusion **8727**.

The protrusion **8727** has a diameter corresponding to that of the inserted portion **8799** and can be inserted into the inserted portion **8799**. Thus, the protrusion **8727** can prevent the water delivering portion **872** from being spaced apart from the connective portion **879**.

In some examples, the channel switching valve **870** can include the sealing member **8773** that prevents leakage of water supplied from the water delivering portion **872** to the connective portion **879**. The connective portion **879** can include the receiving portion **8796** in which the sealing member **8773** is seated.

One end of each of the connective transfer channel **8792** and the receiving channel **8791** can protrude toward the

water delivering portion **872** beyond the connective extension **8793**. The receiving portion **8796** can be disposed on an outer circumferential face of each of the connective transfer channel **8792** and the receiving channel **8791** and face toward the connective extension **8793**. The sealing member **8773** can be disposed between the connective portion **879** and the water delivering portion **872** and be seated in the receiving portion **8796**.

In some examples, the sealing member **8773** can be configured to accommodate an outer circumferential face of each of the connective transfer channel **8792** and the receiving channel **8791**. The sealing member **8773** can serve to seal a space between the connective transfer channel **8792** and the receiving channel **8791**, and the supply channel **8722**.

The first fastening rib **87251a** can extend from an outer circumferential face of the first fastening portion **8725a** toward the first fixing protrusion **8794a** and can accommodate the first fixing protrusion **8794a** therein. The second fastening rib **87251b** can extend from an outer circumferential face of the second fastening portion **8725b** toward the second fixing protrusion **8794b** and accommodate the second fixing protrusion **8794b** therein.

FIG. 27 is an internal cross-sectional view of the connective portion and the water delivering portion in the laundry treating apparatus. Hereinafter, a description of the configuration duplicate with the above configuration is omitted.

A length by which the second fixing protrusion **8794b** extends vertically from the connective extension **8793** can be larger than a length by which the first fixing protrusion **8794a** extends vertically from the connective extension **8793**.

Because the load delivered from the water delivering portion **872** to the connective portion **879** is delivered to the second fixing protrusion **8794b** in a larger amount than an amount in which the load is delivered to the first fixing protrusion **8794a**, a length H11 by which the second fixing protrusion **8794b** extends vertically from the connective extension **8793** can be larger than a length H10 by which the first fixing protrusion **8794a** extends vertically from the connective extension **8793**.

Accordingly, even when the water delivering portion **872** is coupled to the connective portion **879**, the connective portion **879** can stably support the weights of the water delivering portion **872** and the water receiving portion **871**.

Further, a diameter D1 of the supply channel **8722** can be equal to a diameter D2 of each of the connective transfer channel **8792** and the receiving channel **8791**.

Thus, the water discharged from the supply channel **8722** can stably flow to the connective transfer channel **8792** and the receiving channel **8791**. Further, the sealing member **8773** can shield a small gap between the supply channel **8722** and the receiving channel **8791** and the connective transfer channel **8792**. The connective transfer channel **8792** can include a water receiving hole **87921** defined at one end of the connective transfer channel **8792** and a water discharging hole **87922** defined at another end of the connective transfer channel **8792** and configured to discharge the water received through the water receiving hole **87921** toward the water storage tank **120**.

In some examples, the protrusion **8727** can pass through the sealing member **8773** and be inserted into the connective extension **8793**. Thus, this can prevent the sealing member **8773** from being removed from between the water delivering portion **872** and the connective portion **879**.

The receiving channel **8791** can extend in an inclined manner and toward the cleaning water channel **833**, and the receiving channel **8791** can have an inclination angle $\theta 6$ in a range of 10 to 90 degrees.

For example, when the inclination angle $\theta 6$ of the receiving channel **8791** is smaller than 10 degrees, the pressure of the water flowing into the connective portion **879** becomes too low. This can be disadvantageous in terms of the energy efficiency of the pump **861**.

When the inclination angle $\theta 6$ of the receiving channel **8791** exceeds 90 degrees, a length **L6** by which the connective portion **879** extends from the valve connector **838** is too small. Thus, the connective portion **879** may not be able to support the loads of the water delivering portion **872** and the water receiving portion **871**.

However, a distance **H8** between the valve communication hole **8382** and a supply hole **87911** can be appropriately designed based on an extension length of the channel switching valve **870**.

The sealing member **8773** can be accommodated in one end of the receiving channel **8791**, and can be constructed to surround one end of each of the receiving channels **8791**. The supply hole **87911** can be surrounded with the sealing member **8773**, and the supply hole **87911** can protrude toward the water delivering portion **872** relative to the sealing member **8773**.

The connective extension **8793** can be configured to have a larger diameter than a diameter of all of the plurality of receiving channels **8791** and extend from an outer circumferential face of the receiving channel **8791**.

The sealing member **8773** can be seated on the distal end of the connective extension **8793** and surround the plurality of receiving channels **8791**.

FIG. 28A is a perspective view showing a state in which the connective portion, the water delivering portion and the nozzle cover are coupled to each other in the laundry treating apparatus. FIG. 28B is a perspective view viewed in a different direction of a state in which the connective portion, the water delivering portion and the nozzle cover in FIG. 28A are coupled to each other in the laundry treating apparatus.

Hereinafter, a description will be focused on a structure different from that of the water delivering portion **872** and the connective portion **879** shown in FIGS. 26A and 26B.

When a distance by which the connective portion **879** protrudes from the valve connector **838** is too large, a rotational moment applied to a contact point between the connective portion **879** and the valve connector **838** can increase. That is, the structural rigidity of the connective portion **879** can be lowered.

For this reason, the distal end of the valve connector **838** can be configured to protrude beyond the distal end of the connective portion **879**. In other words, the distal end of the connective portion **879** can be positioned below the valve connector **838**.

Thus, the protruding length of the connective portion **879** from the valve connector **838** can be reduced, such that an amount of moment loaded onto the connective portion **879** can be reduced.

Further, as a distal end of the valve connector **838** can be configured to protrude beyond a distal end of the connective portion **879**, the fixing protrusion **8794** can extend from the connective extension **8793** to one side so that the first fixing protrusion **8794a** to which the first fastening portion **8725** is coupled can be omitted.

In this case, the nozzle cover **840** can include a fastening part **8419** coupled to the first fastening portion **8725** or the

second fastening portion **8725b** so as to reinforce structural rigidity of the connective portion **879**.

The fastening part **8419** can be disposed at a position corresponding to that of one of the first fastening portion **8725** or the second fastening portion **8725b** and can protrude from a top face of the nozzle cover **840** in a vertical direction (Z direction). The fastening part **8419** can protrude upwards from the nozzle cover body **841** and extend toward the first fastening portion **8725a**.

Coupling the first fastening portion **8725** to the fastening part **8419** can allow a load applied to the connective portion **879** supporting a weight of the first water collector drain pipe **8911a** and the water delivering portion **872** and the water receiving portion **871** to be reduced. In other words, because the nozzle cover **840** can be coupled to the channel defining portion **834** in a relatively large area, the load applied to the connective portion **879** can be transmitted to the nozzle cover **840** and thus can be distributed. For example, the fastening part **8419** can include a protrusion that protrudes from a top surface of the nozzle cover **840** and a hole defined inside the protrusion.

Further, a length by which the connective portion **879** extends from the valve connector **838** can be reduced. The water delivering portion **872** can be closer to the duct cover **830**, so that an overall extension length of the channel switching valve **870** can be reduced.

Thus, a possibility that the channel switching valve **870** interferes with the drum **200** can be significantly reduced. Furthermore, a length of each of the receiving channel **8791** and the connective transfer channel **8792** can be reduced, so that an amount of residual water inside each of the receiving channel **8791** and the connective transfer channel **8792** can be reduced.

In some examples, a length of the first fastening portion **8725a** can be equal to a length of the second fastening portion **8725b**. In some examples, unlike a configuration shown in FIGS. 26A and 26B, the first fixing protrusion **8794a** can be omitted from the connective portion **879**, such that the length of the first fastening portion **8725a** and that of the second fastening portion **8725b** may not be different from each other.

In some examples, the first fastening portion **8725a** extending upwardly from an outer circumferential face of the receiving portion **8728** can have a length corresponding to that of the second fastening portion **8725b** extending downwardly from an outer circumferential face of the receiving portion **8728**.

Accordingly, manufacturing and repair of the first fastening portion **8725a** and the second fastening portion **8725b** can be facilitated. Further, when assembling the water delivering portion **872**, positions of the first fastening portion **8725a** and the second fastening portion **8725b** can be exchanged with each other such that the second fastening portion **8725b** can be coupled to the fastening portion **8719**. Accordingly, the water delivering portion **872** can be easily assembled to the connective portion **879** and the nozzle cover **840**.

In some examples, the sealing member **8773** can include a first sealing member **8773a** which accommodates an outer circumferential face of the first receiving channel **8791a**, a second sealing member **8773b** which accommodates an outer circumferential face of the second receiving channel **8791b**, a third sealing member **8773c** which accommodates an outer circumferential face of the third receiving channel **8791c**, and a fourth sealing member **8773d** which accommodates the protrusion **8727**.

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The first sealing member **8773a**, the second sealing member **8773b**, and the third sealing member **8773c** can have diameters and thicknesses corresponding to each other, and can be configured to be in contact with each other.

The fourth sealing member **8773d** can be formed in a shape corresponding to that of the protrusion **8727**. The first sealing member **8773a**, the second sealing member **8773b**, and the third sealing member **8773c** can be arranged along a perimeter.

Various implementations of the present disclosure have been described above in detail. However, those of ordinary skill in the art to which the present disclosure belongs can make various modifications to the above-described implementations without deviating from the scope of the present disclosure. Therefore, the scope of the present disclosure should not be limited to the described implementations, and should be defined by the claims as described below as well as the equivalents thereto.

What is claimed is:

1. A laundry treating apparatus comprising:
 - a cabinet having an opening defined at a front surface thereof;
 - a drum rotatably disposed in the cabinet, the drum having a laundry inlet defined at a front surface thereof and configured to introduce laundry into the drum;
 - a base that is disposed below the drum and defines a space configured to receive air discharged from the drum and to guide the air to the drum; and
 - a motor configured to rotate the drum,

wherein the base comprises:

- an air circulating channel configured to fluidly communicate with the drum, the air circulating channel being configured to receive the air discharged from the drum and to guide the air to the drum,
- a heat exchanger comprising (i) a first heat exchanger disposed inside the air circulating channel and configured to cool the air discharged from the drum and (ii) a second heat exchanger spaced apart from the first heat exchanger and configured to heat air that has been cooled by the first heat exchanger,
- a water collector body that is disposed outside the air circulating channel and in fluid communication with the air circulating channel, the water collector body being configured to collect water condensed from the air in the first heat exchanger,
- a cleaning water channel disposed above the air circulating channel and configured to receive the water from the water collector body and to discharge water received from the water collector body to the first heat exchanger,
- a pump configured to move the water collected in the water collector body to the cleaning water channel, and
- a channel switching valve connected to the pump and configured to selectively supply the water from the pump to the cleaning water channel,

wherein the channel switching valve comprises:

- a water receiving portion connected to the pump and configured to receive the water from the pump, and
 - a connective portion connected to the water receiving portion and disposed at a lateral side of the air circulating channel, the connective portion being configured to deliver the water from the water receiving portion to the cleaning water channel, and
- wherein at least portion of the connective portion is located lower than a top surface of the air circulating channel.

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2. The laundry treating apparatus of claim 1, wherein the air circulating channel comprises:

- an air flow duct that extends upward from the base and accommodates the first heat exchanger and the second heat exchanger; and
 - a duct cover that is coupled to the air flow duct and covers the first heat exchanger and the second heat exchanger, the cleaning water channel being disposed at a top surface of the duct cover, and
- wherein the connective portion is disposed at a side surface of the duct cover, the at least portion of the connective portion being located lower than the top surface of the duct cover.

3. The laundry treating apparatus of claim 2, wherein the duct cover defines:

- a cover through-hole that passes through the top surface of the duct cover and faces at least a portion of the first heat exchanger; and
 - a valve communication hole that passes through a surface of the cleaning water channel and fluidly communicates the cleaning water channel with the connective portion, and
- wherein the cleaning water channel extends from the valve communication hole to the cover through-hole and is configured discharge the water toward the first heat exchanger through the cover through-hole.

4. The laundry treating apparatus of claim 2, wherein the connective portion is integrally formed with the duct cover such that the water is delivered to the cleaning water channel without leakage at the connective portion.

5. The laundry treating apparatus of claim 4, wherein the connective portion defines:

- a supply hole connected to the water receiving portion and configured to receive the water from the water receiving portion; and
 - a valve communication hole that passes through a bottom surface of the cleaning water channel and is configured to deliver the water supplied from the supply hole to the cleaning water channel, and
- wherein the supply hole and the valve communication hole are spaced apart from each other and face different directions from each other.

6. The laundry treating apparatus of claim 5, wherein the connective portion further comprises a receiving channel configured to guide the water from the water receiving portion to the cleaning water channel, the receiving channel having a first end that defines the supply hole and a second end that defines the valve communication hole, and

- wherein the receiving channel extends in an inclined direction with respect to the top surface of the duct cover.

7. The laundry treating apparatus of claim 6, further comprising a water storage tank spaced from the base and connected to the connective portion, the water storage tank being configured to receive the water from the water collector body, and

wherein the connective portion further defines:

- a water receiving hole connected to the water receiving portion and configured to receive the water from the water receiving portion, and
- a water discharge hole connected to the water storage tank and configured to discharge the water received through the water receiving hole toward the water storage tank, the water discharge hole being spaced apart from the water receiving hole and facing away from the water receiving hole.

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8. The laundry treating apparatus of claim 7, wherein the connective portion further defines a connective transfer channel that has a first end defining the water receiving hole and a second end defining the water discharge hole, the connective transfer channel being configured to guide the water from the water receiving portion to the water storage tank, and

wherein the connective transfer channel and the receiving channel are defined in a single body.

9. The laundry treating apparatus of claim 4, wherein the channel switching valve further comprises a water delivering portion disposed between the water receiving portion and the connective portion and configured to guide the water supplied from the water receiving portion to the connective portion, and

wherein the connective portion is coupled to the water delivering portion and configured to receive the water from the water receiving portion through the water delivering portion.

10. The laundry treating apparatus of claim 9, wherein the channel switching valve further comprises a sealing member disposed between the connective portion and the water delivering portion and configured to block the water guided from the water delivering portion from leaking out between the connective portion and the water delivering portion.

11. The laundry treating apparatus of claim 9, further comprising a nozzle cover that is coupled to the top surface of the duct cover and covers the cleaning water channel,

wherein the nozzle cover comprises a fastening part that protrudes from a top surface of the nozzle cover and is coupled to the water delivering portion, and wherein the water delivering portion comprises:

a first fastening portion that extends from one side of the water delivering portion and is coupled to the fastening part of the nozzle cover, and

a second fastening portion that extends from another side of the water delivering portion and is coupled to the connective portion.

12. The laundry treating apparatus of claim 9, wherein the connective portion comprises a first fixing protrusion that protrudes upward from a top of the connective portion and a second fixing protrusion that protrudes downward from a bottom of the connective portion, and

wherein the water delivering portion comprises:

a first fastening portion that extends from one side of the water delivering portion and receives the first fixing protrusion, and

a second fastening portion that extends from another side of the water delivering portion and receives the second fixing protrusion.

13. The laundry treating apparatus of claim 4, wherein the cabinet comprises:

a first side panel that defines a first side of the cabinet and faces one side of the drum; and

a second side panel that defines a second side of the cabinet and faces another side of the drum,

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wherein the air flow duct and the duct cover are located closer to the second side panel than to the first side panel, and

wherein the connective portion extends from the duct cover toward the first side panel.

14. The laundry treating apparatus of claim 13, wherein the connective portion extends toward the first side panel along an inclined direction relative to an extension direction of the duct cover.

15. The laundry treating apparatus of claim 13, wherein the water collector body is located between the first side panel and the air flow duct, and

wherein the channel switching valve is located between the drum and the water collector body and overlaps with the water collector body along a vertical direction.

16. The laundry treating apparatus of claim 1, wherein a top of the water receiving portion is located lower than a bottom of the drum and is spaced apart from the bottom of the drum.

17. The laundry treating apparatus of claim 6, wherein the cleaning water channel comprises a plurality of cleaning water channels,

wherein the receiving channel comprises a plurality of receiving channels that are connected to the plurality of cleaning water channels, respectively, and

wherein a number of the plurality of receiving channels is equal to a number of the plurality of cleaning water channels.

18. The laundry treating apparatus of claim 17, wherein the channel switching valve further comprises a scroll accommodated in the water receiving portion and configured to selectively supply the water in the water receiving portion to the receiving channel, and

wherein the scroll is configured to, based on rotating about an axis, selectively communicate the water receiving portion with one of the plurality of receiving channels.

19. The laundry treating apparatus of claim 18, wherein the water receiving portion comprises:

a valve rotatable shaft coupled to the scroll and configured to rotate the scroll; and

a valve driver coupled to the valve rotatable shaft and configured to rotate the valve rotatable shaft,

wherein the scroll defines a scroll communication hole, a width of the scroll communication hole being equal to a diameter of one of the plurality of receiving channels, and

wherein the scroll communication hole is configured to selectively communicate with one of the plurality of receiving channels based on a rotation angle of the valve rotatable shaft.

20. The laundry treating apparatus of claim 19, wherein the scroll comprises a scroll plate that defines the scroll communication hole and a scroll coupling groove, the scroll coupling groove being defined at a rotation center of the scroll plate and coupled to the valve rotatable shaft.

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